(Deep) Neural Networks

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Fundamentals of Machine Learning

22/10/2024

OUTLINE

- Introduction
- From neural networks to deep learning
- In practice:
  - Transfer learning
  - Standard libraries
- Limitations & future
- Focus: few-shot learning

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Introduction

└ What does deep learning mean?

https://en.wikipedia.org/wiki/Deep_learning

Deep learning (also known as deep structured learning or differential programming) is part of a broader family of **machine learning methods** based on **artificial neural networks** with representation learning. Learning can be **supervised, semi-supervised or unsupervised**.

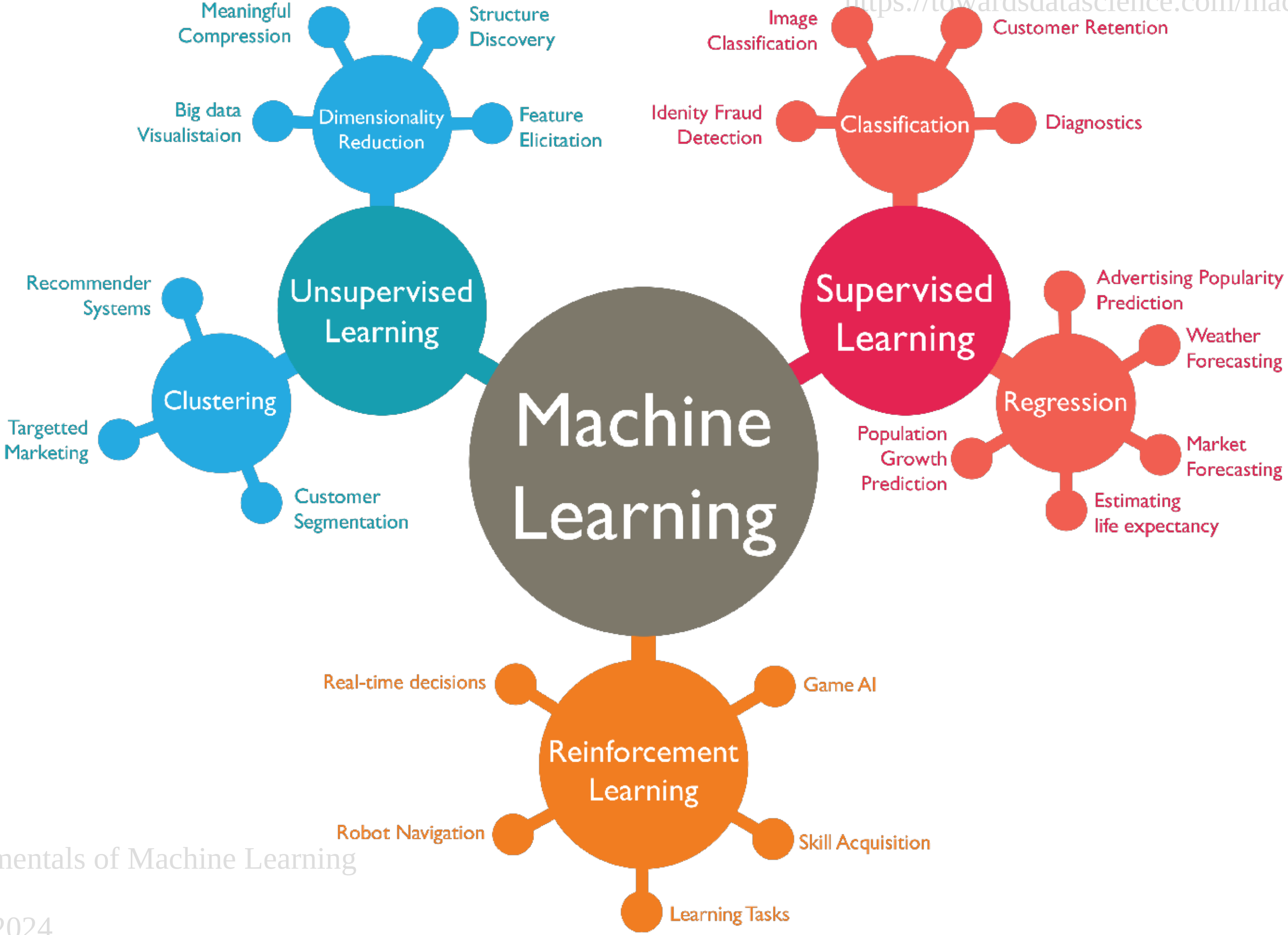
[...]

A deep neural network (DNN) is an artificial neural network (ANN) with multiple layers between the input and output layers. The DNN finds the correct mathematical manipulation to turn the input into the output, whether it be a linear relationship or a non-linear relationship. The network moves through the layers calculating the probability of each output. [...] Each mathematical manipulation as such is considered a layer, and complex DNN have many layers, hence the name "deep" networks.

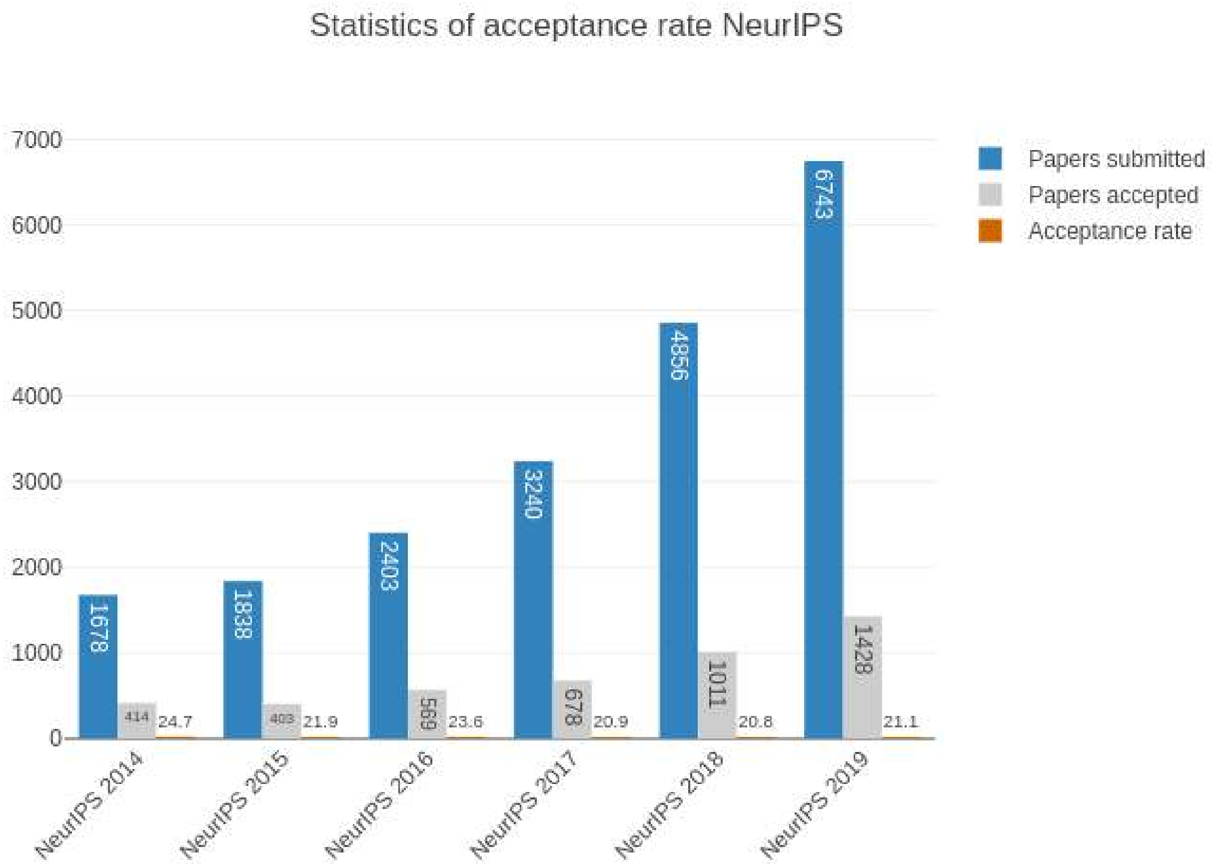
Introduction

└ Supervised, unsupervised, reinforcement

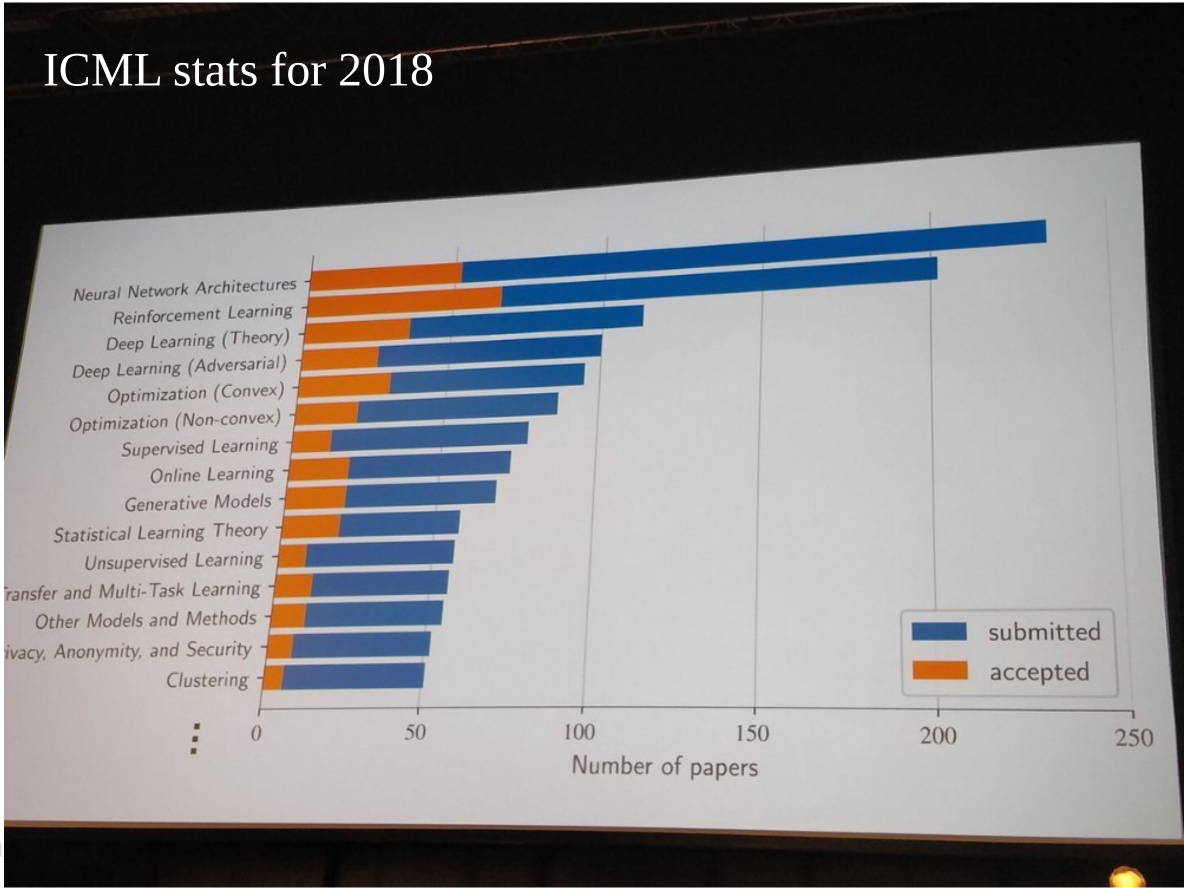
https://towardsdatascience.com/machine-learning-types-2-c1291d4f04b1



https://medium.com/@dcharrezt/neurips-2019-stats-c91346d31c8f



https://www.kdnuggets.com/2018/10/recent-advances-deep-learning.html



https://awards.acm.org/about/2018-turing

Yoshua Bengio is a Professor at the **University of Montreal**, and the Scientific Director of both Mila (Quebec's Artificial Intelligence Institute) and IVADO (the Institute for Data Valorization). He is Co-director (with Yann LeCun) of CIFAR's Learning in Machines and Brains program. Bengio received a Bachelor's degree in electrical engineering, a Master's degree in computer science and a Doctoral degree in computer science from McGill University.

Bengio's honors include being named an Officer of the Order of Canada, Fellow of the Royal Society of Canada and the Marie-Victorin Prize. His work in founding and serving as Scientific Director of the Quebec Artificial Intelligence Institute (Mila) is also recognized as a major contribution to the field. Mila, an independent nonprofit organization, now counts 300 researchers and 35 faculty members among its ranks. It is the largest academic center for deep learning research in the world, and has helped put Montreal on the map as a vibrant AI ecosystem, with research labs from major companies as well as AI startups.

Turing awards in 2018 (2/6)

https://awards.acm.org/about/2018-turing

Probabilistic models of sequences: In the 1990s, Bengio combined neural networks with probabilistic models of sequences, such as hidden Markov models. These ideas were incorporated into a system used by AT&T/NCR for reading handwritten checks, were considered a pinnacle of neural network research in the 1990s, and modern deep learning speech recognition systems are extending these concepts.

High-dimensional word embeddings and attention: In 2000, Bengio authored the landmark paper, “A Neural Probabilistic Language Model,” that introduced high-dimension word embeddings as a representation of word meaning. Bengio’s insights had a huge and lasting impact on natural language processing tasks including langage translation, question answering, and visual question answering. His group also introduced a form of attention mechanism which led to breakthroughs in machine translation and form a key component of sequential processing with deep learning.

Generative adversarial networks: Since 2010, Bengio’s papers on generative deep learning, in particular the Generative Adversarial Networks (GANs) developed with Ian Goodfellow, have spawned a revolution in computer vision and computer graphics. In one fascinating application of this work, computers can actually create original images, reminiscent of the creativity that is considered a hallmark of human intelligence.

https://awards.acm.org/about/2018-turing

Geoffrey Hinton is VP and Engineering Fellow of **Google**, Chief Scientific Adviser of The Vector Institute and a University Professor Emeritus at the University of Toronto. Hinton received a Bachelor's degree in experimental psychology from Cambridge University and a Doctoral degree in artificial intelligence from the University of Edinburgh. He was the founding Director of the Neural Computation and Adaptive Perception (later Learning in Machines and Brains) program at CIFAR.

Hinton's honors include Companion of the Order of Canada (Canada's highest honor), Fellow of the Royal Society (UK), foreign member of the National Academy of Engineering (US), the International Joint Conference on Artificial Intelligence (IJCAI) Award for Research Excellence, the NSERC Herzberg Gold medal, and the IEEE James Clerk Maxwell Gold medal. He was also selected by Wired magazine for "The Wired 100—2016's Most Influential People" and by Bloomberg for the 50 people who changed the landscape of global business in 2017.

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└ Turing awards in 2018 (3.5/6)

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https://awards.acm.org/about/2018-turing

Backpropagation: In a 1986 paper, “Learning Internal Representations by Error Propagation,” co-authored with David Rumelhart and Ronald Williams, Hinton demonstrated that the backpropagation algorithm allowed neural nets to discover their own internal representations of data, making it possible to use neural nets to solve problems that had previously been thought to be beyond their reach. The backpropagation algorithm is standard in most neural networks today.

Boltzmann Machines: In 1983, with Terrence Sejnowski, Hinton invented Boltzmann Machines, one of the first neural networks capable of learning internal representations in neurons that were not part of the input or output.

Improvements to convolutional neural networks: In 2012, with his students, Alex Krizhevsky and Ilya Sutskever, Hinton improved convolutional neural networks using rectified linear neurons and dropout regularization. In the prominent ImageNet competition, Hinton and his students almost halved the error rate for object recognition and reshaped the computer vision field.

Introduction

└ Turing awards in 2018 (5/6)

https://awards.acm.org/about/2018-turing

Yann LeCun is Silver Professor of the Courant Institute of Mathematical Sciences at New York University, and VP and Chief AI Scientist at **Facebook**. He received a Diplôme d'Ingénieur from the Ecole Superieure d'Ingénieur en Electrotechnique et Electronique (ESIEE), and a PhD in computer science from Université Pierre et Marie Curie.

His honors include being a member of the US National Academy of Engineering; Doctorates Honoris Causa, from IPN Mexico and École Polytechnique Fédérale de Lausanne (EPFL); the Pender Award, University of Pennsylvania; the Holst Medal, Technical University of Eindhoven & Philips Labs; the Nokia-Bell Labs Shannon Luminary Award; the IEEE PAMI Distinguished Researcher Award; and the IEEE Neural Network Pioneer Award. He was also selected by Wired magazine for “The Wired 100—2016’s Most Influential People” and its “25 Geniuses Who are Creating the Future of Business.” LeCun was the founding director of the NYU Center of Data Science, and is a Co-director (with Yoshua Bengio) of CIFAR’s Learning in Machines and Brains program. LeCun is also a co-founder and former Member of the Board of the Partnership on AI, a group of companies and nonprofits studying the societal consequences of AI.

https://awards.acm.org/about/2018-turing

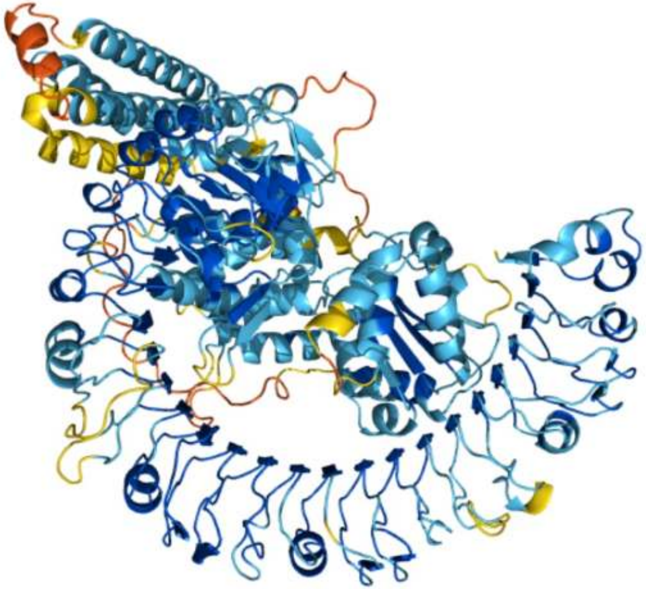
Convolutional neural networks: In the 1980s, LeCun developed convolutional neural networks, a foundational principle in the field, which, among other advantages, have been essential in making deep learning more efficient. In the late 1980s, while working at the University of Toronto and Bell Labs, LeCun was the first to train a convolutional neural network system on images of handwritten digits. Today, convolutional neural networks are an industry standard in computer vision, as well as in speech recognition, speech synthesis, image synthesis, and natural language processing. They are used in a wide variety of applications, including autonomous driving, medical image analysis, voice-activated assistants, and information filtering.

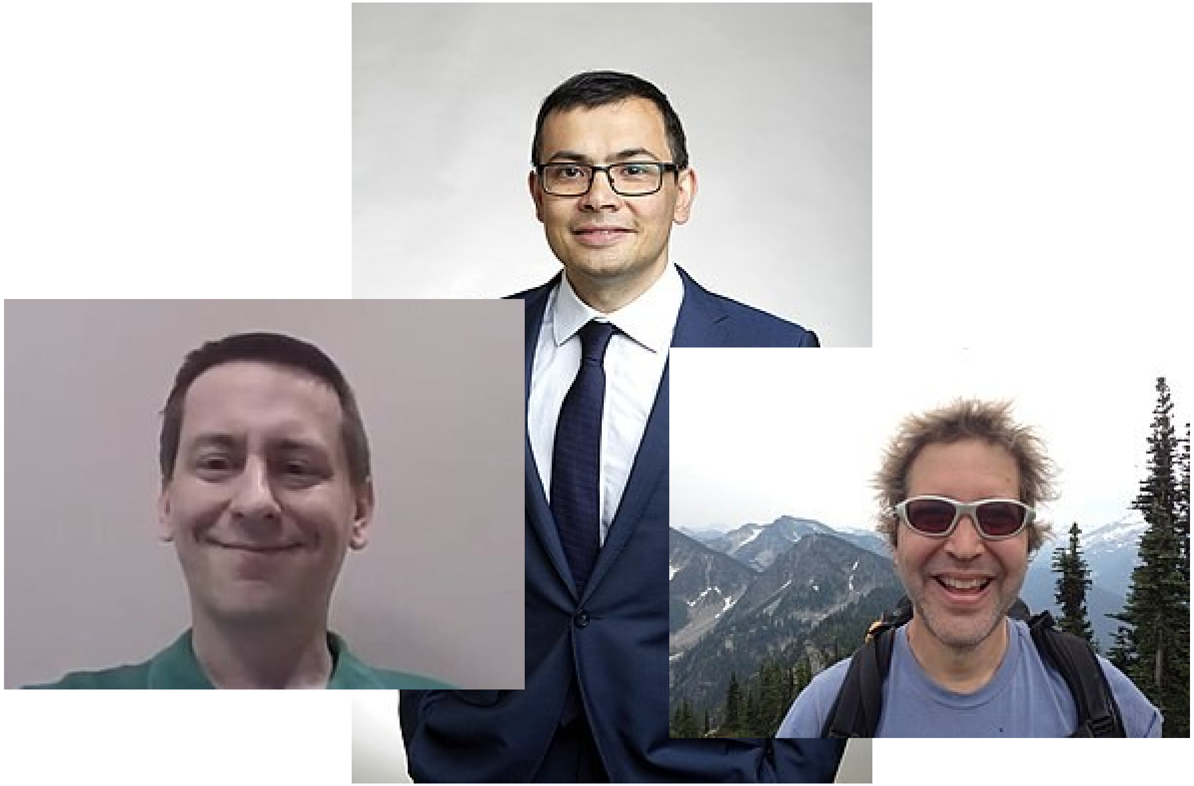
Improving backpropagation algorithms: LeCun proposed an early version of the backpropagation algorithm (backprop), and gave a clean derivation of it based on variational principles. His work to speed up backpropagation algorithms included describing two simple methods to accelerate learning time.

Broadening the vision of neural networks: LeCun is also credited with developing a broader vision for neural networks as a computational model for a wide range of tasks, introducing in early work a number of concepts now fundamental in AI. For example, in the context of recognizing images, he studied how hierarchical feature representation can be learned in neural networks—a concept that is now routinely used in many recognition tasks. Together with Léon Bottou, he proposed the idea, used in every modern deep learning software, that learning systems can be built as complex networks of modules where backpropagation is performed through automatic differentiation. They also proposed deep learning architectures that can manipulate structured data, such as graphs.

Introduction

└ Nobel prizes in 2024 (chemistry)





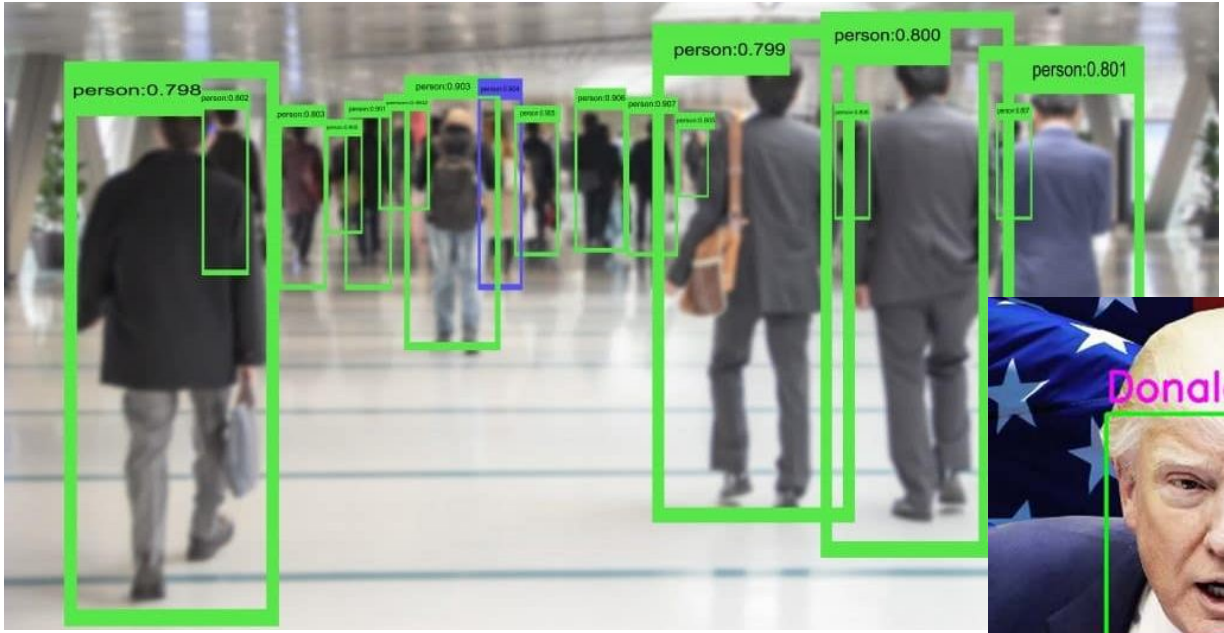
John Michael Jumper, Demis Hassabis, David Baker https://www.youtube.com/watch?v=gLoI9hAX9dw

Countless applications - Defense

https://www.clubic.com/drone/actualite-866669-essaims-drones-prochaine-etape-reconnaissance-test-armee-americaine.html
https://talwaq1998.wordpress.com/2019/06/16/artificial-intelligence-in-the-military/

https://www.kaggle.com/c/prudential-life-insurance-assessment
https://www.hockani.com/blog/62/the-most-evil-insurance-companies/

https://www.intelligence-artificielle-school.com/actualite/deep-learning-intelligence-artificielle/
https://morioh.com/p/2ccee92141ef

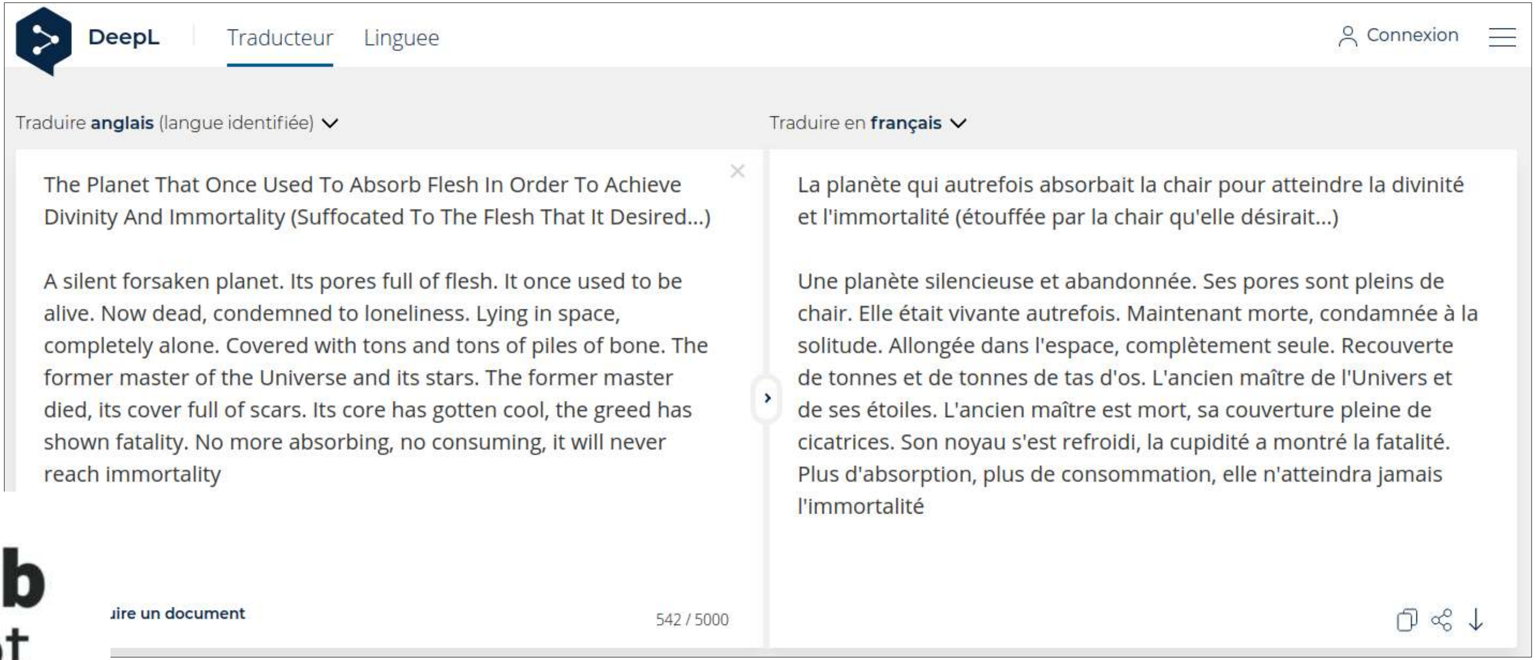


https://towardsdatascience.com/neural-networks-for-music-generation-97c983b50204
https://www.lebigdata.fr/midjourney-concours-art

Introduction

└ Countless applications - Natural language processing

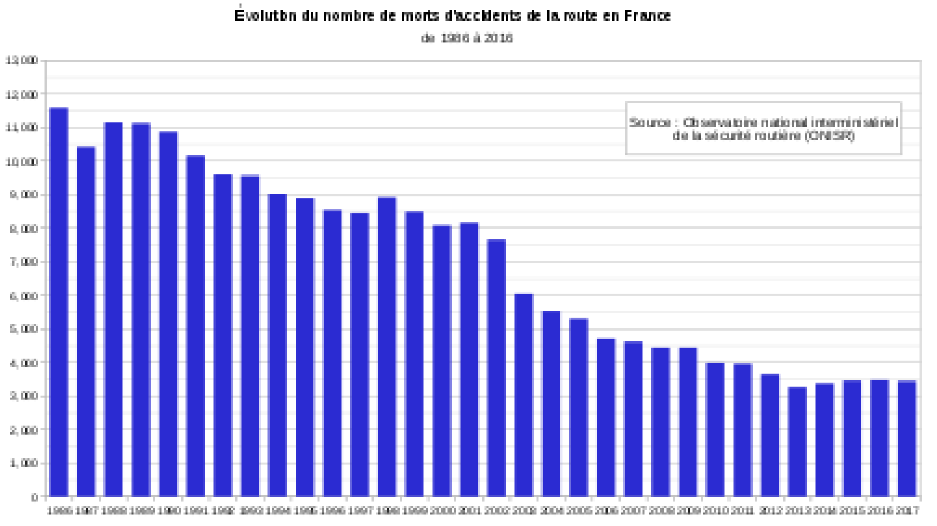
https://www.deepl.com/

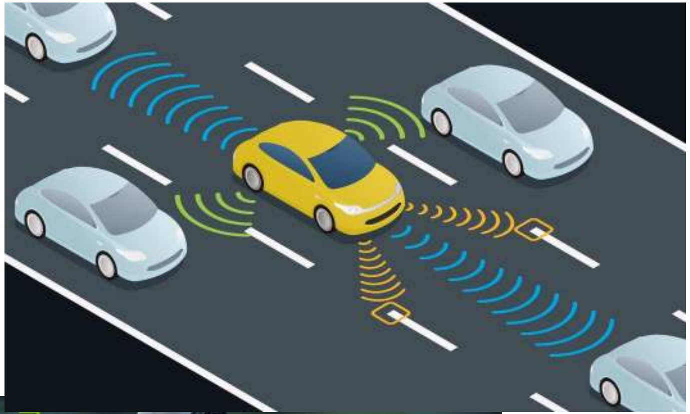


https://actu.meilleurmobile.com/google-assistant-alexa-siri-les-assistants-vocaux-au-coeur-dun-drole-de-test-de-piratage_200907
https://gmail.googleblog.com/2015/07/the-mail-you-want-not-spam-you-dont.html
s://www.amazon.fr/SPAM-Classic-12-Ounce-Cans-Foods/dp/B001EQ5NHE
https://www.1min30.com/social-media-marketing/2148-2148
s-D%C3%A9di%C3%A9s-Navigation-Embarqu%C3%A9e/dp/B01H0I0GUI

└ Countless applications - Self-driving cars

https://www.eltis.org/in-brief/news/trust-and-safety-among-top-priorities-autonomous-vehicles
https://www.fia.com/fr/node/18761
https://fr.wikipedia.org/wiki/Accident_de_la_route_en_France



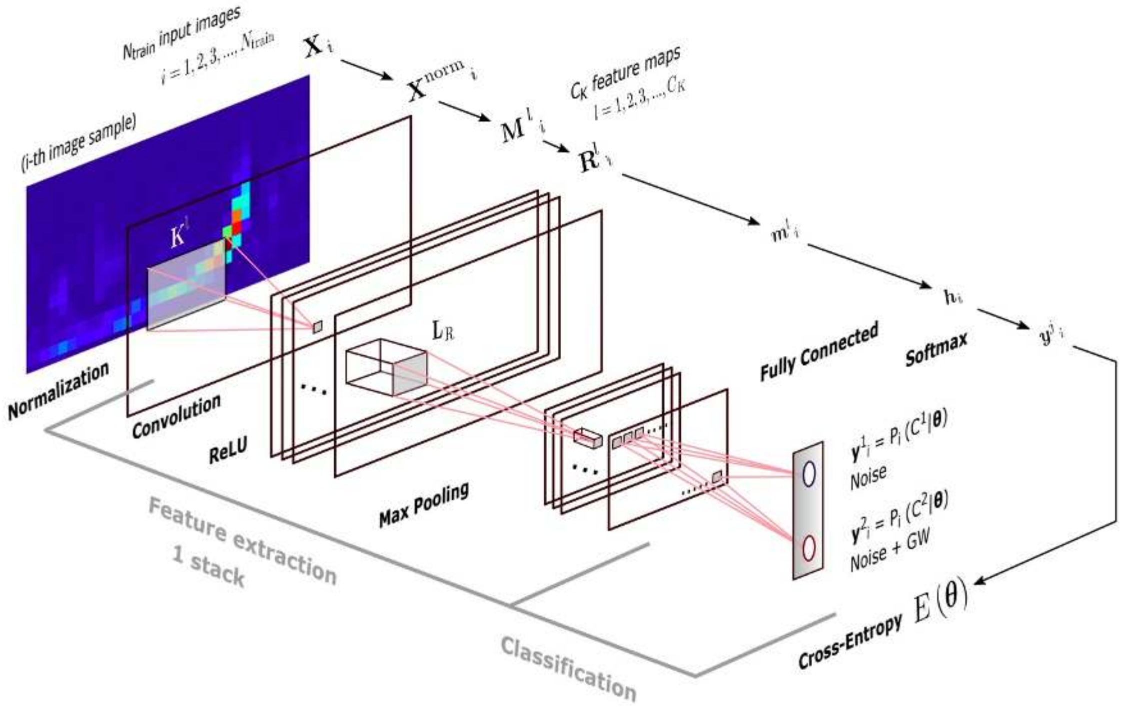


Introduction

Countless applications - Robotics

https://emerj.com/ai-sector-overviews/machine-learning-in-robotics/

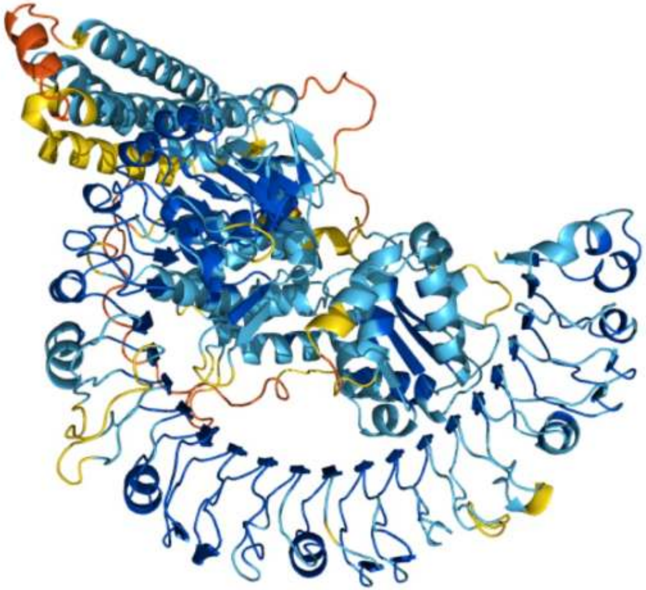
Morales et al. **Deep learning for gravitational-wave data analysis: A resampling white-box approach**. *Sensors 21*. 2021
https://hyperight.com/how-nasa-uses-ai-and-machine-learning-for-space-exploration/
https://www.elliptika.com/fr/

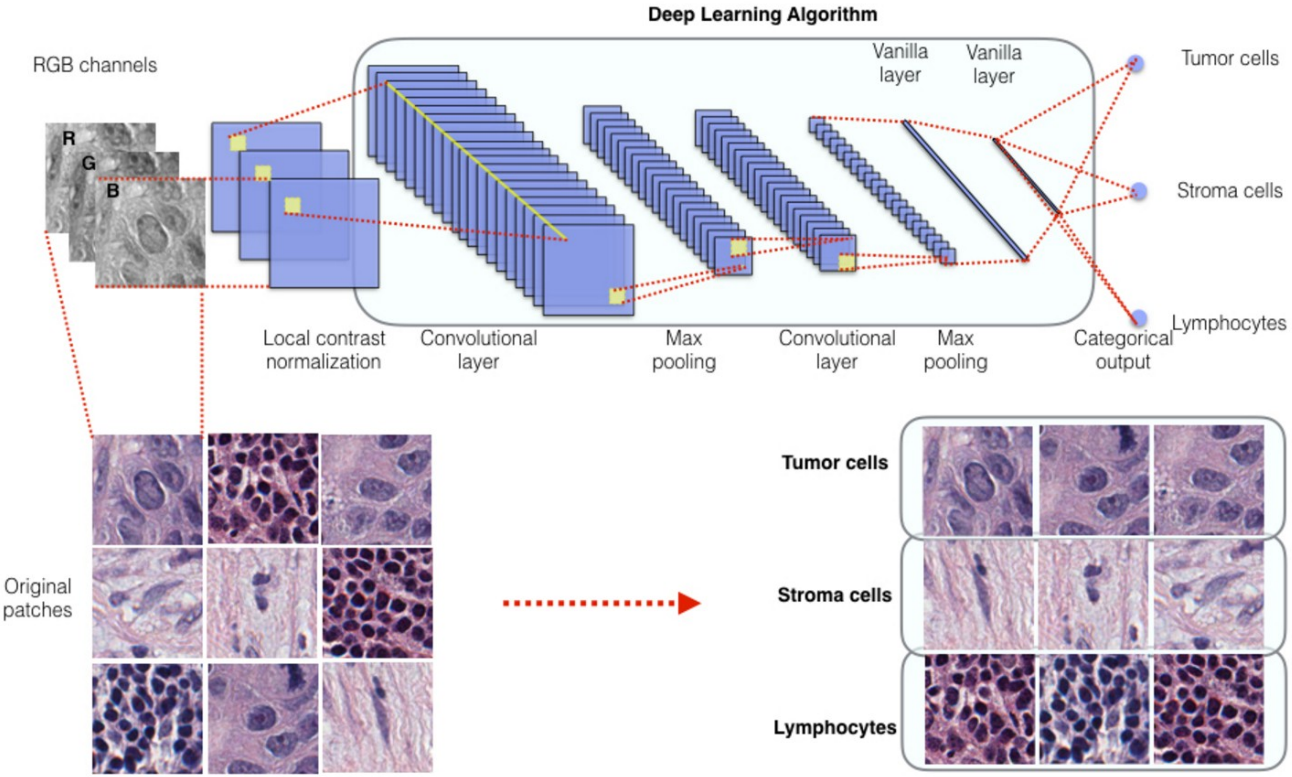


Introduction

└ Countless applications - Health

https://qbrc.swmed.edu/projects/cnn
https://alphafold.ebi.ac.uk/
s://www.wired.com/2014/07/everything-you-need-to-know-about-the-10-brain-myth-explained-in-60-seconds/





Lu et al. **The Expressive Power of Neural Networks: A View from the Width**. *Advances in Neural Information Processing Systems 30*. 2017

Theorem 1 (Universal Approximation Theorem for Width-Bounded ReLU Networks). *For any Lebesgue-integrable function $f : \mathbb{R}^n \to \mathbb{R}$ and any $\epsilon > 0$, there exists a fully-connected ReLU network $\mathscr{A}$ with width $d_m \leq n + 4$, such that the function $F_{\mathscr{A}}$ represented by this network satisfies*

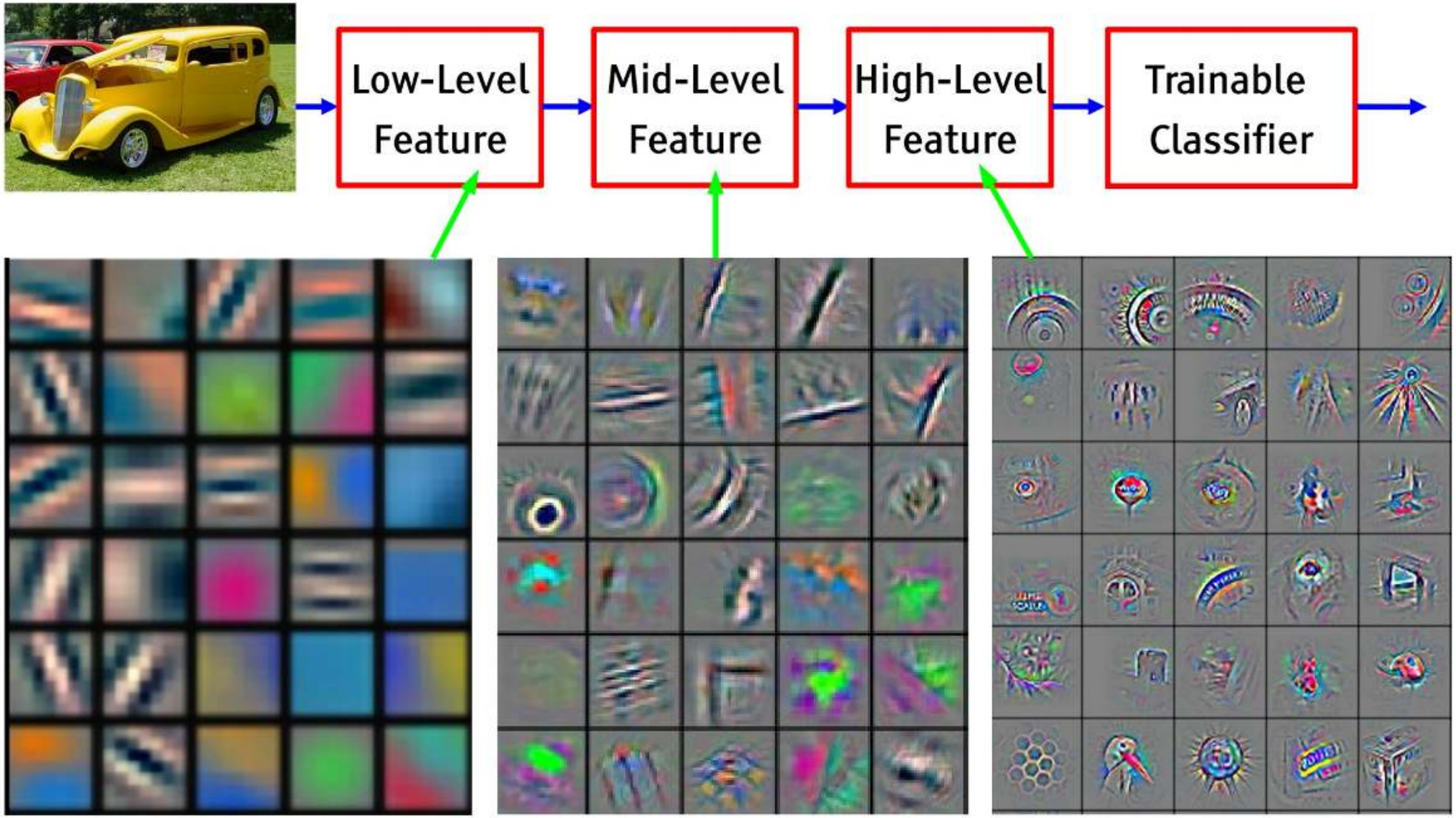
$$\int_{\mathbb{R}^n} |f(x) - F_{\mathscr{A}}(x)| \mathrm{d}x < \epsilon. \tag{3}$$

<u>In brief:</u> Neural networks can be seen as universal function approximators.
i.e., they will succeed in most application cases

Introduction

└ Why does deep learning work? - Learns to abstract

https://towardsdatascience.com/three-reasons-that-you-should-not-use-deep-learning-15bec517b622

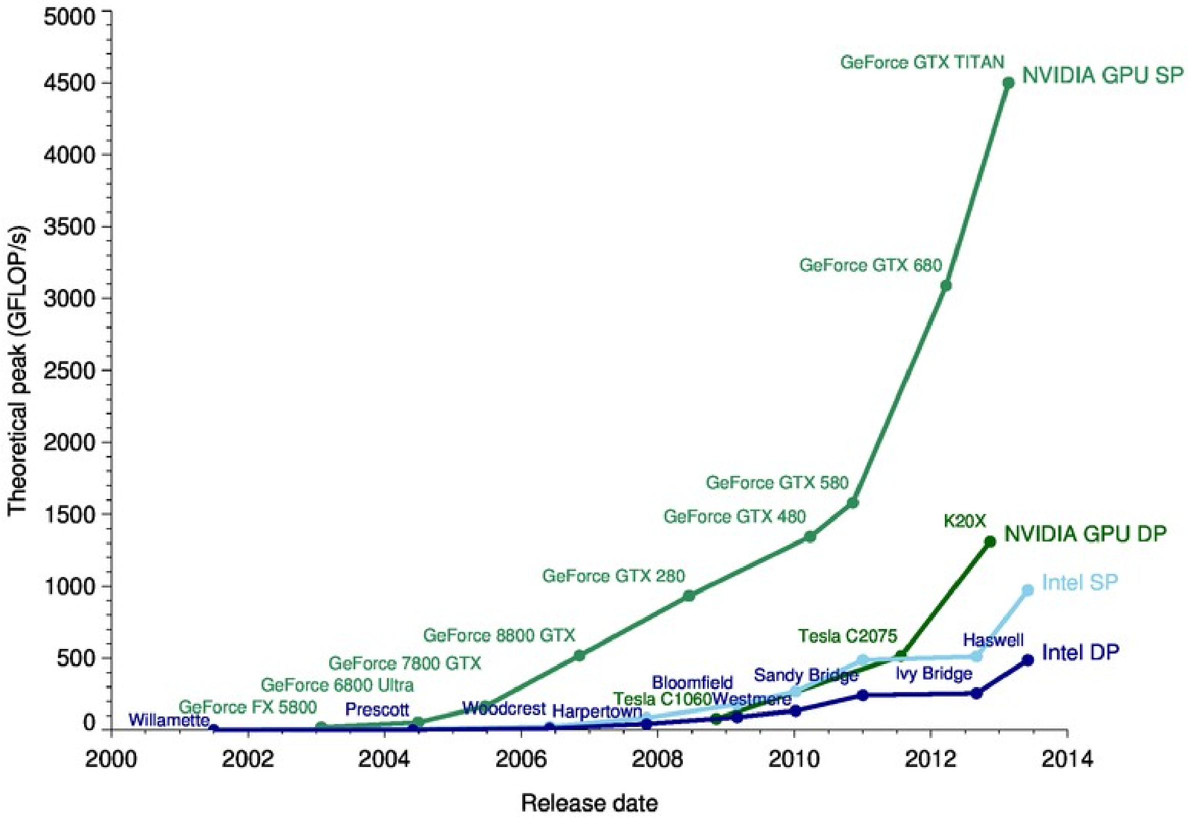


Feature visualization of convolutional net trained on ImageNet from [Zeiler & Fergus 2013]

Introduction

Why does deep learning work? - The rise of GPUs

https://bit.ly/2Vq3HDl
https://www.cybertek.fr/carte-graphique/asus/89643.aspx



OUTLINE

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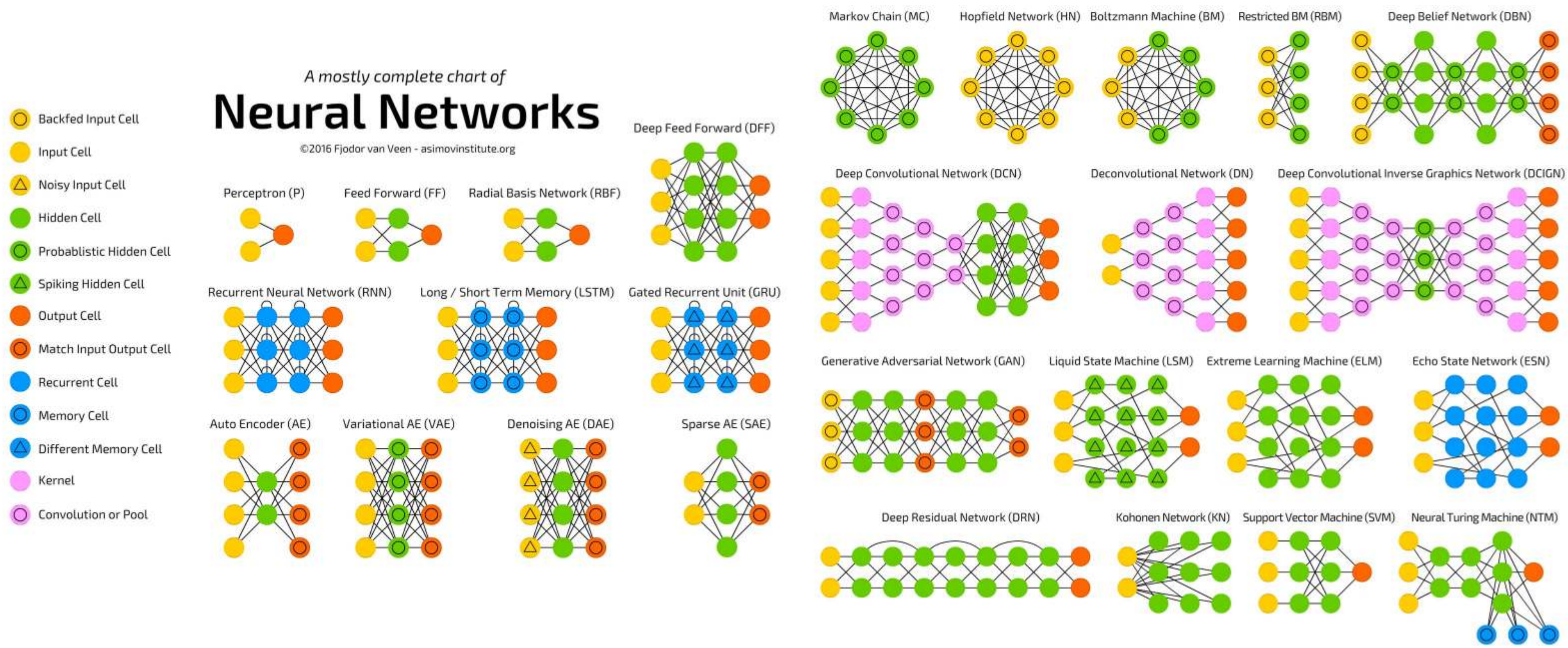
From neural networks to deep learning

Which ones do you know? (1/2)

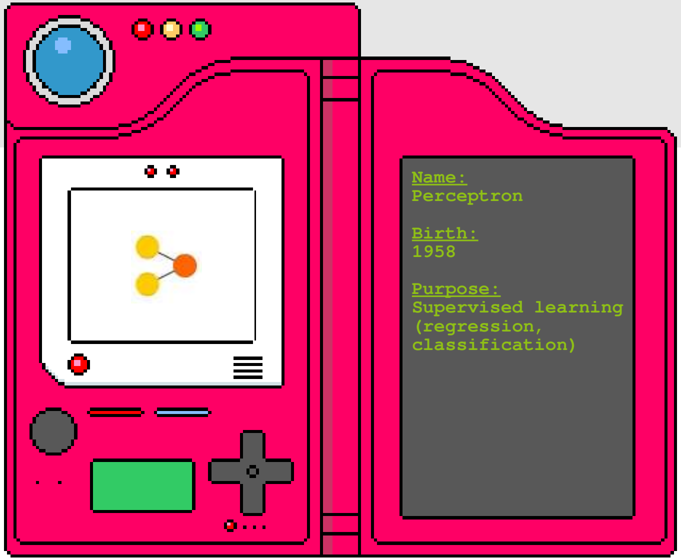
From neural networks to deep learning

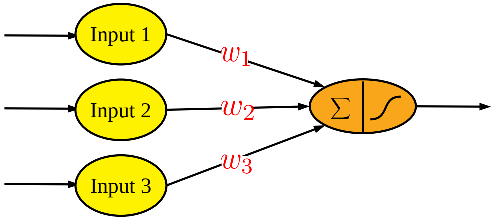
Which ones do you know? (2/2)

https://www.asimovinstitute.org/author/fjodorvanveen/



https://kamaleshjicse.files.wordpress.com/2018/11/pokedex-kamalesh.gif





- Building block of everything that comes afterward
- Weights are trained by backpropagating the error
- Learns weights (coeficients) of a hyperplane to separate data points
- Handles linearly separable datasets only

From neural networks to deep learning
└ Taxonomy of neural networks - Perceptron (2/3)

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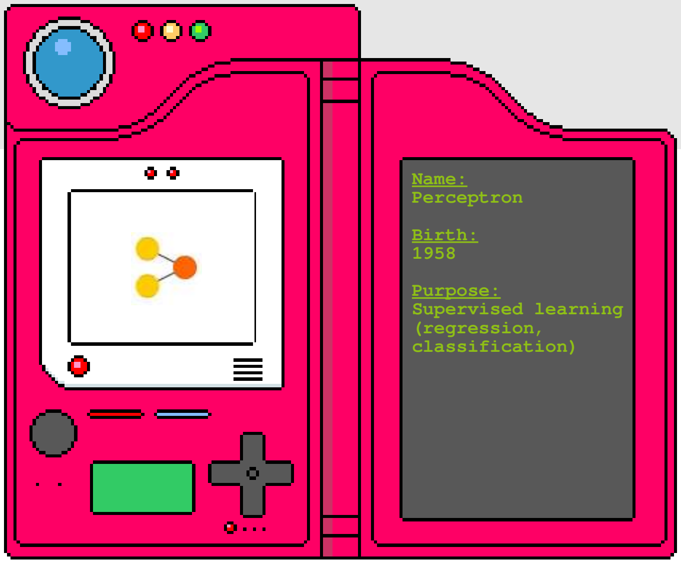

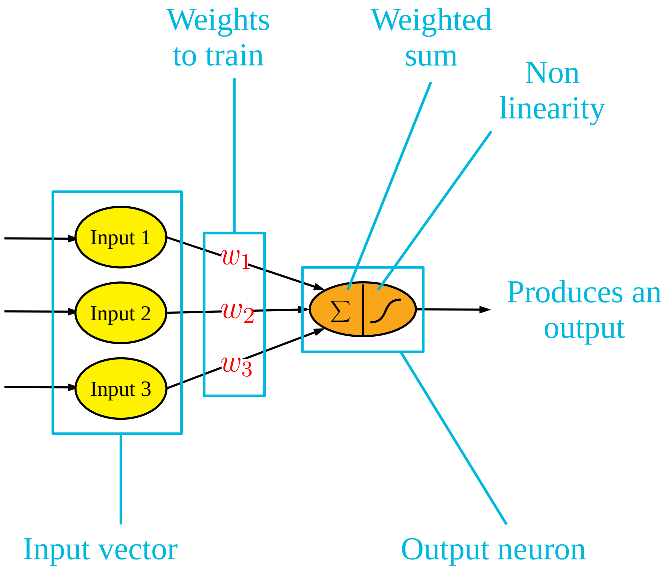

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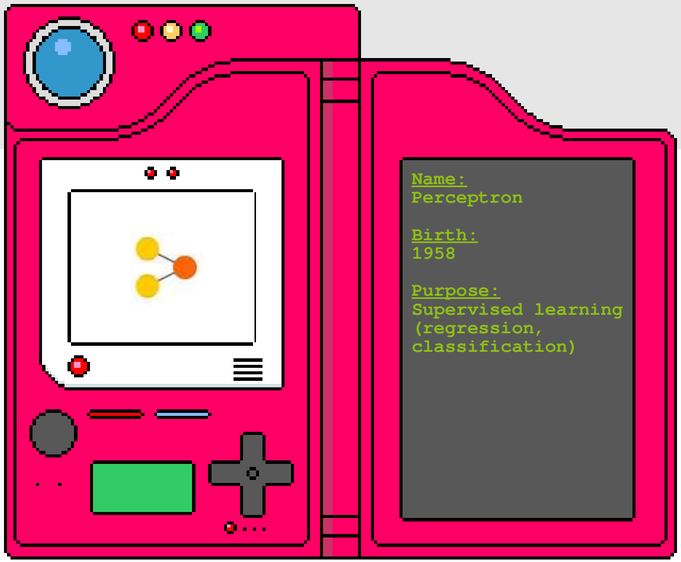
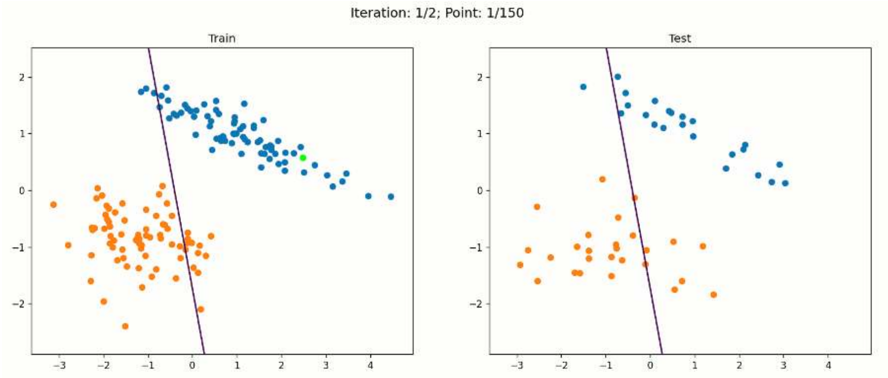
https://miro.medium.com/max/1800/1*c9WXUJdx00_fniK8h--k1A.gif





Example: Linear separability of points

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From neural networks to deep learning

└ Taxonomy of neural networks - MLP (1/3)

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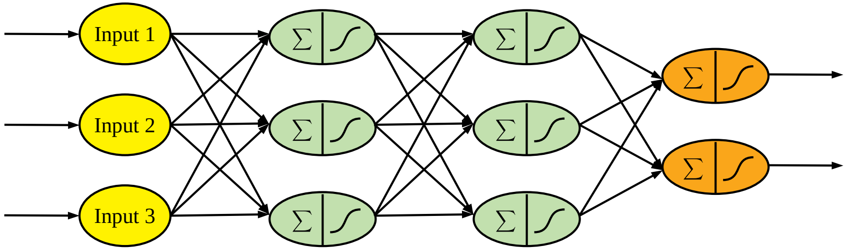
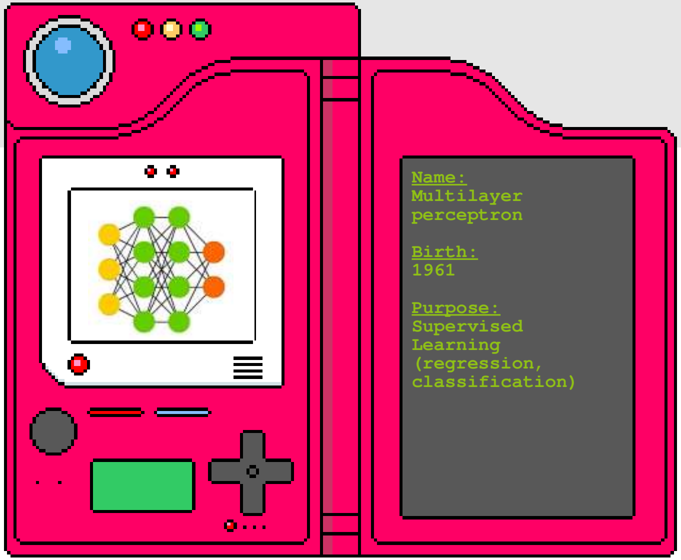
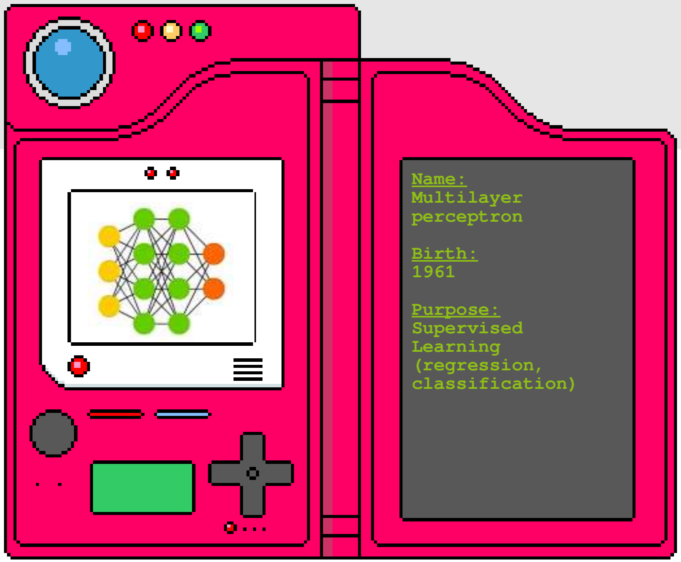
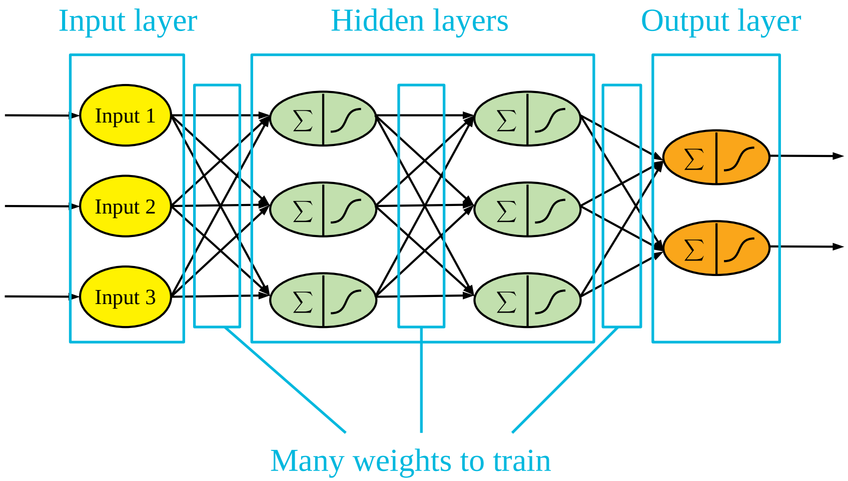




- ➢ Direct extension of the perceptron
- ➢ Also named « feedforward networks »
- ➢ Vertical and/or horizontal duplication of neurons
- ➢ Handles non-linearly separable datasets

https://kamaleshjicse.files.wordpress.com/2018/11/pokedex-kamalesh.gif





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From neural networks to deep learning

└ Taxonomy of neural networks - MLP (3/3)

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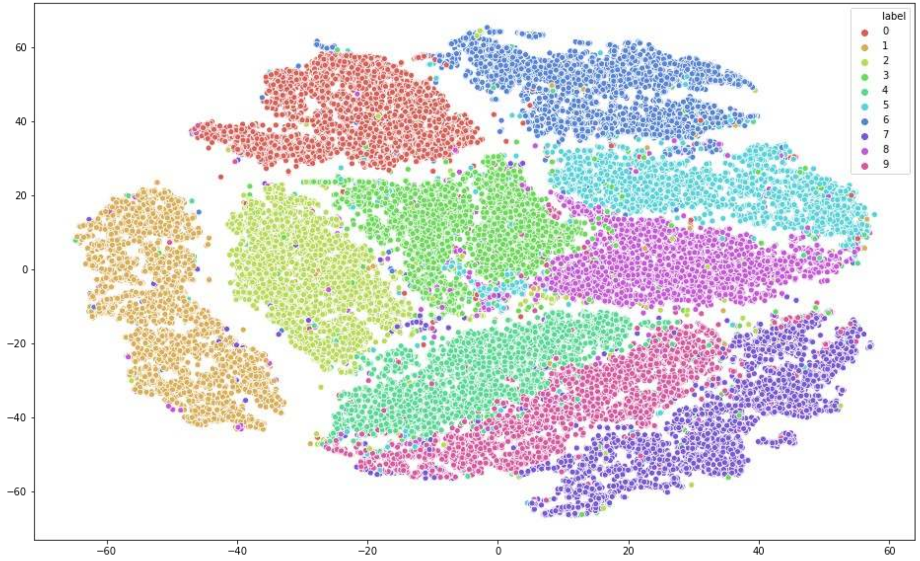
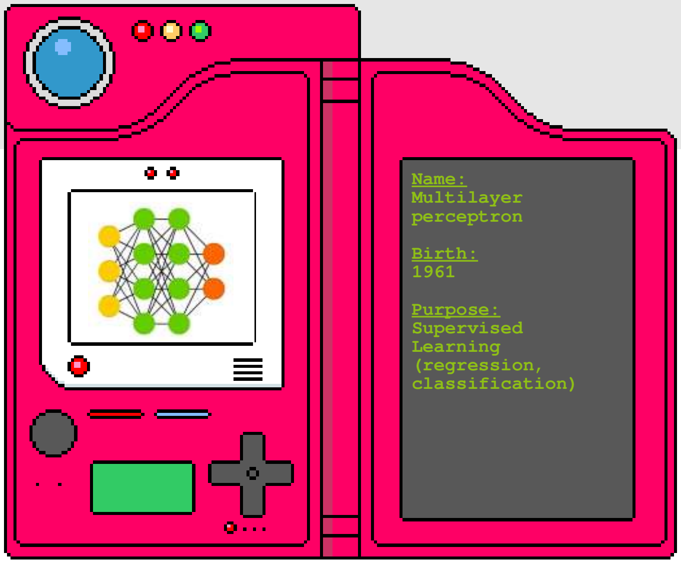


Example: MNIST 2D projection



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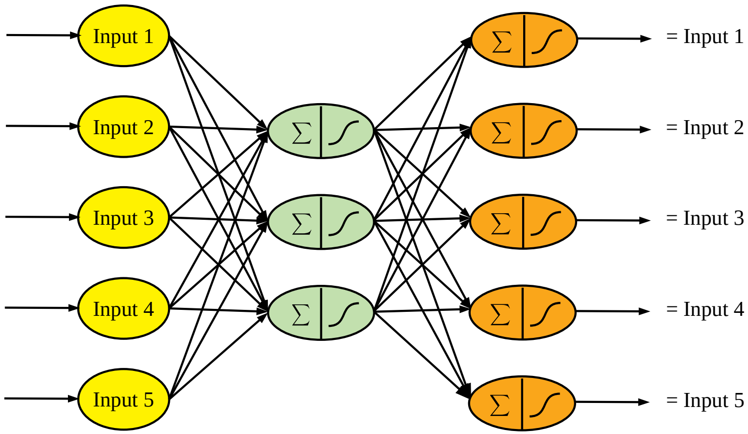
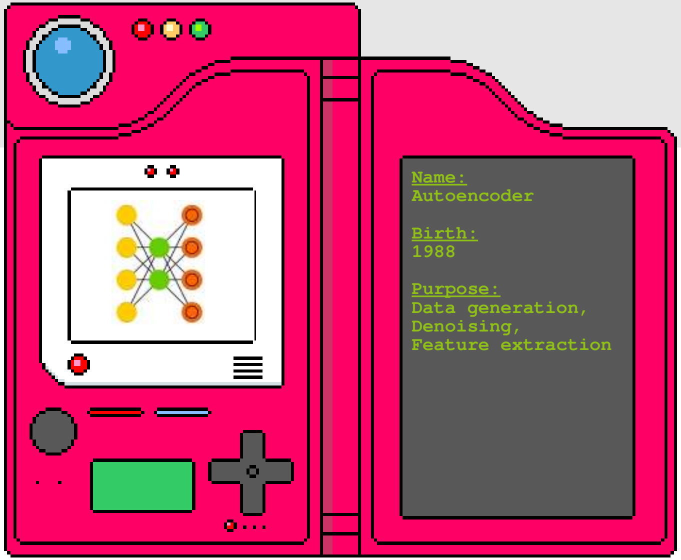
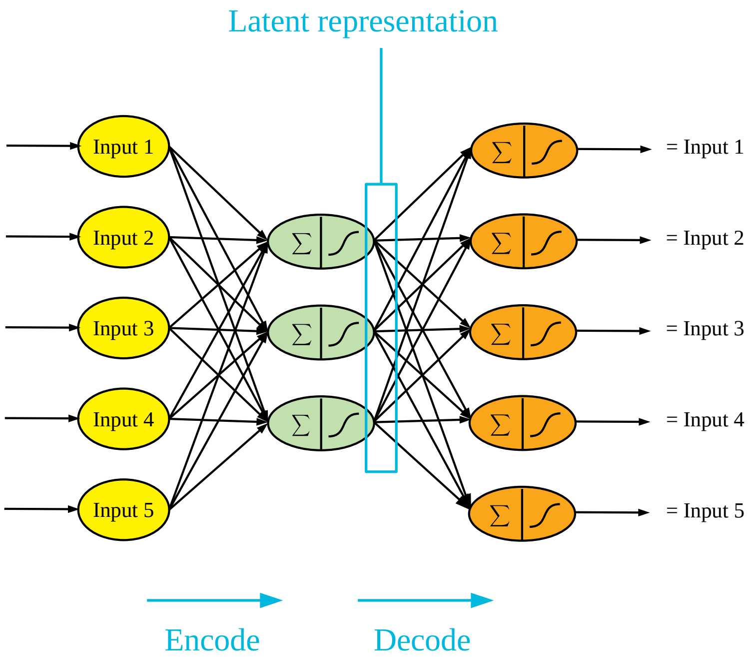
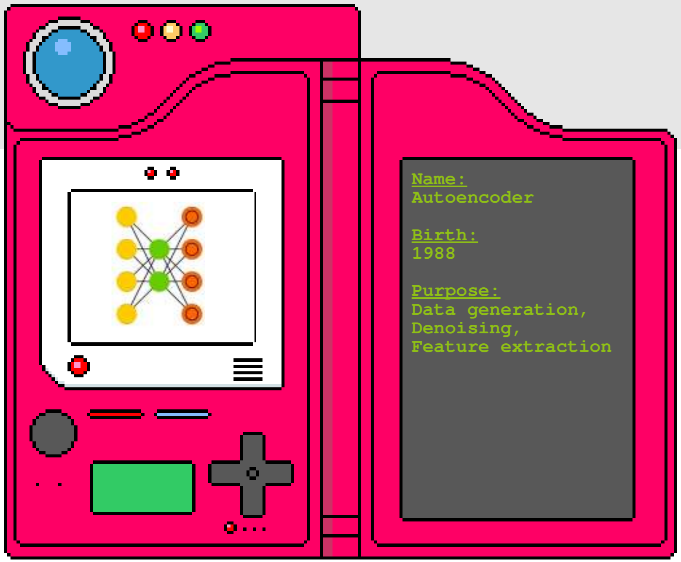




- Output is trained to reproduce input
- Does not require labeled data → unsupervised
- Goes through a compressed representation of data
- Many extensions exist

https://kamaleshjicse.files.wordpress.com/2018/11/pokedex-kamalesh.gif





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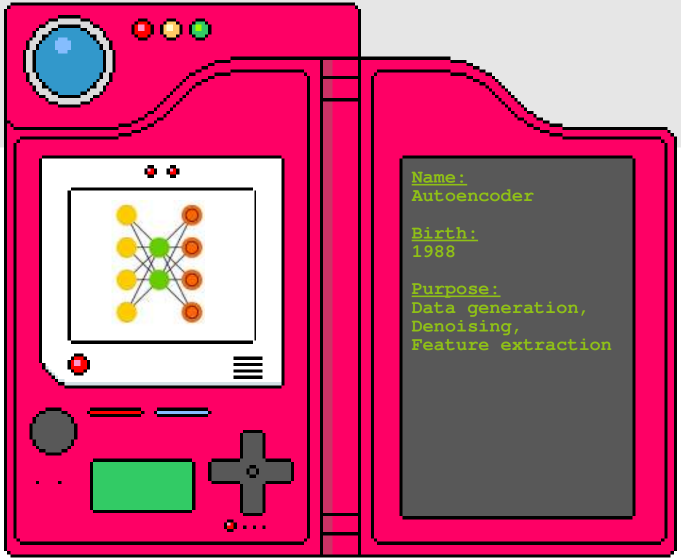
Taxonomy of neural networks - Autoencoder (3/3)

https://kamaleshjicse.files.wordpress.com/2018/11/pokedex-kamalesh.gif

Razavi et al. **Generating diverse high-fidelity images with vq-vae-2.** *Advances in Neural Information Processing Systems.* 2019



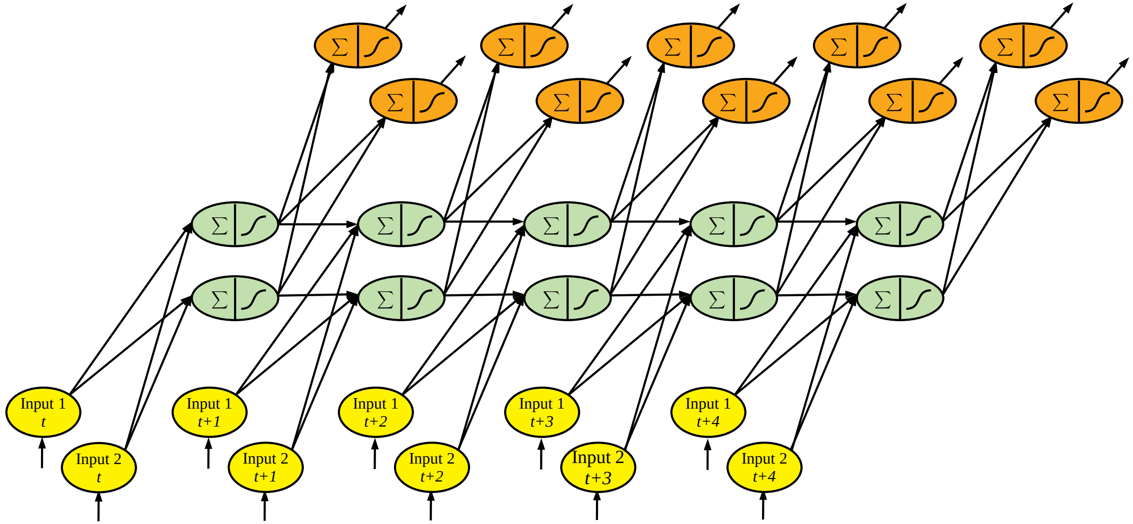
<u>Example:</u> Face generation

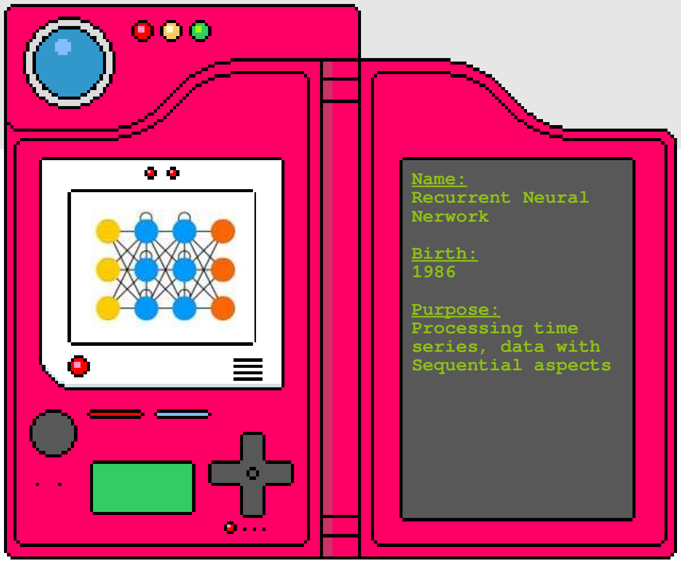
- Output is trained to reproduce input
- Does not require labeled data → unsupervised
- Goes through a compressed representation of data
- Many extensions exist

From neural networks to deep learning

└ Taxonomy of neural networks - RNN (1/3)

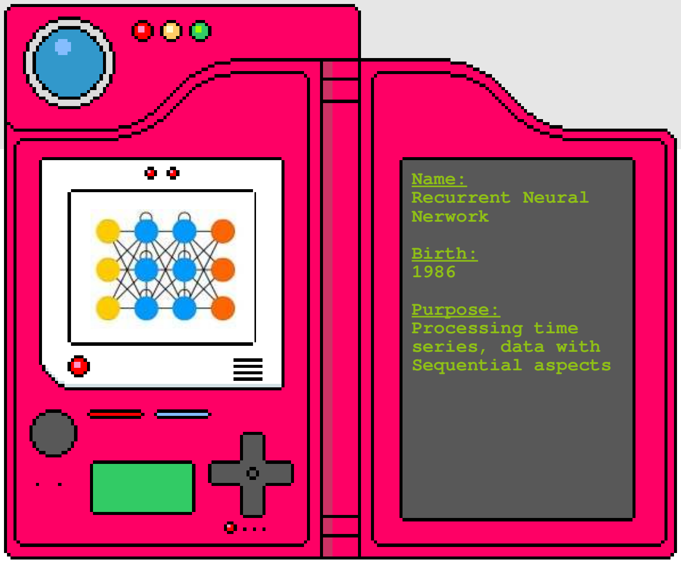
https://kamaleshjicse.files.wordpress.com/2018/11/pokedex-kamalesh.gif
https://medium.com/@jon.froiland/advanced-use-of-recurrent-neural-networks-part-1-91ff1b36af25

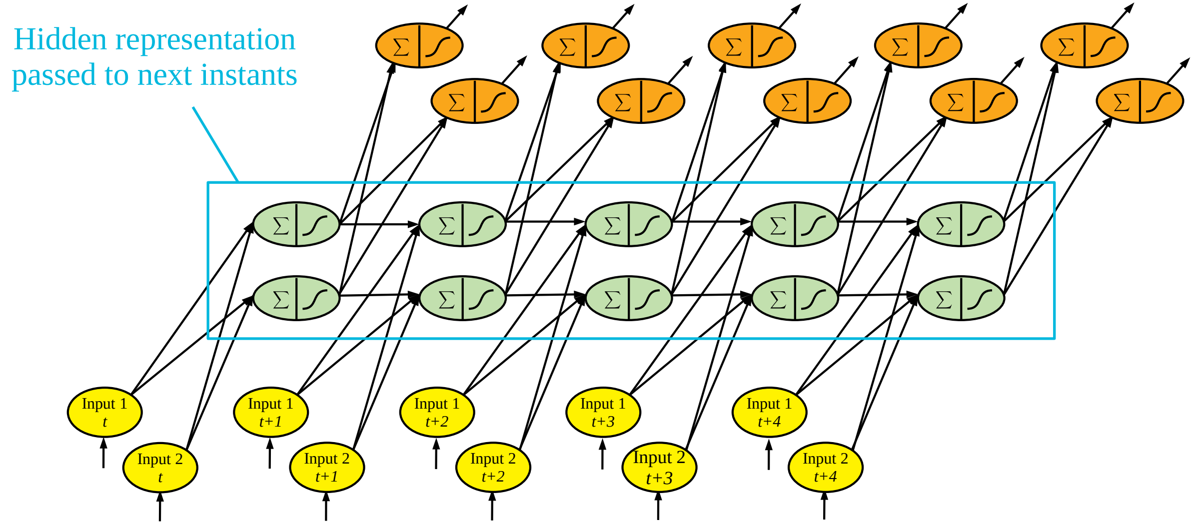




- Takes into account sequentiality of information (time series, texts…)
- Suffers a from gradient vanishing problem
- Base model for LSTMs or GRUs

https://kamaleshjicse.files.wordpress.com/2018/11/pokedex-kamalesh.gif
https://medium.com/@jon.froiland/advanced-use-of-recurrent-neural-networks-part-1-91ff1b36af25



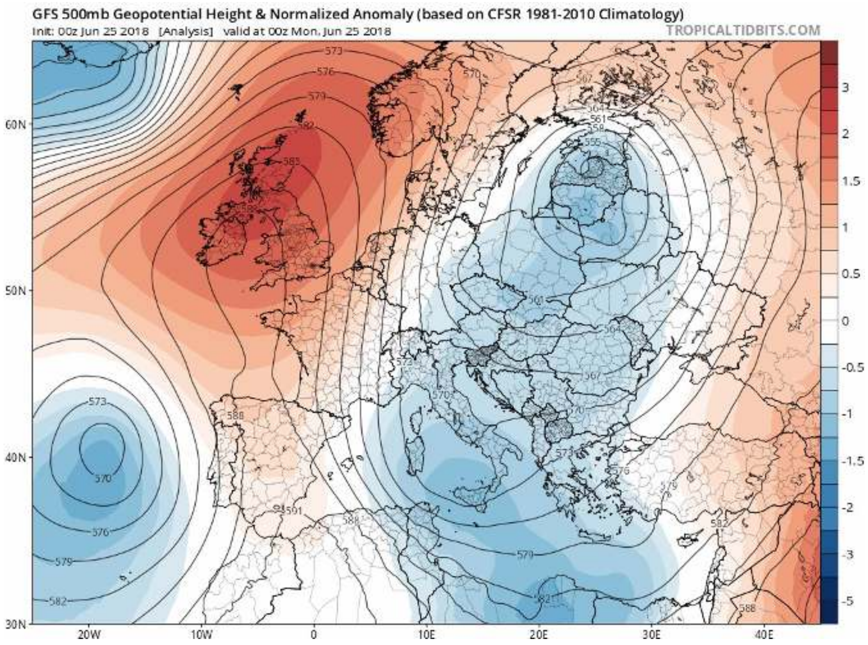


- ➢ Takes into account sequentiality of information (time series, texts…)
- ➢ Suffers a from gradient vanishing problem
- ➢ Base model for LSTMs or GRUs

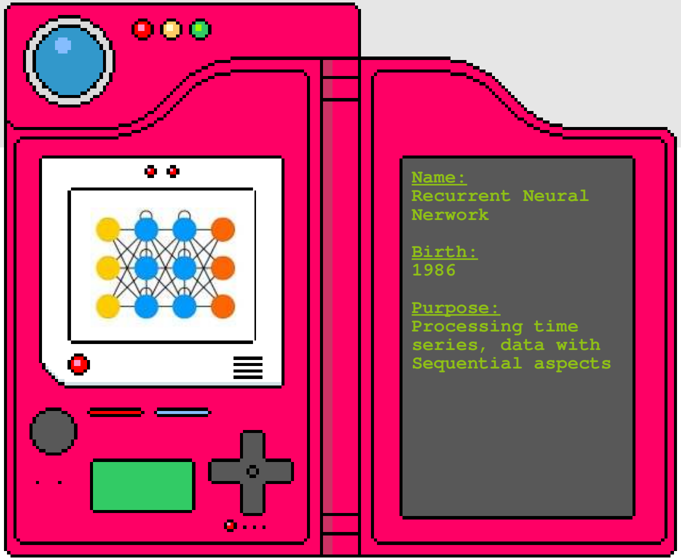
From neural networks to deep learning

└ Taxonomy of neural networks - RNN (3/3)

https://kamaleshjicse.files.wordpress.com/2018/11/pokedex-kamalesh.gif
https://medium.com/@jon.froiland/advanced-use-of-recurrent-neural-networks-part-1-91ff1b36af25



Example: Temperature forecasting

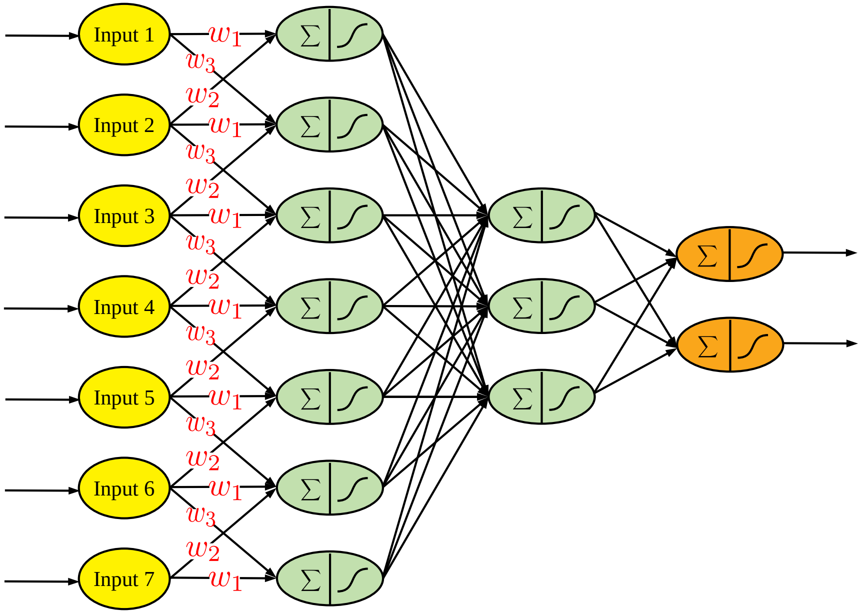


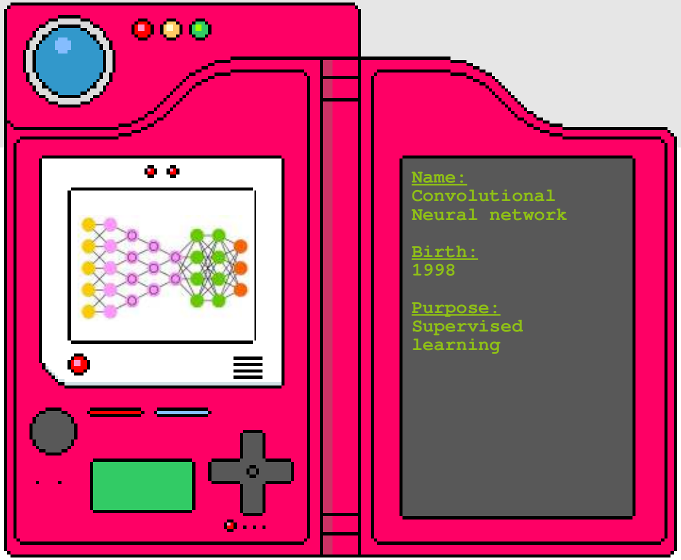
- Takes into account sequentiality of information (time series, texts…)
- Suffers a from gradient vanishing problem
- Base model for LSTMs or GRUs

From neural networks to deep learning

└ Taxonomy of neural networks - CNN (1/4)

https://kamaleshjicse.files.wordpress.com/2018/11/pokedex-kamalesh.gif

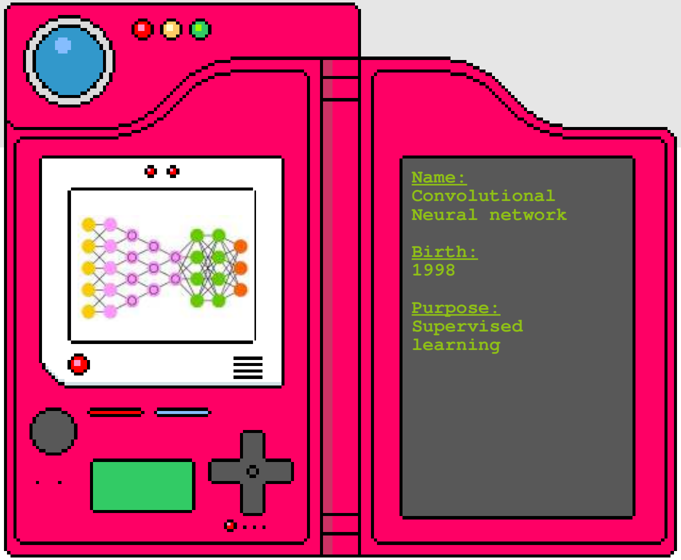




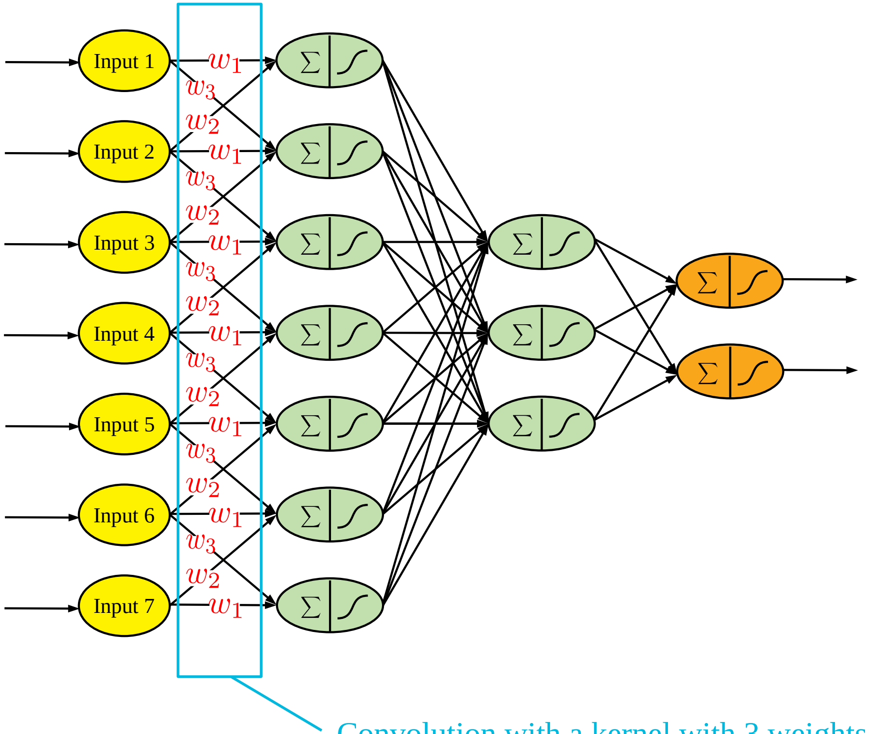
- Linear combination is replaced by convolution (captures locality of features)
- Some weights are shared (translation invariance)
- Ideal for data lying on Euclidean spaces
- Graph CNN extend this idea to irregular domains

From neural networks to deep learning

└ Taxonomy of neural networks - CNN (2/4)



https://kamaleshjicse.files.wordpress.com/2018/11/pokedex-kamalesh.gif

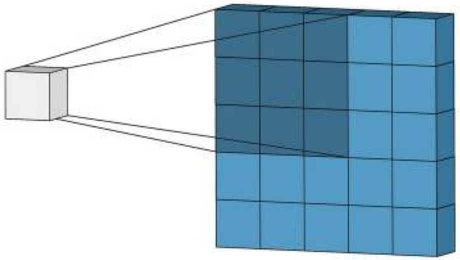


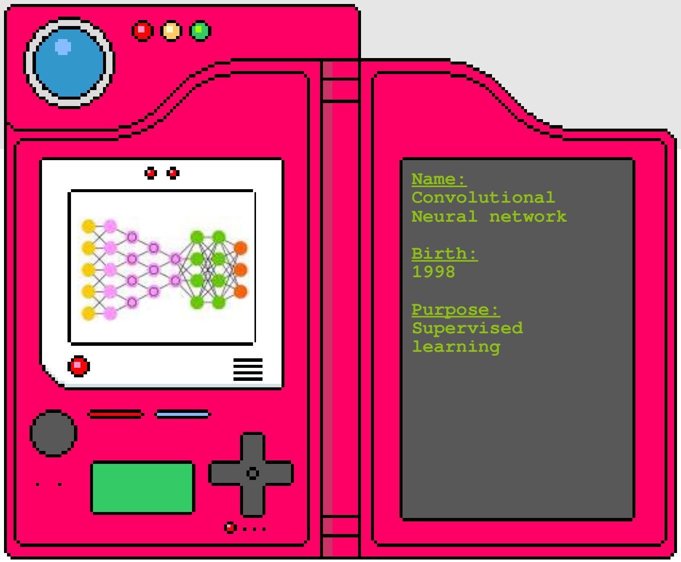
Convolution with a kernel with 3 weights

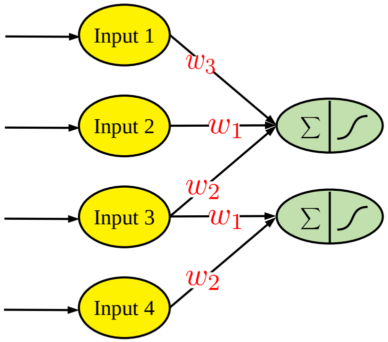
- Linear combination is replaced by convolution (captures locality of features)
- Some weights are shared (translation invariance)
- Ideal for data lying on Euclidean spaces
- Graph CNN extend this idea to irregular domains

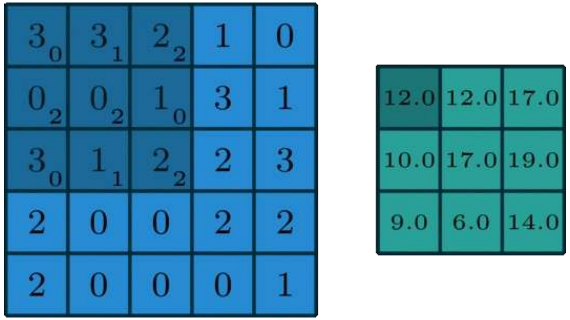
From neural networks to deep learning
└ Taxonomy of neural networks - CNN (3/4)

https://kamaleshjicse.files.wordpress.com/2018/11/pokedex-kamalesh.gif
https://towardsdatascience.com/intuitively-understanding-convolutions-for-deep-learning-1f6f42faee1







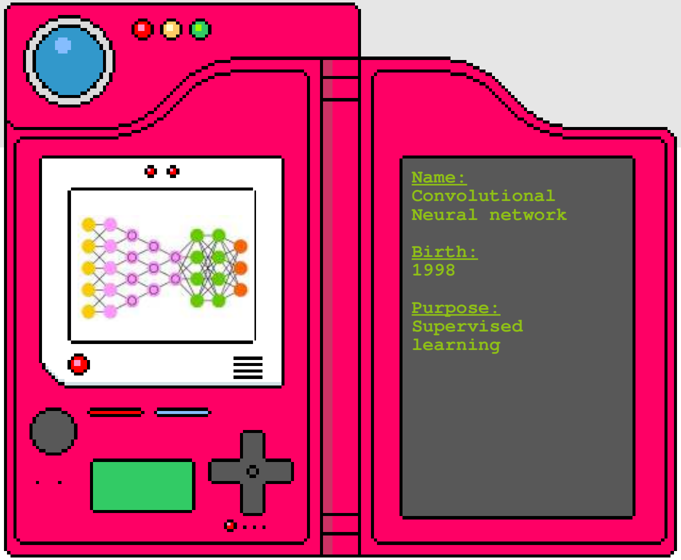


- ➢ Linear combination is replaced by convolution (captures locality of features)
- ➢ Some weights are shared (translation invariance)
- ➢ Ideal for data lying on Euclidean spaces
- ➢ Graph CNN extend this idea to irregular domains

From neural networks to deep learning

└ Taxonomy of neural networks - CNN (4/4)

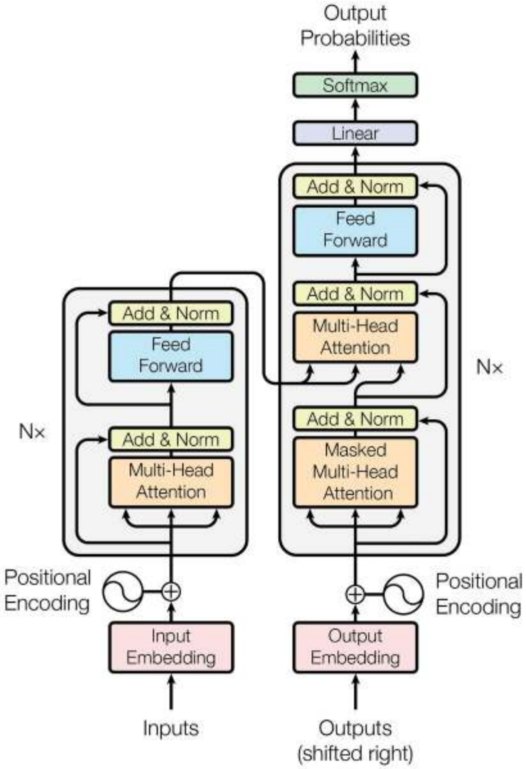
https://kamaleshjicse.files.wordpress.com/2018/11/pokedex-kamalesh.gif
https://jonathan-hui.medium.com/image-segmentation-with-mask-r-cnn-ebe6d793272

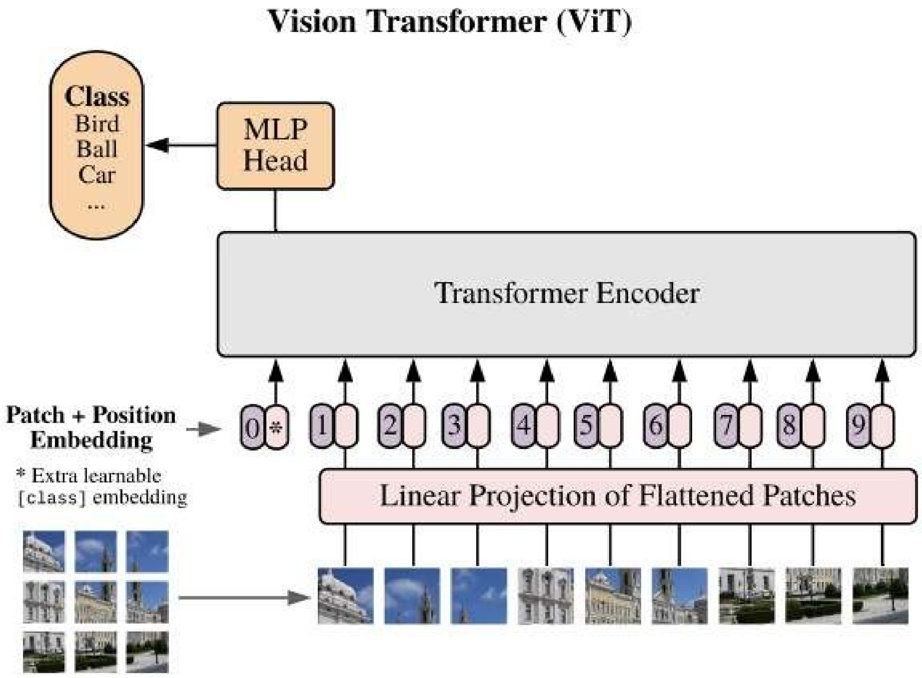


Example: Segmentation of images

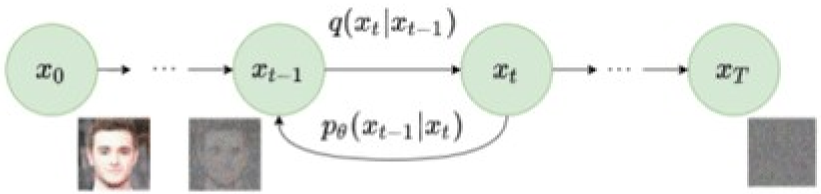
- ➢ Linear combination is replaced by convolution (captures locality of features)
- ➢ Some weights are shared (translation invariance)
- ➢ Ideal for data lying on Euclidean spaces
- ➢ Graph CNN extend this idea to irregular domains

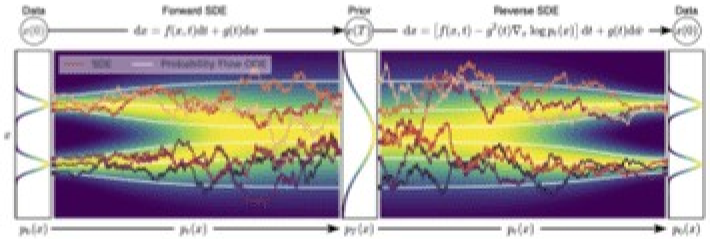
https://machinelearningmastery.com/wp-content/uploads/2021/08/attention_research_1.png
https://production-media.paperswithcode.com/methods/Screen_Shot_2021-01-26_at_9.43.31_PM_uI4jjMq.png

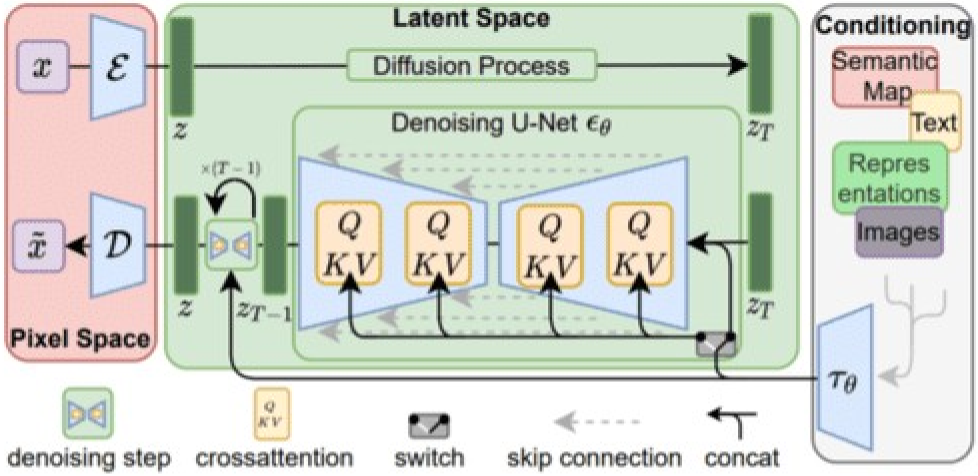




https://theaisummer.com/diffusion-models/







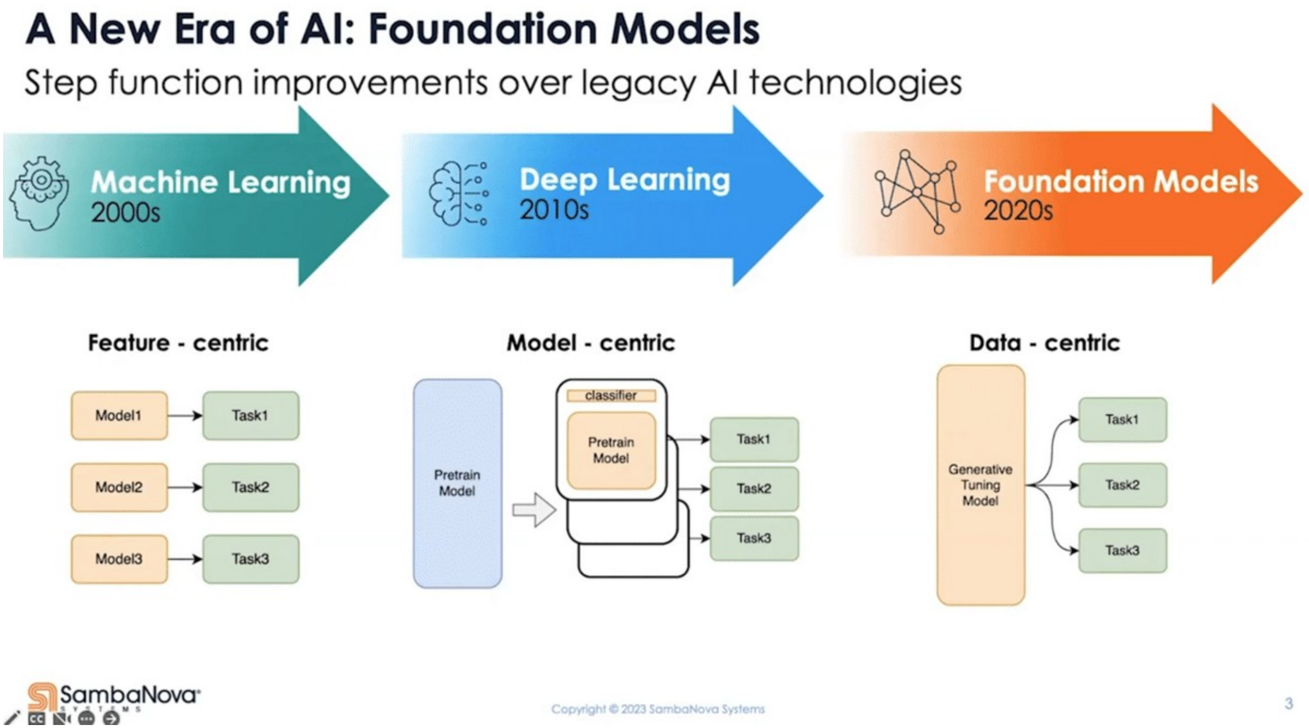
Algorithm 1 Training

repeat
  $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
  $t \sim \mathrm{Uniform}(\{1, \dots, T\})$
  $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
  Take gradient descent step on
  $\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar\alpha_t}\mathbf{x}_0 + \sqrt{1-\bar\alpha_t}\boldsymbol{\epsilon}, t) \right\|^2$
until converged

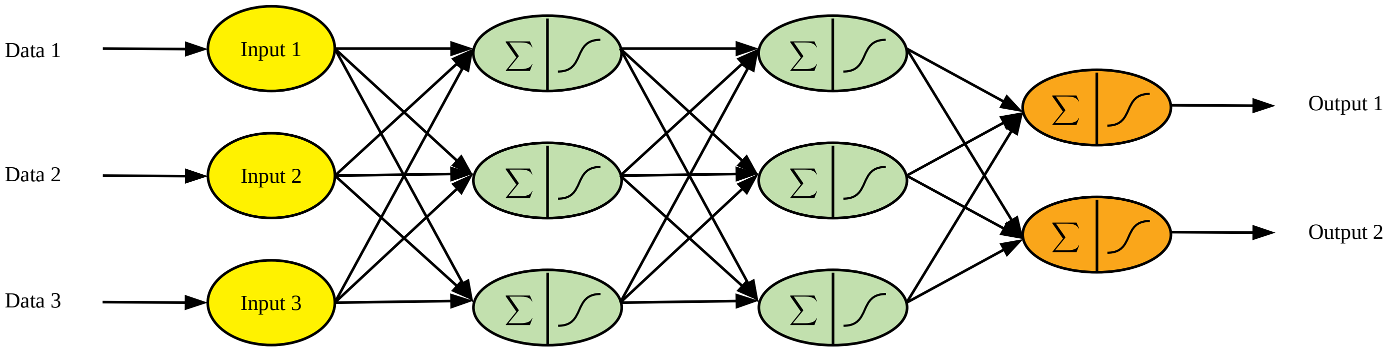
Algorithm 2 Sampling

$\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
for $t = T, \dots, 1$ **do**
  $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
  $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar\alpha_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$
end for
return $\mathbf{x}_0$

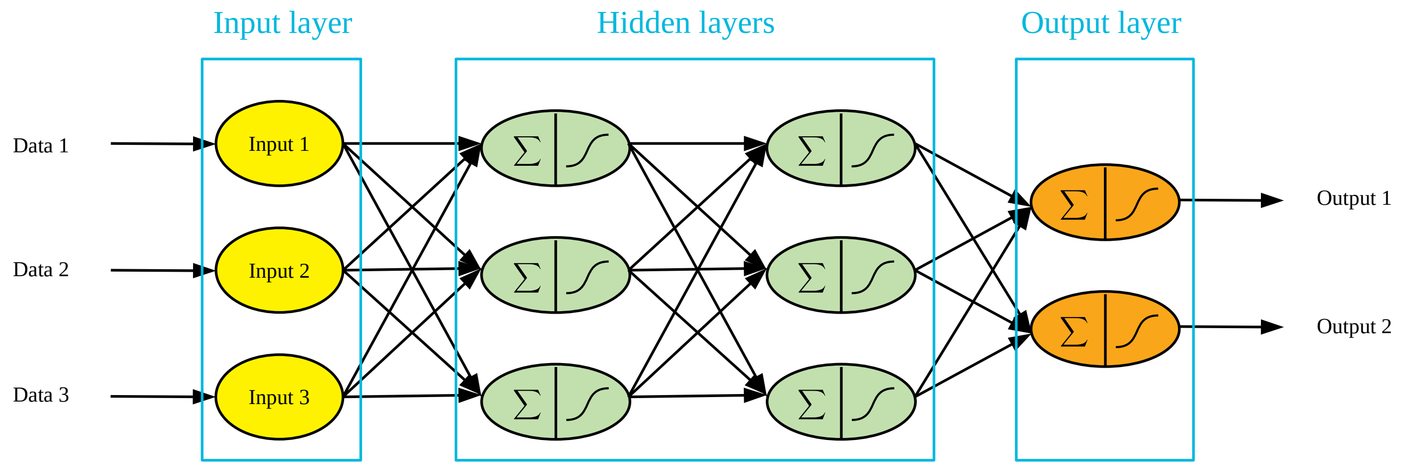
https://s46486.pcdn.co/wp-content/uploads/2023/03/YJCHsy_J6B4-00-02-42.png



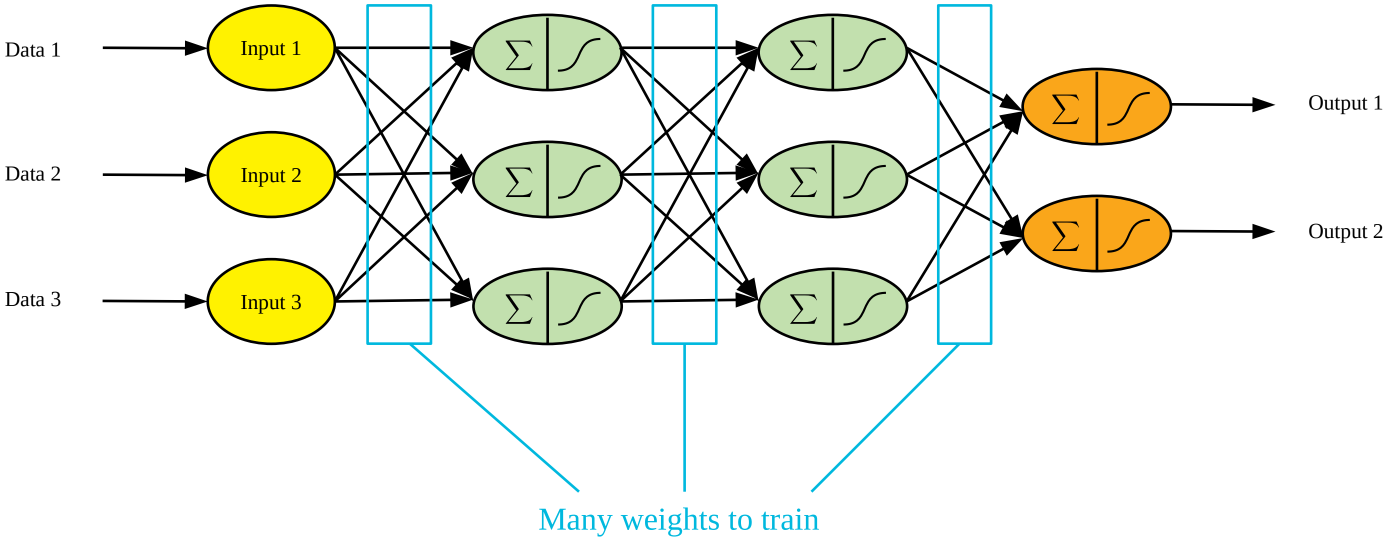
https://github.com/brain-bzh/tutorial-foundation-models
https://www.youtube.com/watch?v=0-pVVtLo4zY&list=PLfm_tIfDbIKvKCHs9nCXIgKFDBKpa6kdG

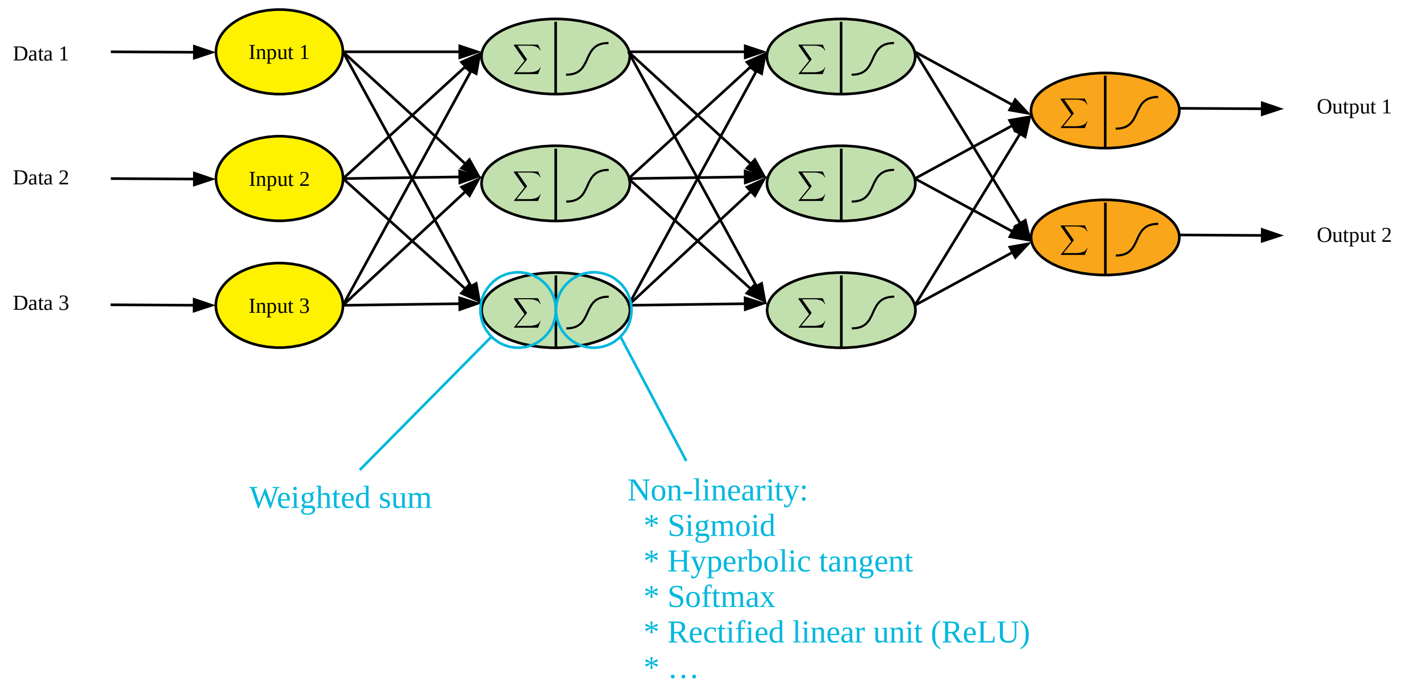


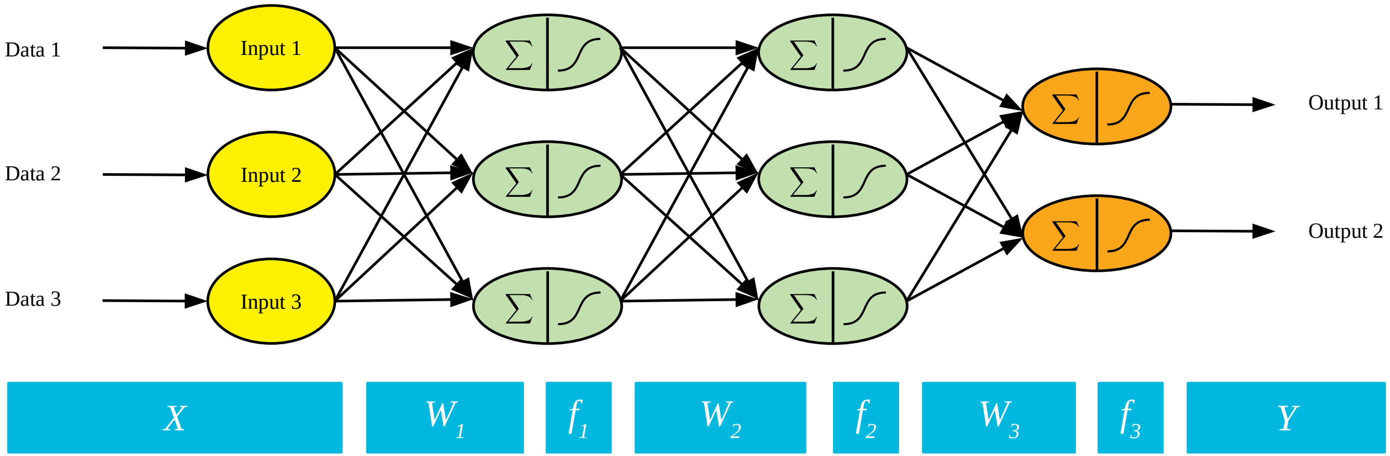
Data 1
Data 2
Data 3
Input 1
Input 2
Input 3
Output 1
Output 2



Input layer
Hidden layers
Output layer
Data 1
Data 2
Data 3
Input 1
Input 2
Input 3
Output 1
Output 2


Data 1
Data 2
Data 3
Input 1
Input 2
Input 3
Output 1
Output 2
Many weights to train

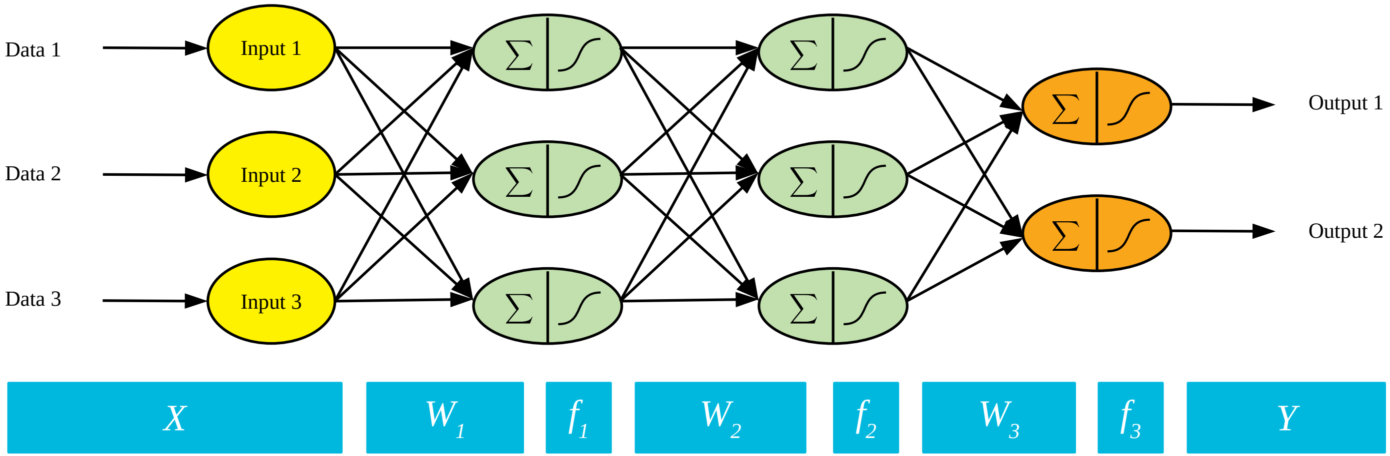

Data 1
Data 2
Data 3
Input 1
Input 2
Input 3
Output 1
Output 2
Weighted sum
Non-linearity:
* Sigmoid
* Hyperbolic tangent
* Softmax
* Rectified linear unit (ReLU)
* …



$$Y = f_3(W_3 \cdot f_2(W_2 \cdot f_1(W_1 \cdot X)))$$

From neural networks to deep learning

└ Building blocks of a neural networks (6/6)

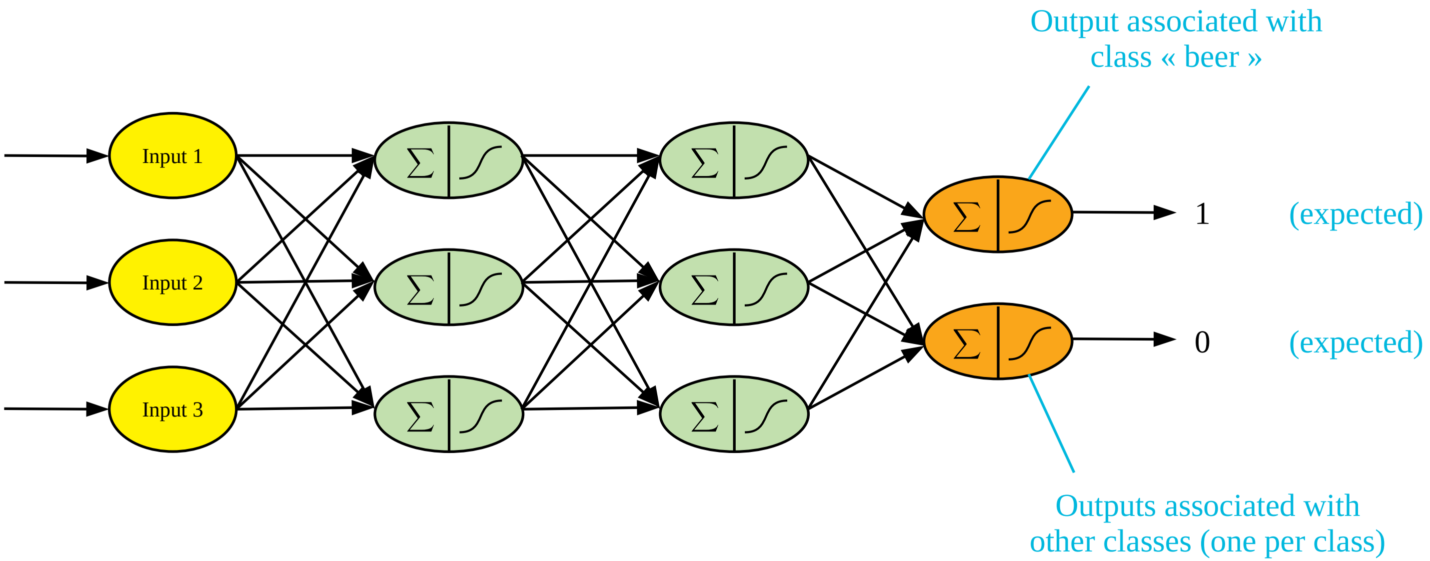


$$Y = f_3(W_3 \cdot f_2(W_2 \cdot f_1(W_1 \cdot X)))$$

Derivable → Backpropagation

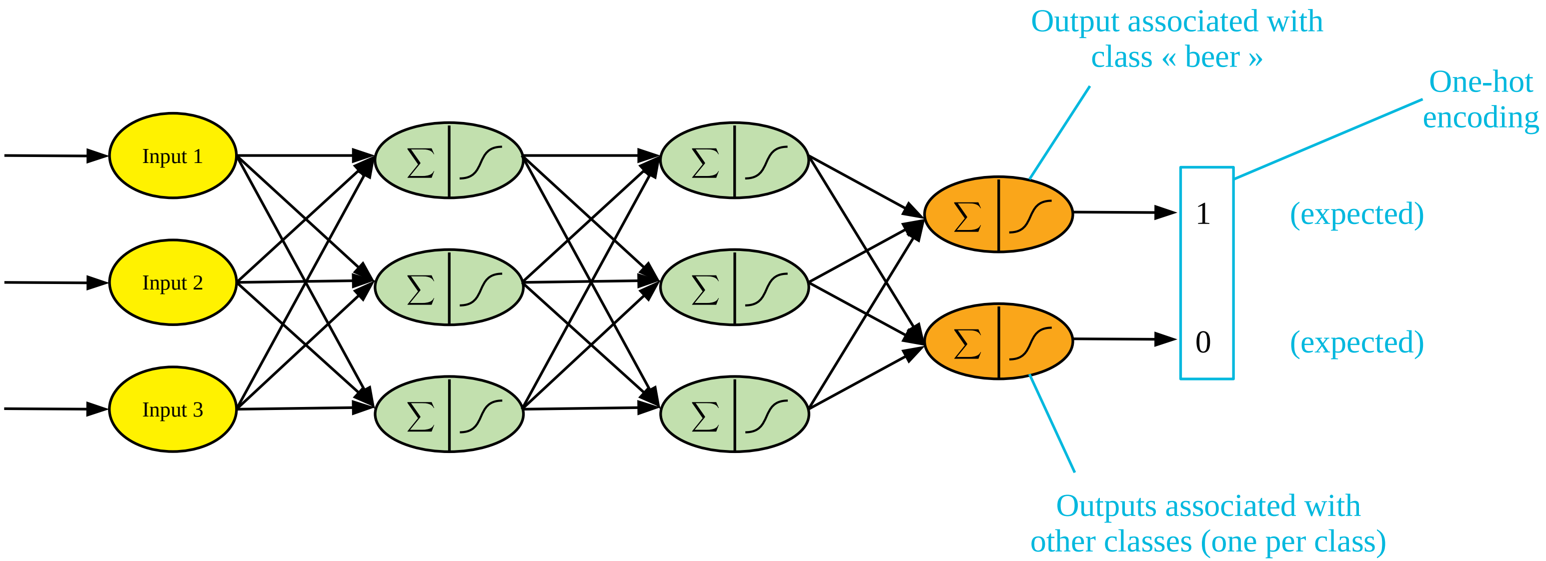
http://www.image-net.org/

Raw input
(or sometimes features)

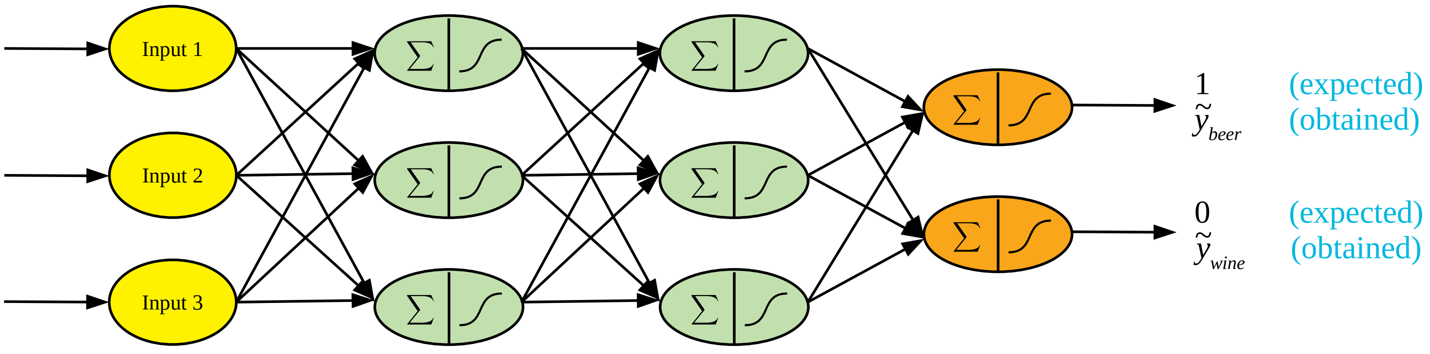


http://www.image-net.org/

Raw input
(or sometimes features)

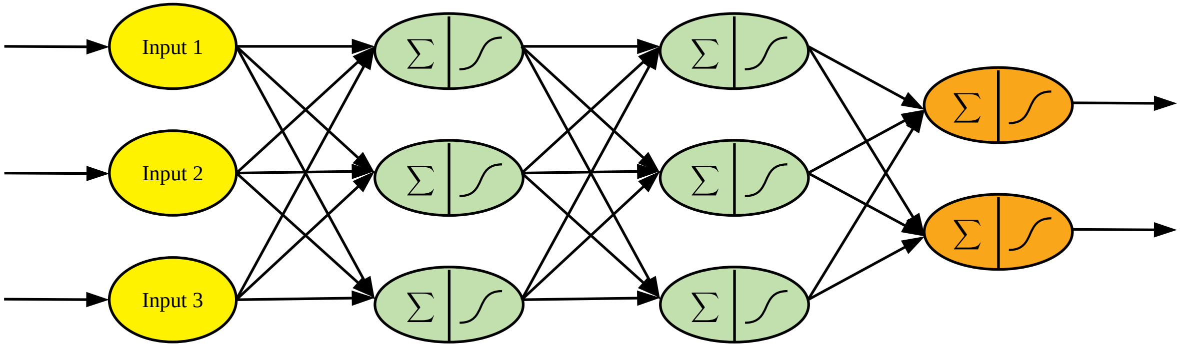


http://www.image-net.org/



(1) Forward pass

http://www.image-net.org/



1 (expected)
$\tilde{y}_{beer}$ (obtained)

0 (expected)
$\tilde{y}_{wine}$ (obtained)

(1) Forward pass

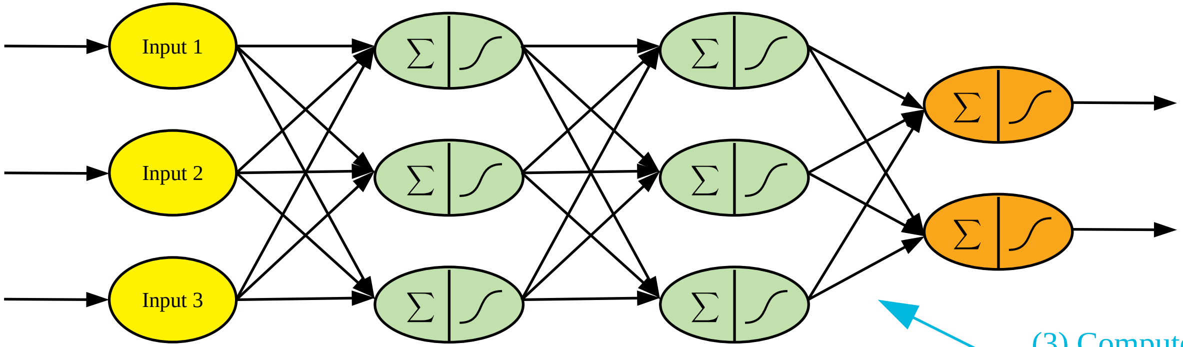
$$\varepsilon = loss(Y, \tilde{Y})$$

(2) Compute error using *loss function*

e.g., cross-entropy: $-\log(\mathbf{Y}^T\tilde{\mathbf{Y}})$

From neural networks to deep learning
└ Backpropagation (5/9)

http://www.image-net.org/



1 (expected)
$\tilde{y}_{beer}$ (obtained)

0 (expected)
$\tilde{y}_{wine}$ (obtained)

(3) Compute gradient
with respect to
individual weights

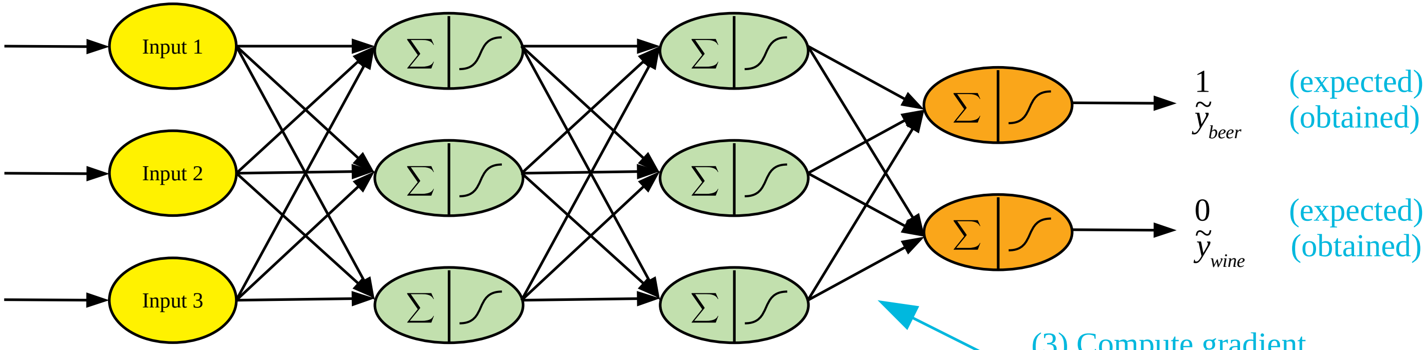
$$\varepsilon = loss(\mathrm{Y}, \tilde{\mathrm{Y}})$$

(1) Forward pass

(2) Compute error
using *loss function*

e.g., cross-entropy: $-\log(\mathbf{Y}^T\tilde{\mathbf{Y}})$

http://www.image-net.org/



(3) Compute gradient with respect to individual weights

(4) Chain rule

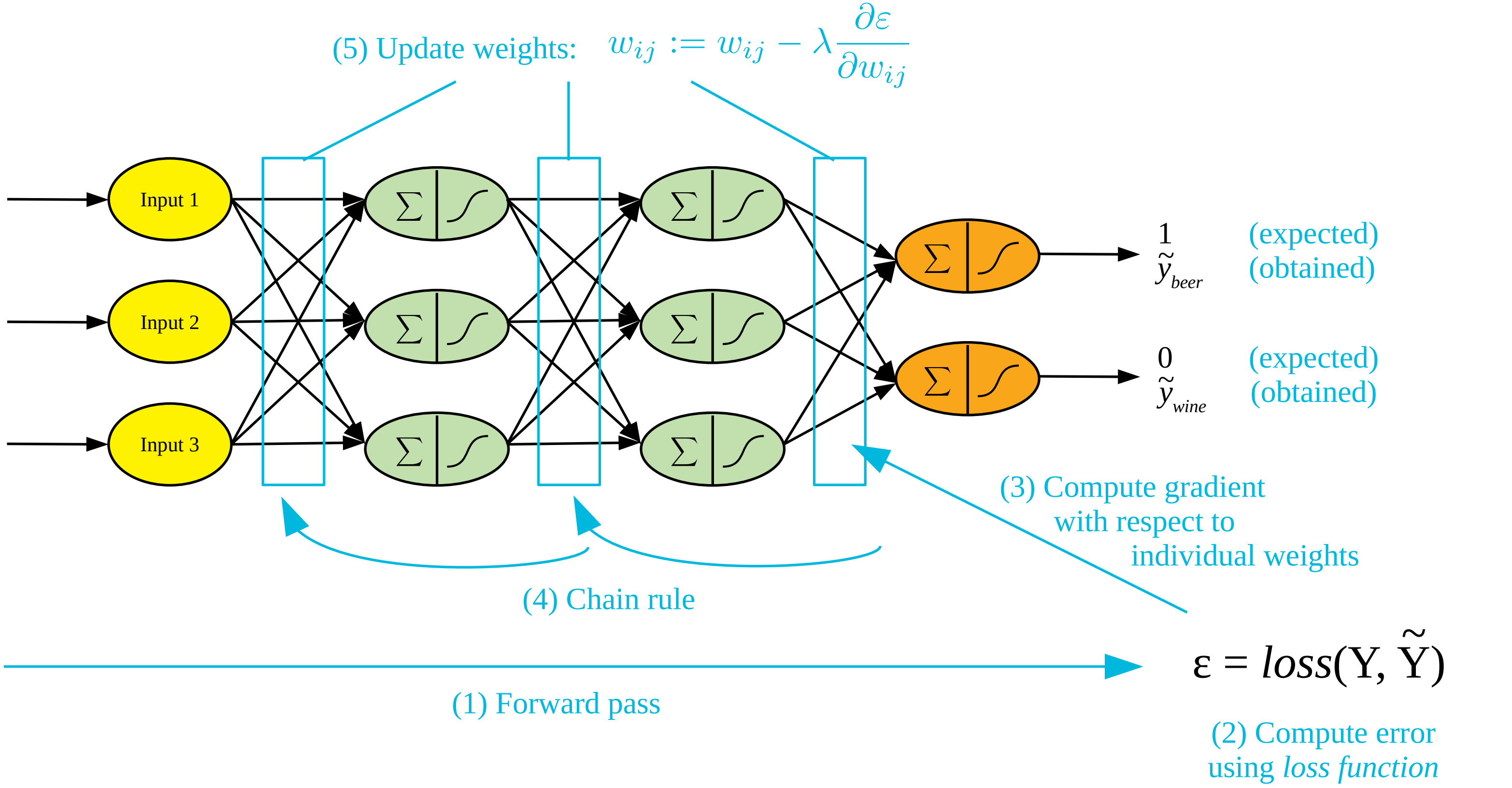
(1) Forward pass

$$\varepsilon = loss(\mathrm{Y}, \tilde{\mathrm{Y}})$$

(2) Compute error using *loss function*

e.g., cross-entropy: $-\log(\mathbf{Y}^{T}\tilde{\mathbf{Y}})$

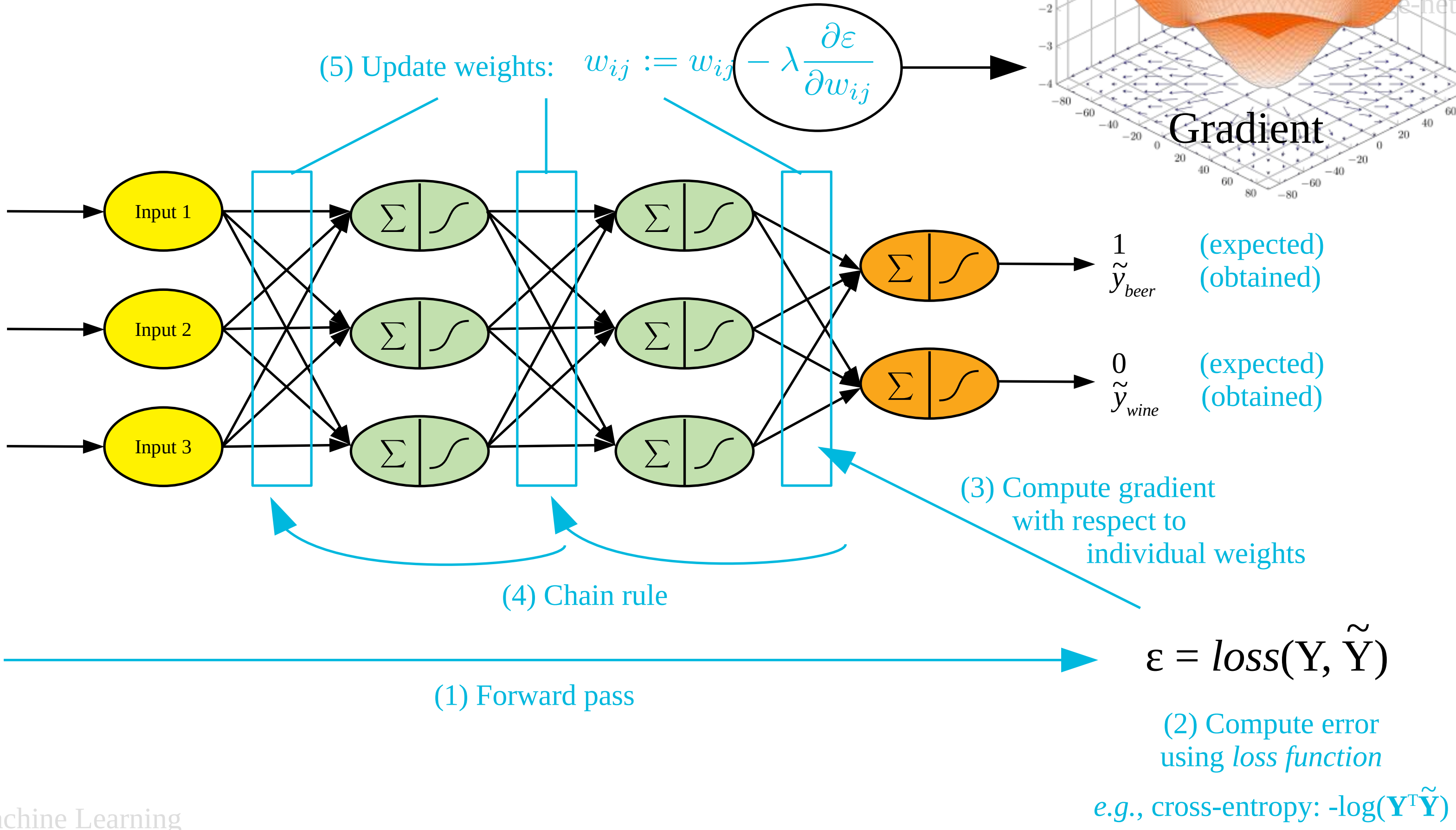
http://www.image-net.org/

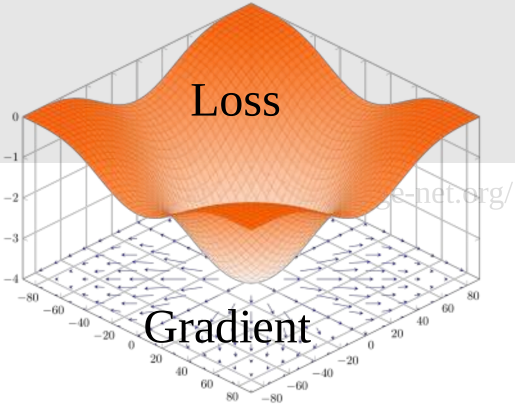


e.g., cross-entropy: $-\log(\mathbf{Y}^{T}\tilde{\mathbf{Y}})$

From neural networks to deep learning

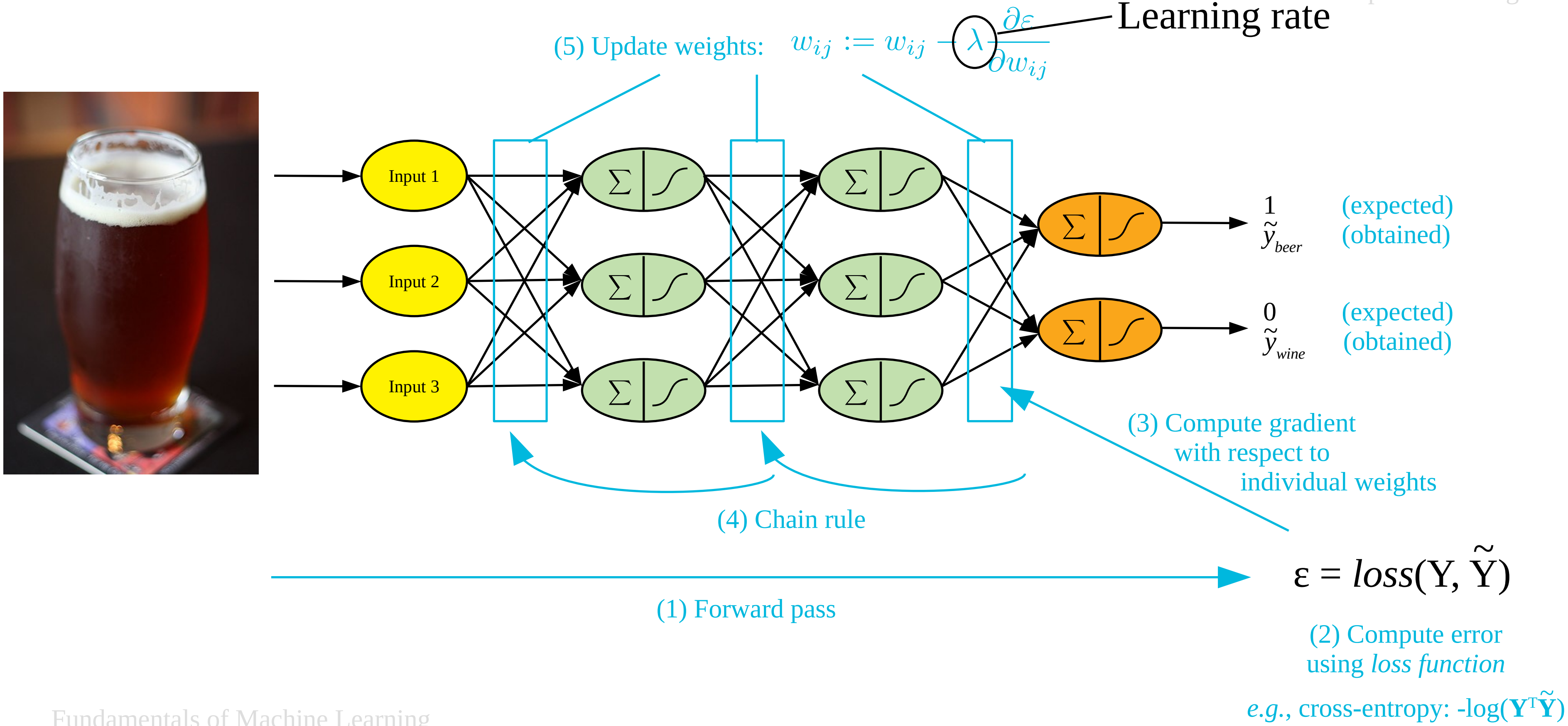
└ Backpropagation (8/9)



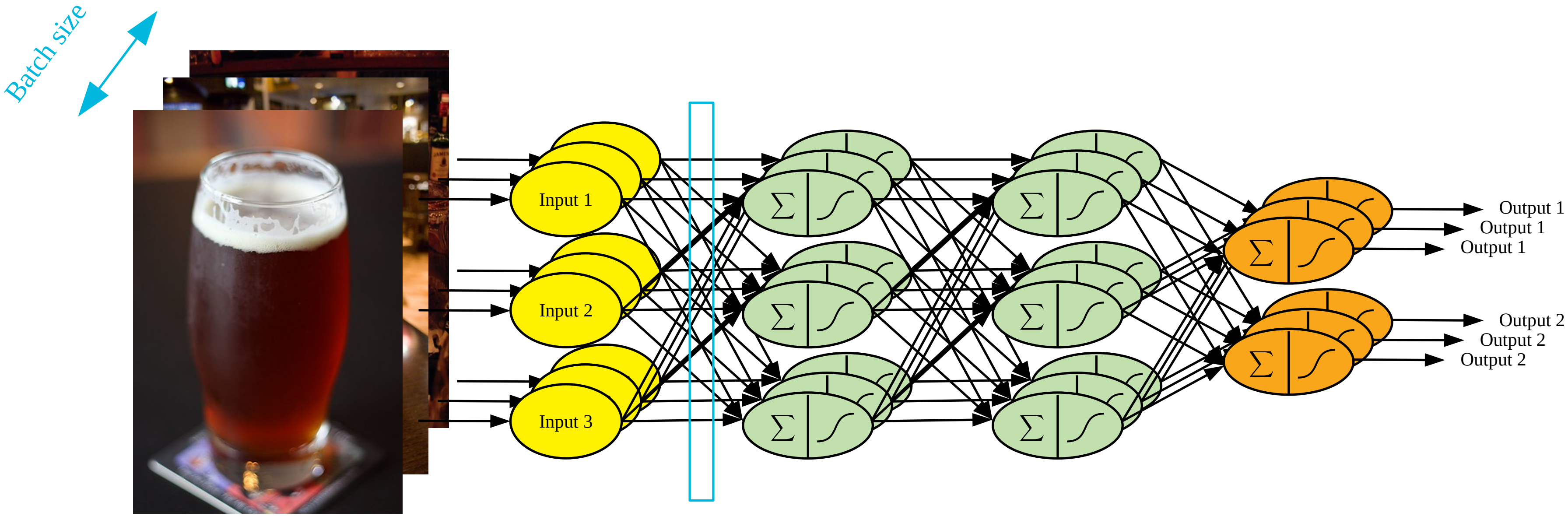


└ Backpropagation (9/9)

http://www.image-net.org/



http://www.image-net.org/



Batch normalization layer
to fix mean and variance
per input neuron

From neural networks to deep learning

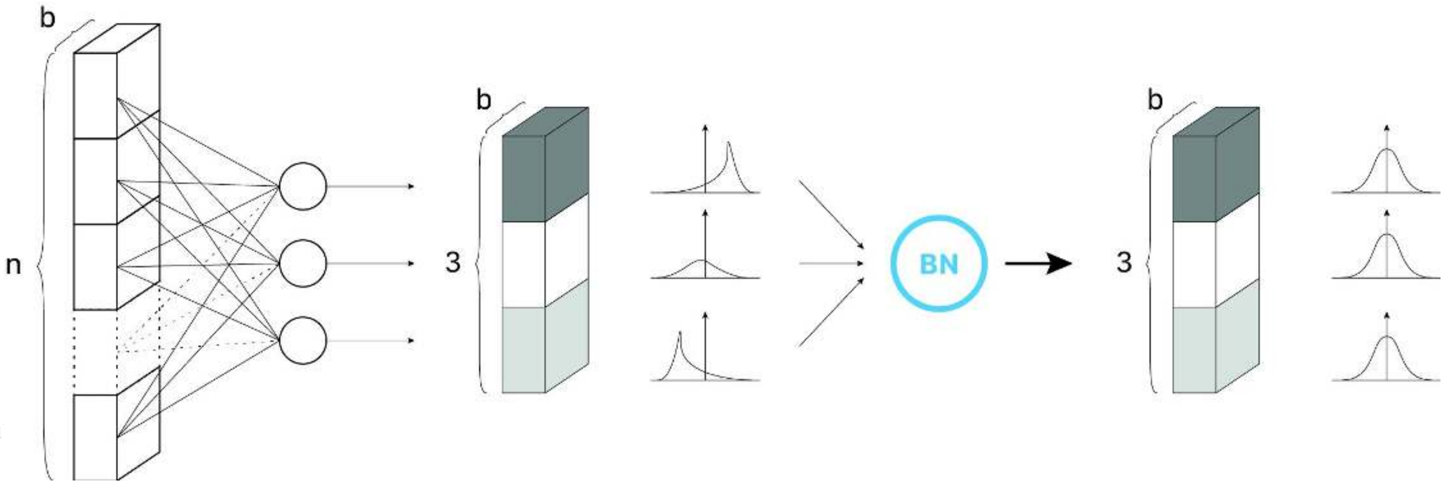
Batch normalization

Loffe et al. **Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift**. *International conference on machine learning*. 2015
https://towardsdatascience.com/batch-normalization-in-3-levels-of-understanding-14c2da90a338#b93c

Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift

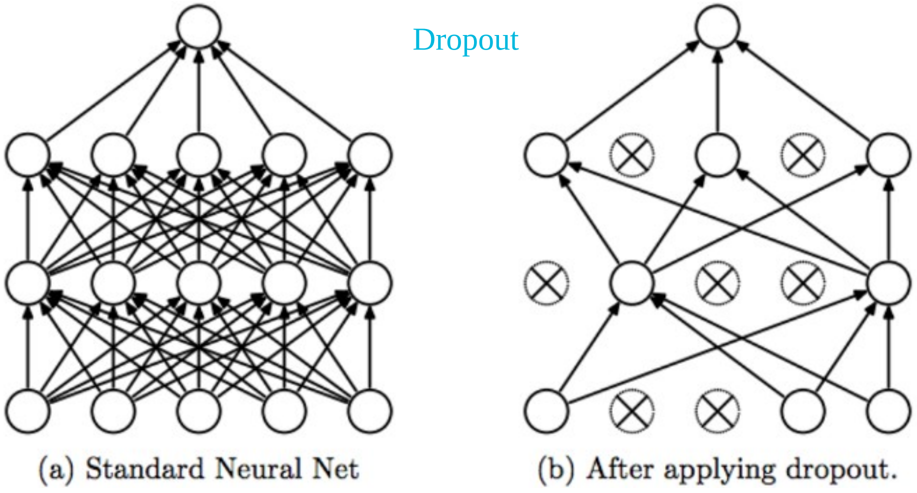
Sergey Ioffe, Christian Szegedy

Training Deep Neural Networks is complicated by the fact that the distribution of each layer's inputs changes during training, as the parameters of the previous layers change. This slows down the training by requiring lower learning rates and careful parameter initialization, and makes it notoriously hard to train models with saturating nonlinearities. We refer to this phenomenon as internal covariate shift, and address the problem by normalizing layer inputs. Our method draws its strength from making normalization a part of the model architecture and performing the normalization for each training mini-batch. Batch Normalization allows us to use much higher learning rates and be less careful about initialization. It also acts as a regularizer, in some cases eliminating the need for Dropout. Applied to a state-of-the-art image classification model, Batch Normalization achieves the same accuracy with 14 times fewer training steps, and beats the original model by a significant margin. Using an ensemble of batch-normalized networks, we improve upon the best published result on ImageNet classification: reaching 4.9% top-5 validation error (and 4.8% test error), exceeding the accuracy of human raters.

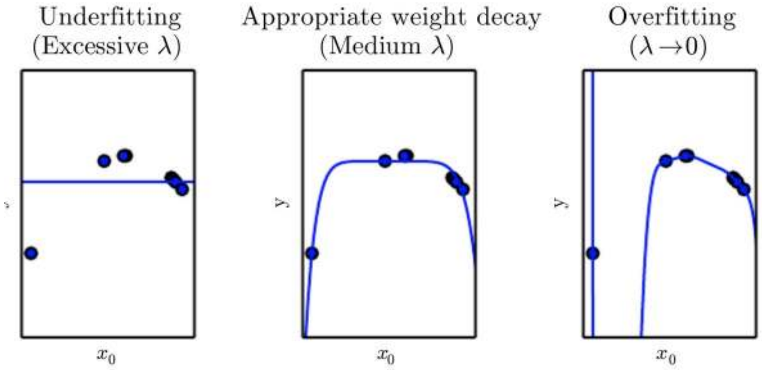


https://medium.com/@amarbudhiraja/https-medium-com-amarbudhiraja-learning-less-to-learn-better-dropout-in-deep-machine-learning-74334da4bfc5
https://paperswithcode.com/method/weight-decay#

Dropout



Weight decay



Regularization refers to training a model well
so that it can generalize to data it hasn't seen before

Generalization usually prevents the model
to converge too quickly

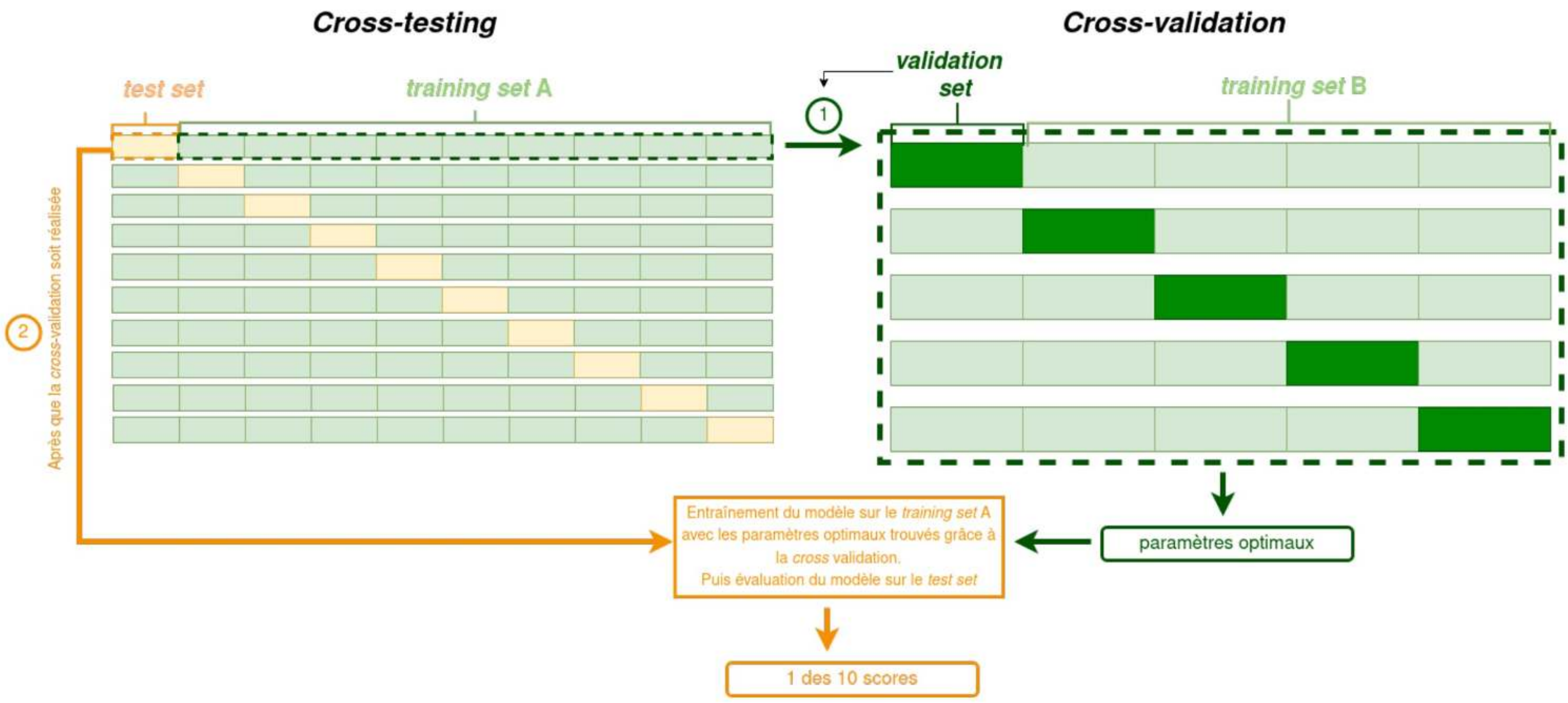
$$L_{new}(w) = L_{original}(w) + \lambda w^T w$$

From neural networks to deep learning

Training a neural network - Use a validation set

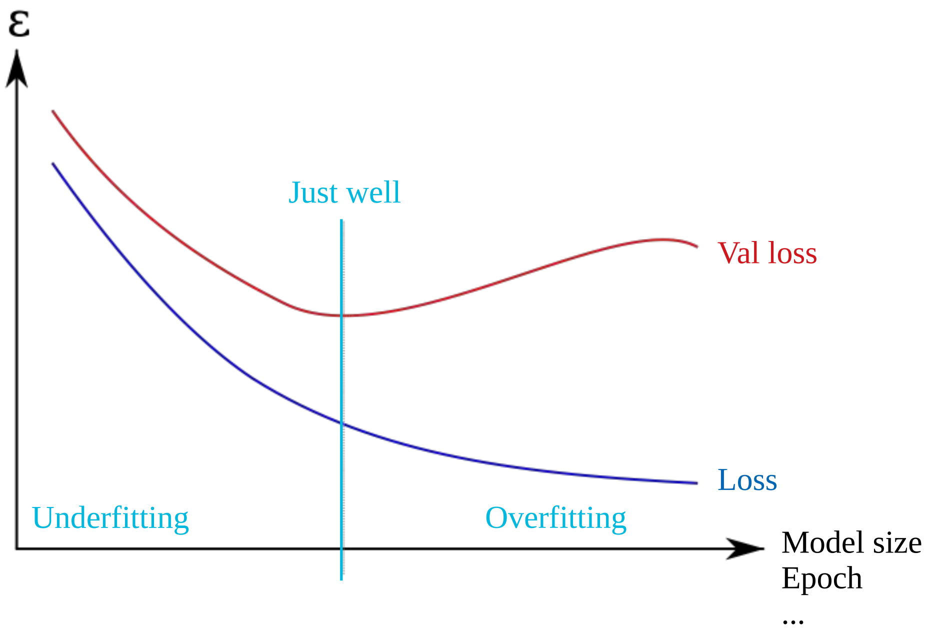
http://www.cs.toronto.edu/~kriz/cifar.html

Credit: Amine El Ouahidi (Master report)

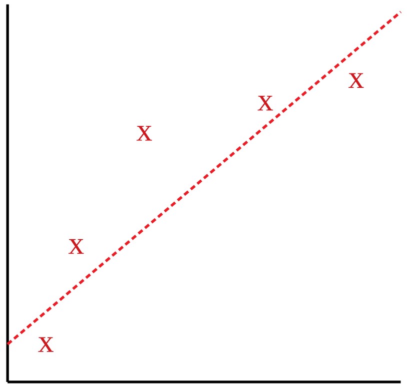


Training a neural network - Beware of overfitting

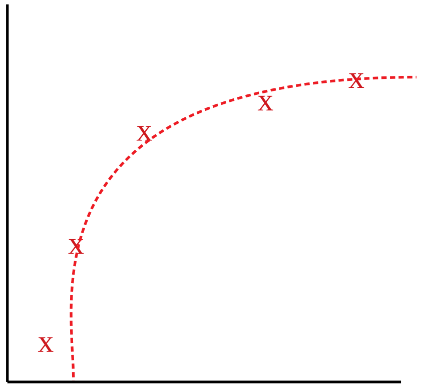
https://upload.wikimedia.org/wikipedia/commons/1/1f/Overfitting_svg.svg



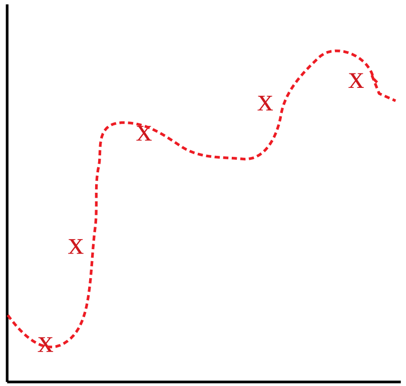

```
Train on 16000 samples, validate on 4000 samples
Epoch 1/200
16000/16000 [==============================] - 3s 190us/step - loss: 0.1749 - val_loss: 0.1459
Epoch 00001: val_loss improved from inf to 0.14589, saving model to ...

Epoch 2/200
16000/16000 [==============================] - 3s 173us/step - loss: 0.1282 - val_loss: 0.1159
Epoch 00002: val loss improved from 0.14589 to 0.11591, saving model to ...

Epoch 3/200
16000/16000 [==============================] - 3s 181us/step - loss: 0.0927 - val_loss: 0.0849
Epoch 00003: val loss improved from 0.11591 to 0.08487, saving model to ...

Epoch 4/200
16000/16000 [==============================] - 3s 181us/step - loss: 0.0608 - val_loss: 0.0594
Epoch 00004: val loss improved from 0.08487 to 0.05943, saving model to ...

Epoch 5/200
16000/16000 [==============================] - 3s 174us/step - loss: 0.0391 - val_loss: 0.0429
Epoch 00005: val loss improved from 0.05943 to 0.04287, saving model to ...

Epoch 6/200
16000/16000 [==============================] - 3s 177us/step - loss: 0.0258 - val_loss: 0.0324
Epoch 00006: val_loss improved from 0.04287 to 0.03238, saving model to ...

Epoch 7/200
15872/16000 [============================>.] - ETA: 0s - loss: 0.0175
```



High bias
(underfit)

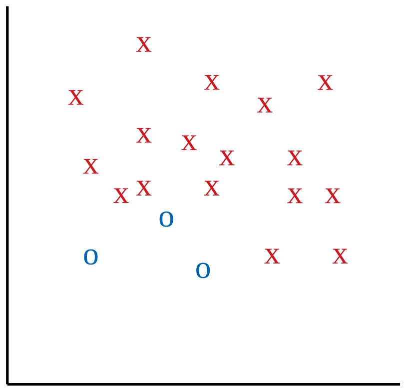


Well-trained



High variance
(overfit)

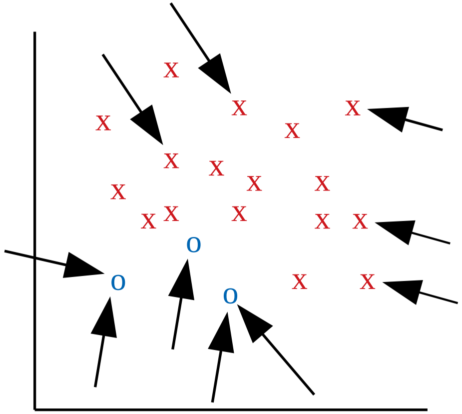
Training a neural network - The problem of unbalanced classes (1/3)

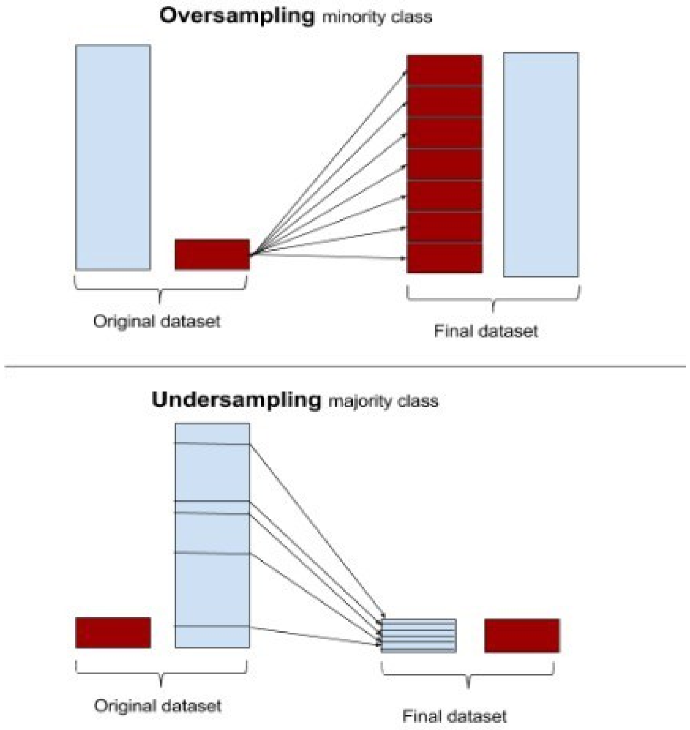


From neural networks to deep learning

Training a neural network - The problem of unbalanced classes (2/3)

https://towardsdatascience.com/breaking-the-curse-of-small-datasets-in-machine-learning-part-1-36f28b0c044d

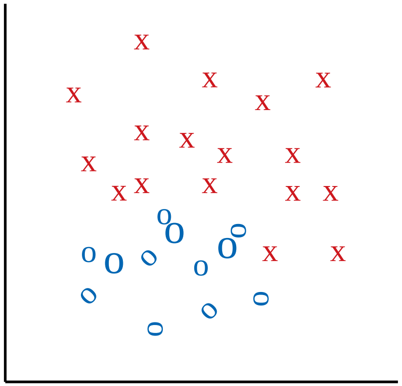


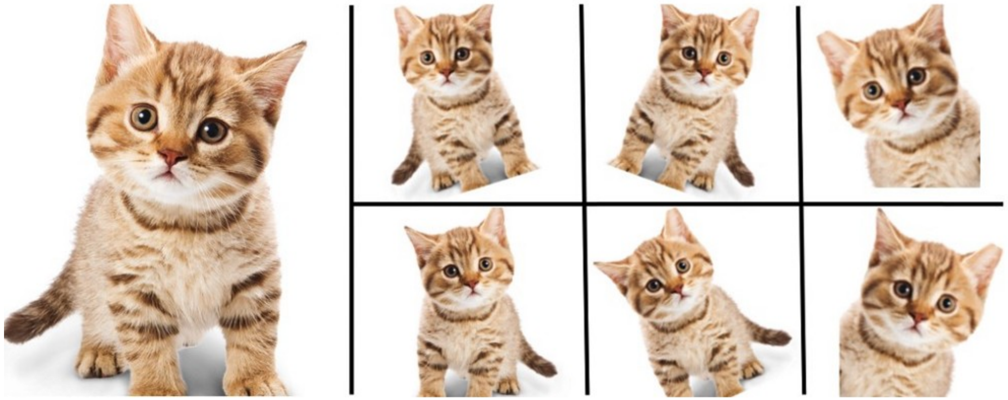


From neural networks to deep learning

└ Training a neural network - The problem of unbalanced classes (3/3)

https://www.kdnuggets.com/2018/05/data-augmentation-deep-learning-limited-data.html





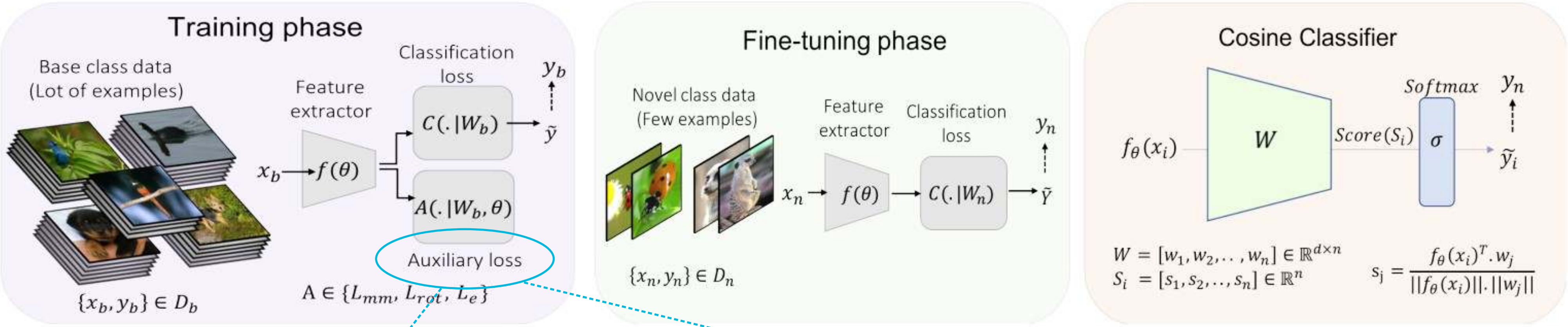
Data augmentation
(rotating, scaling, brightness, color shift…)

https://neovision.fr/data-augmentation-solutions-manque-donnees/

From neural networks to deep learning

└ Self-supervised learning

Mangla et al. **Charting the Right Manifold: Manifold Mixup for Few-shot Learning**. *IEEE/CVF Winter Conference on Applications of Computer Vision*. 2020
https://www.kdnuggets.com/2018/05/data-augmentation-deep-learning-limited-data.html



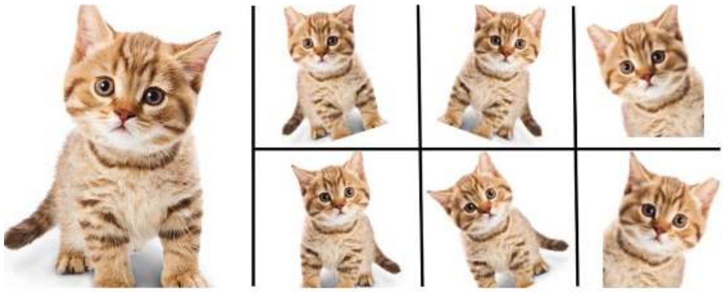
Example 1: learn to quantify rotation

Adds *false* problems to solve

Trains more generic features

Can be done unsupervisedly

Example 2: learn to retrieve original image



OUTLINE

- Introduction
- From neural networks to deep learning
- In practice:
  - Transfer learning
  - Standard libraries
- Limitations & future
- Focus: few-shot learning

https://fr.wikipedia.org/wiki/Fichier:Lilith_(computer).jpg
https://www.piqsels.com/en/search?q=money&page=2
https://github.com/cazala/mnist
https://medium.com/zylapp/review-of-deep-learning-algorithms-for-image-classification-5fdbca4a05e2
https://medium.com/ai-research-lab-kampala/alexnet-a-brief-review-14979ce7cc84
https://picsou.fandom.com/fr/wiki/Balthazar_Picsou
https://www.shutterstock.com/search/poor+student
https://bit.ly/3jsNit5

<u>You</u>

<u>Those big companies you all know</u>

In practice – Transfer learning

Quiz (2/7)

You

Those big companies you all know

https://fr.wikipedia.org/wiki/Fichier:Lilith_(computer).jpg
ps://www.piqsels
https://github.com/cazala/mnist
https://medium.com/zylap -algorithms-for-image-classi e2
https: lab-kampala/alexnet-a-brief-review-14979ce7cc84
ttps://picsou.fandom.com/fr/wiki/Balthazar_Picsou
https://www.shutterstock.com/search/poor+student
https://bit.ly/3jsNit5

In practice – Transfer learning

└ Quiz (3/7)

https://fr.wikipedia.org/wiki/Fichier:Lilith_(computer).jpg
s://www.piqsels
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https ... lab-kampala/alexnet-a-brief-review-14979ce7cc84

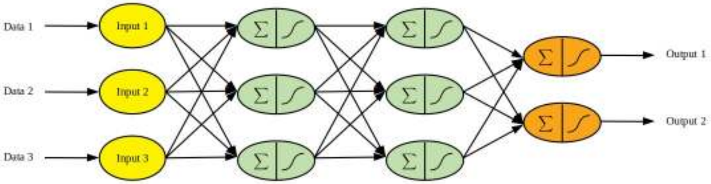
You

Those big companies
you all know

In practice – Transfer learning

└ Quiz (4/7)

<u>You</u>



<u>Those big companies you all know</u>

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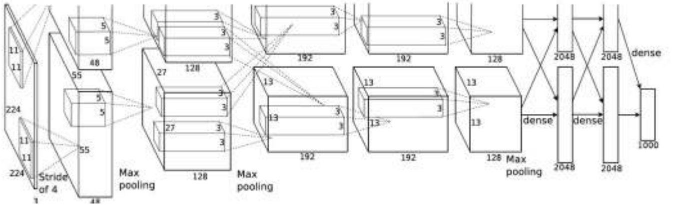

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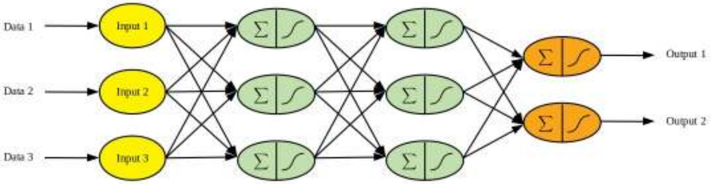

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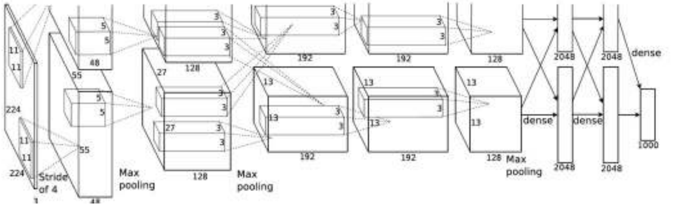

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https://github.com/cazala/mnist
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https... lab-kampala/alexnet-a-brief-review-14979ce7cc84

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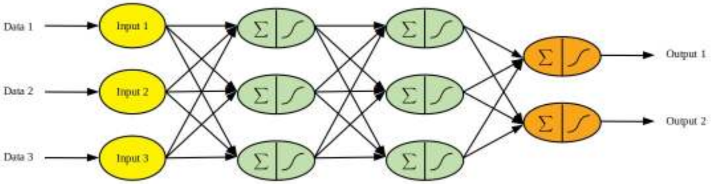
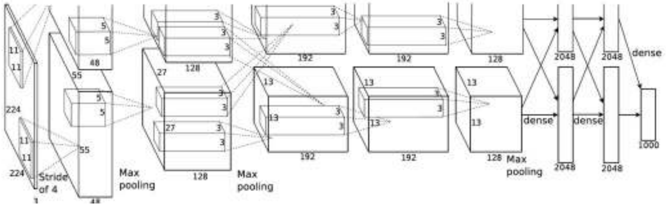


Those big companies
you all know

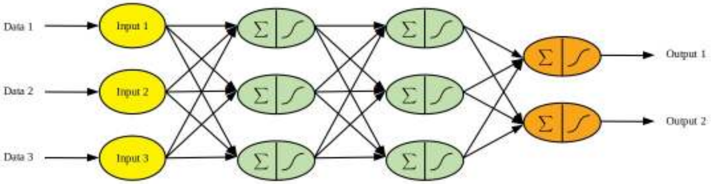


Question 1: Who gets the best performance?

In practice – Transfer learning

└ Quiz (7/7)

You



Those big companies you all know

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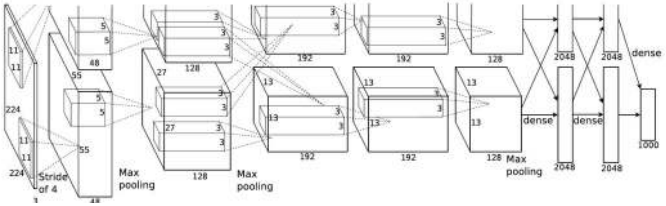


Question 1: Who gets the best performance?
Question 2: Why not profit from what others have done?

In practice – Transfer learning

One more reason why you should consider transfer learning

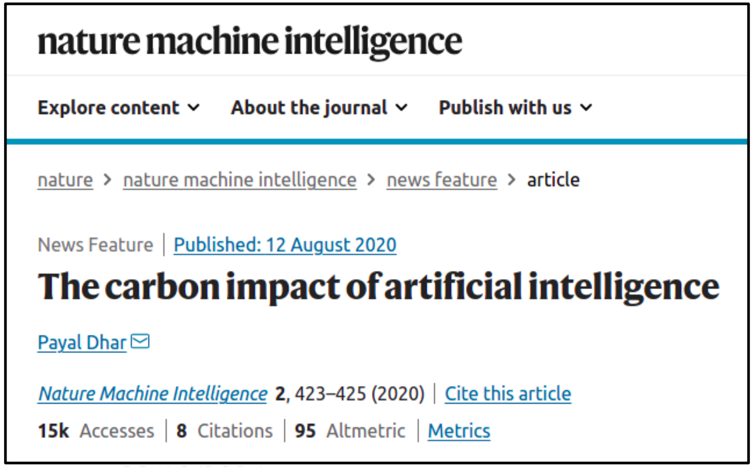
https://www.nature.com/articles/s42256-020-0219-9?proof=t
https://www.forbes.com/sites/robtoews/2020/06/17/deep-learnings-climate-change-problem/
https://towardsdatascience.com/deep-learning-and-carbon-emissions-79723d5bc86e


Forbes

Jun 17, 2020, 11:54am EDT | 17,351 views

Deep Learning's Carbon Emissions Problem

Rob Toews Contributor
AI
I write about the big picture of artificial intelligence.


nature machine intelligence

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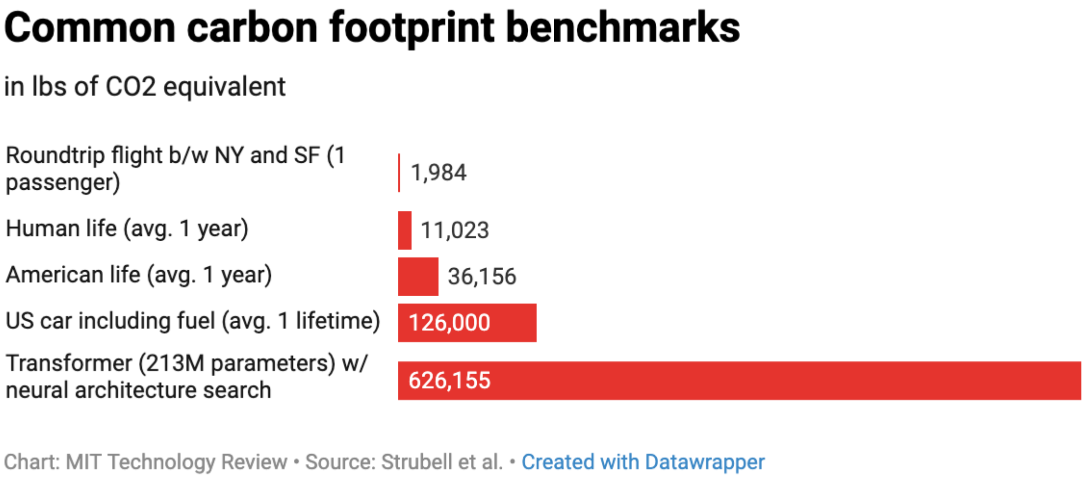
News Feature | Published: 12 August 2020

The carbon impact of artificial intelligence

Payal Dhar

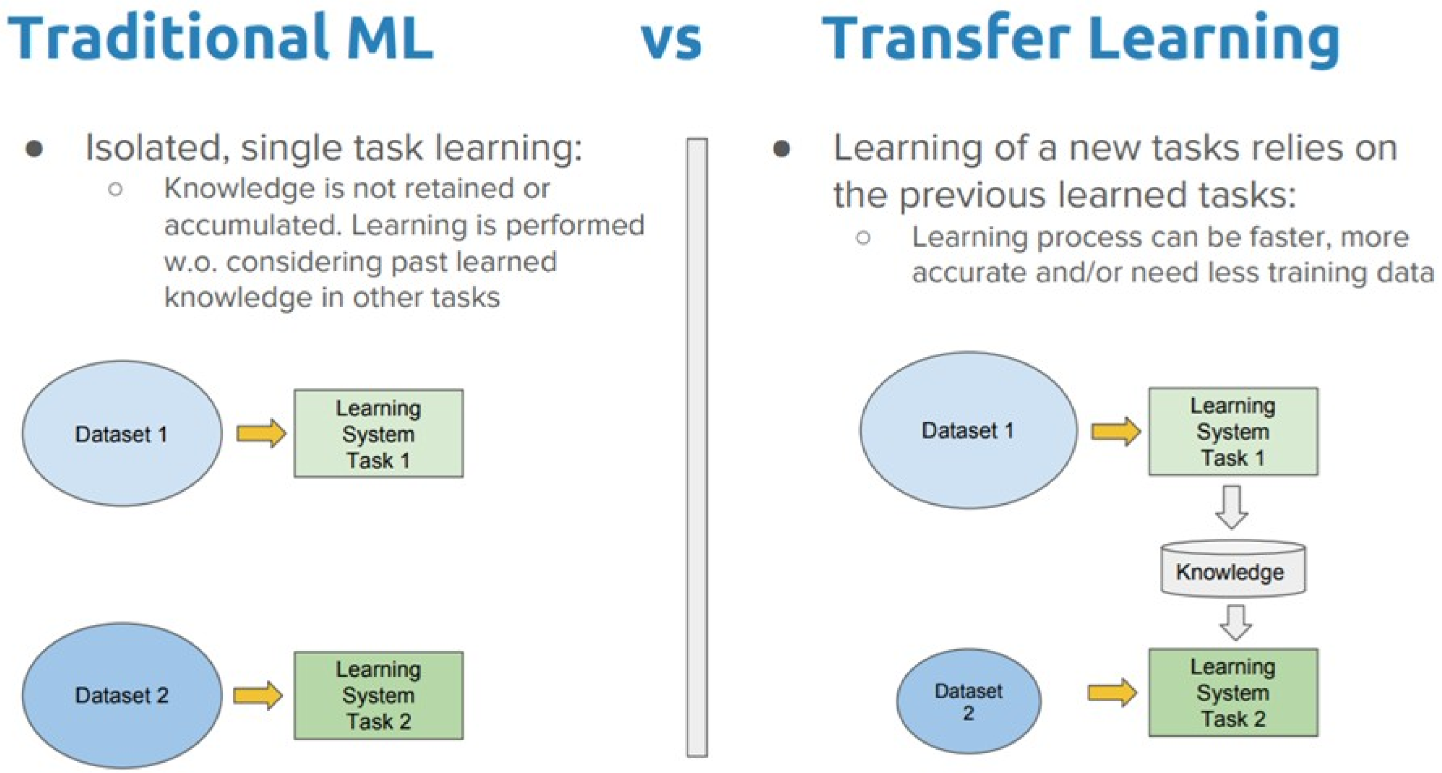
Nature Machine Intelligence **2**, 423–425 (2020) | Cite this article

15k Accesses | **8** Citations | **95** Altmetric | Metrics



└ What is transfer learning? (1/2)

https://towardsdatascience.com/deep-misconceptions-about-deep-learning-f26c41faceec



https://towardsdatascience.com/a-comprehensive-hands-on-guide-to-transfer-learning-with-real-world-applications-in-deep-learning-212bf3b2f27a

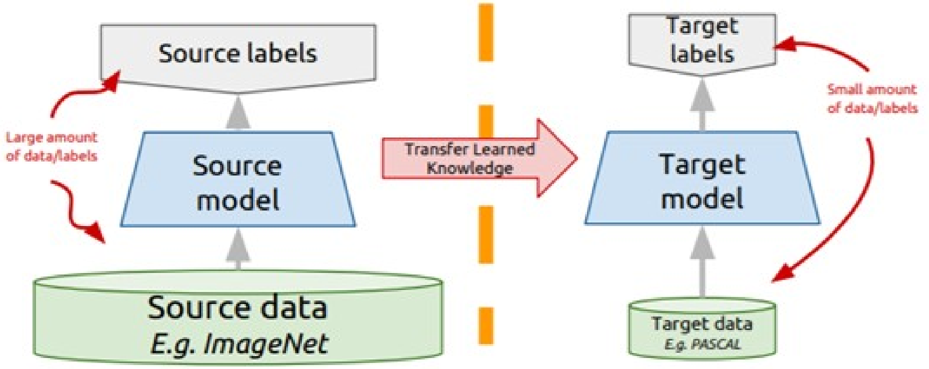
Transfer learning: idea

Instead of training a deep network from scratch for your task:

- Take a network trained on a different domain for a different **source task**
- Adapt it for your domain and your **target task**

Variations:

- Same domain, different task
- Different domain, same task



In practice – Transfer learning

└ Transfer learning in deep learning (1/3)

https://hackernoon.com/traffic-sign-classification-6e7113d9c4d5

Example: same domain, different task https://medium.com/zylapp/review-of-deep-learning-algorithms-for-image-classification-5fdbca4a05e2

https://pytorch.org/tutorials/beginner/transfer_learning_tutorial.html

In practice – Transfer learning

└ Transfer learning in deep learning (2/3)

<u>Example:</u> same domain, different task

https://hackernoon.com/traffic-sign-classification-6e7113d9c4d5
https://medium.com/zylapp/review-of-deep-learning-algorithms-for-image-classification-5fdbca4a05e2
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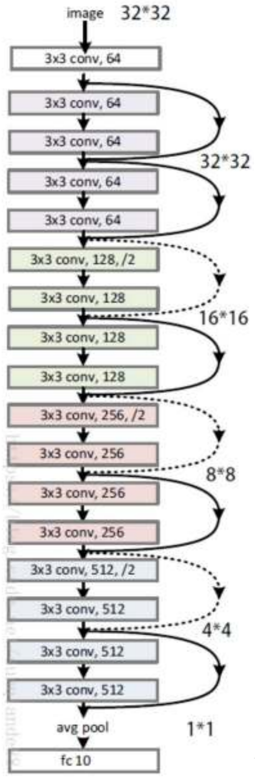

ResNet18 is a powerful image recognition network that has been trained on Imagenet 1000

└ Transfer learning in deep learning (3/3)

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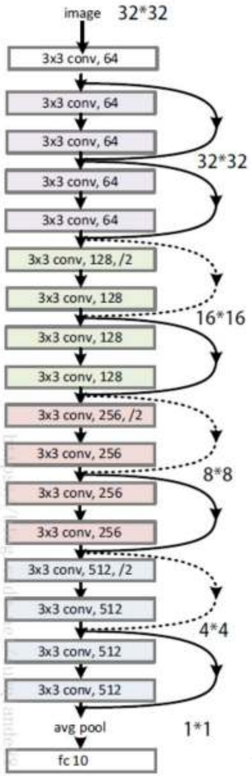
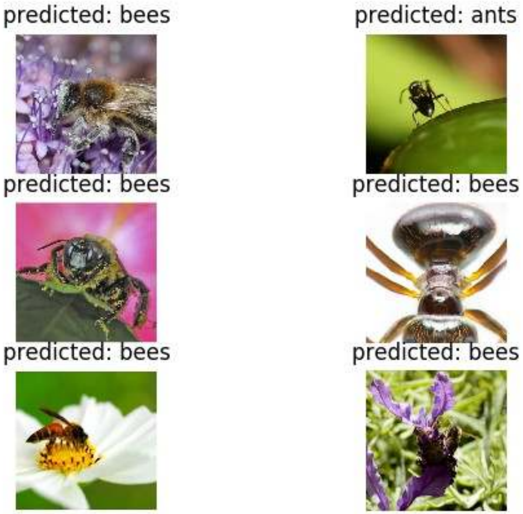
<u>Example:</u> same domain, different task



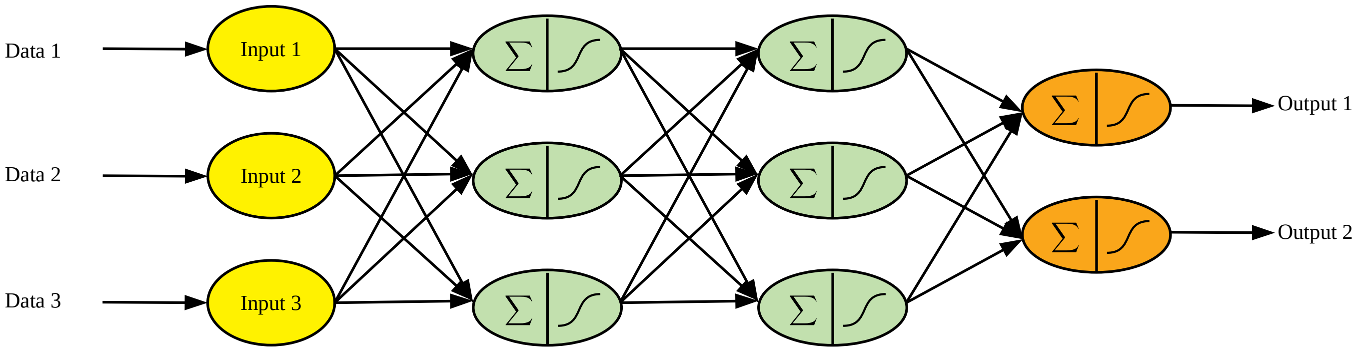
ResNet18 is a powerful image recognition network that has been trained on Imagenet 1000

Objective: use its recognition power on our small dataset of bees and ants



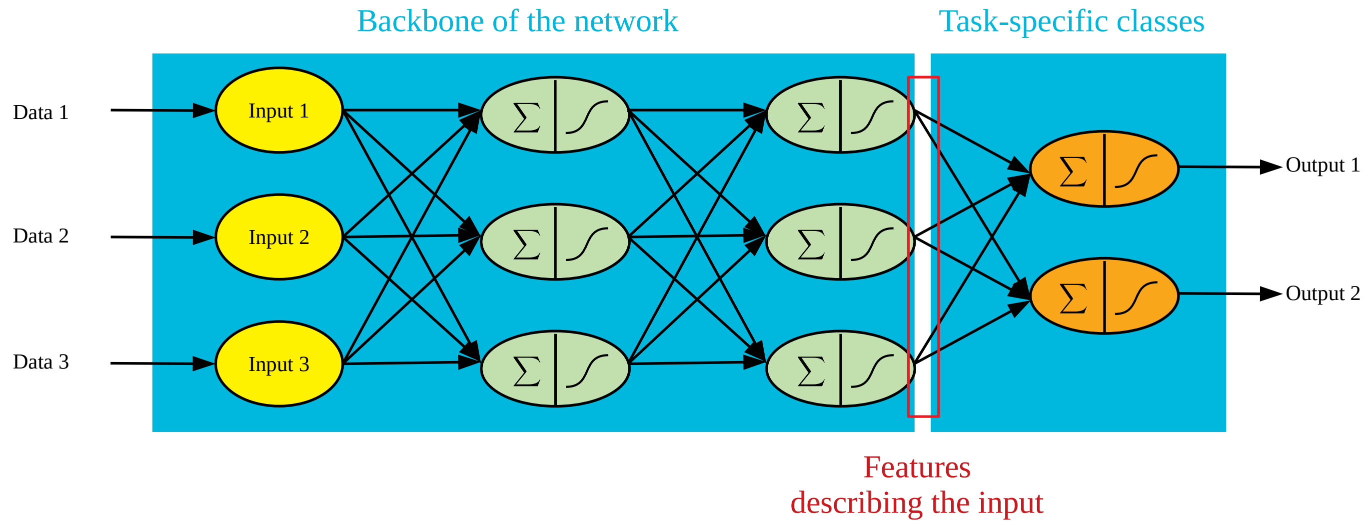
In practice – Transfer learning

Exploiting a pre-trained model (1/7)

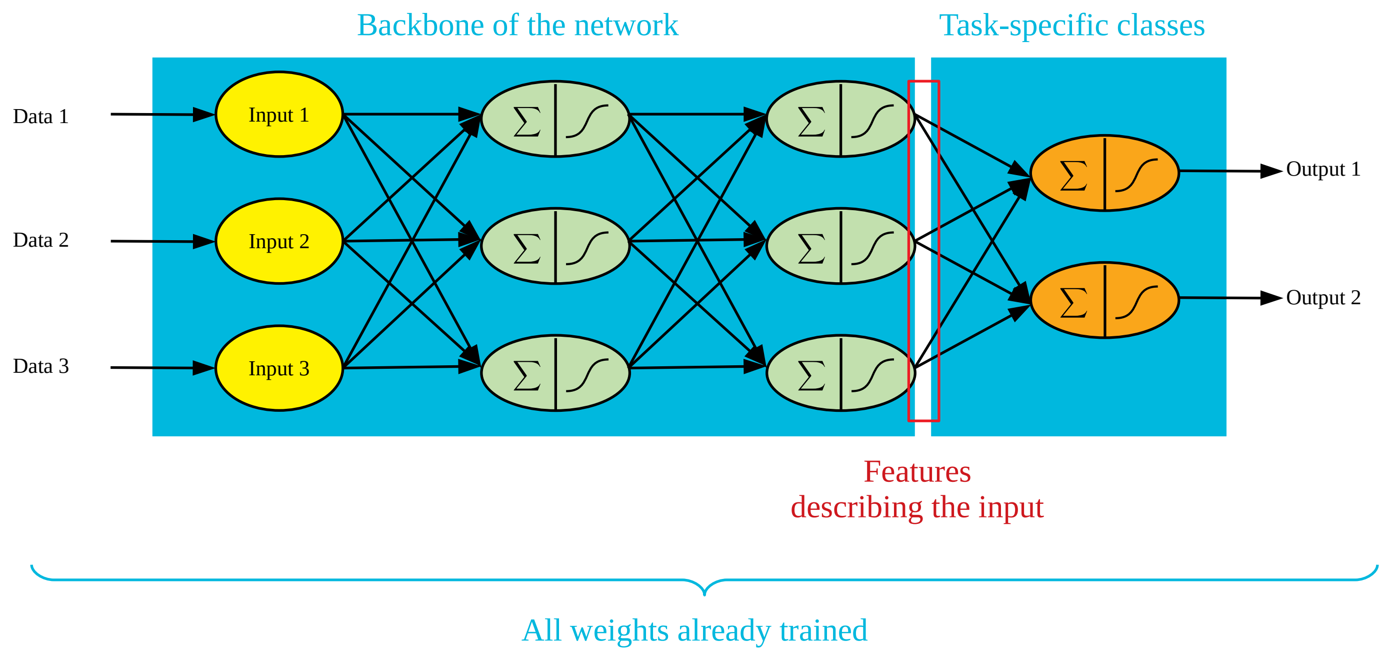


In practice – Transfer learning

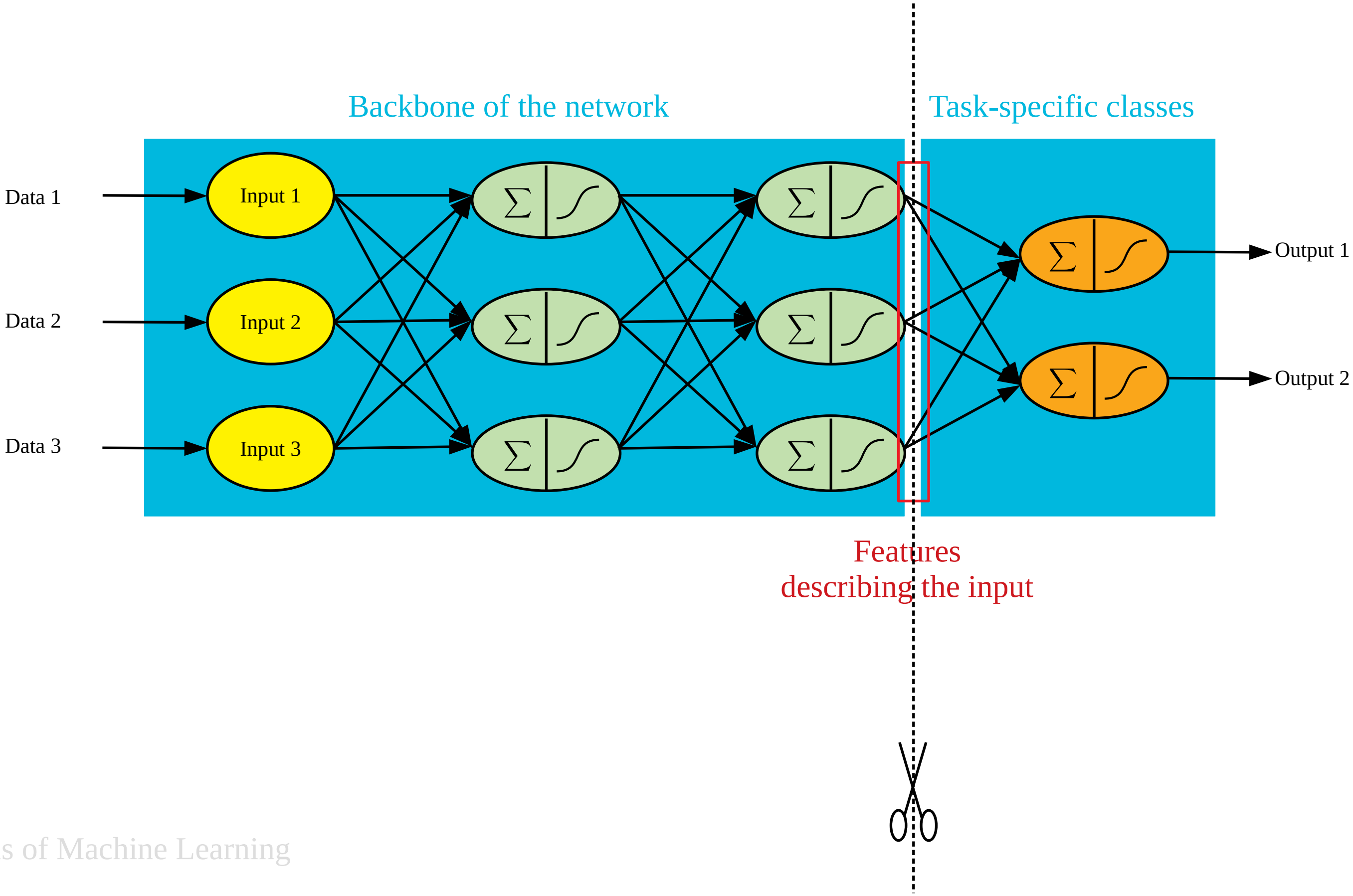
Exploiting a pre-trained model (2/7)



Exploiting a pre-trained model (3/7)

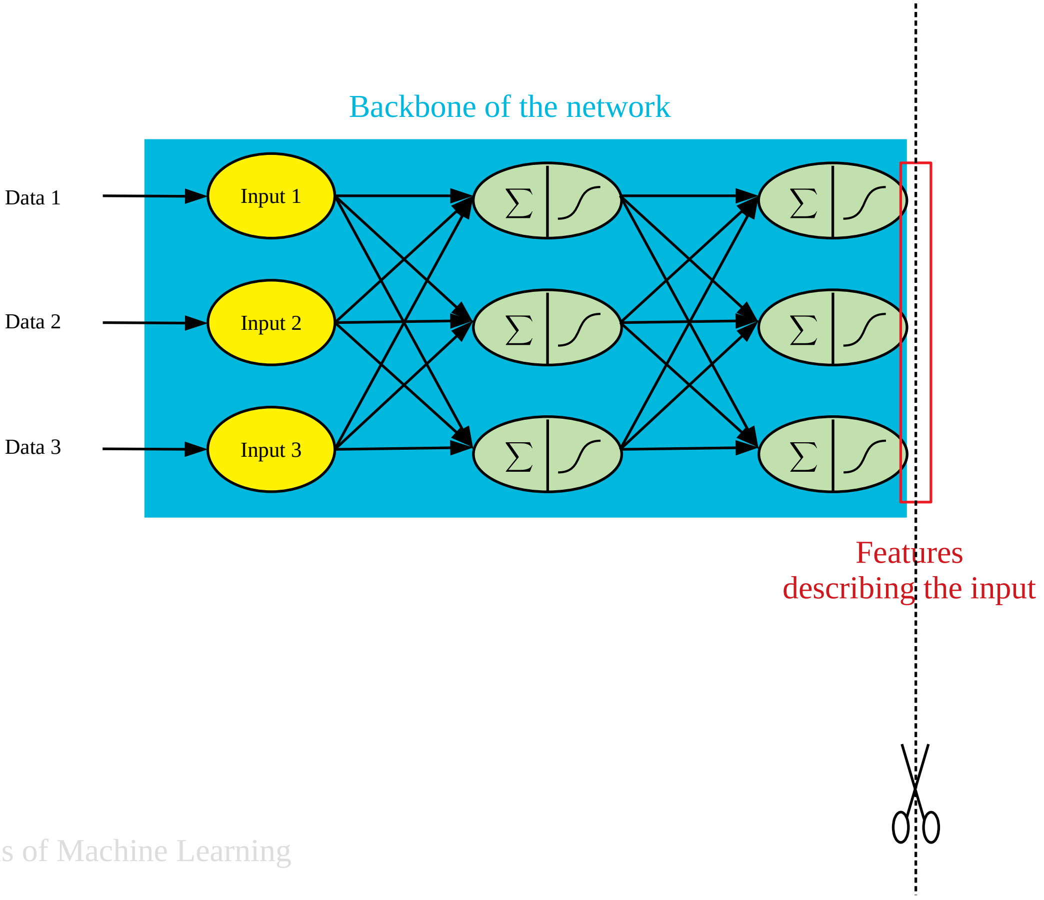


In practice – Transfer learning
Exploiting a pre-trained model (4/7)



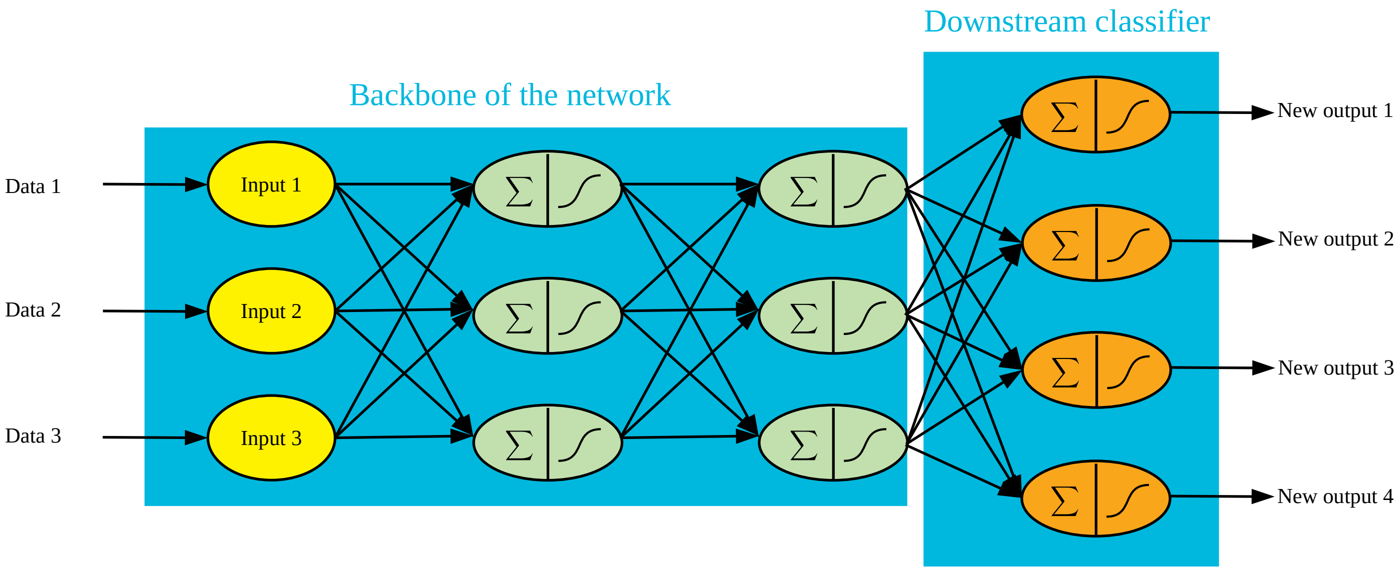
In practice – Transfer learning

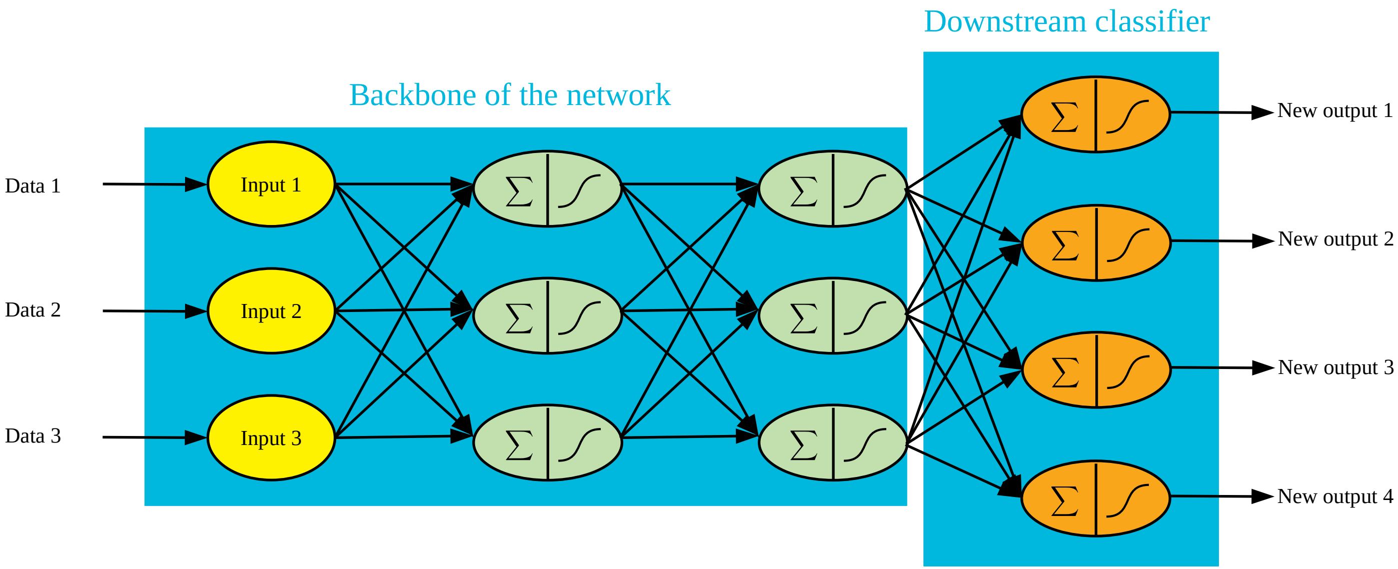
Exploiting a pre-trained model (5/7)



In practice – Transfer learning

Exploiting a pre-trained model (6/7)





Now, we need to train the network a bit to make things work:

Solution 1: Freeze weights from the backbone and train the last layer
Solution 2 (fine-tuning): Train weights from both the backbone and the downstream

In practice – Transfer learning

Freezing layers or fine tuning?

https://towardsdatascience.com/a-comprehensive-hands-on-guide-to-transfer-learning-with-real-world-applications-in-deep-learning-212bf3b2f27a

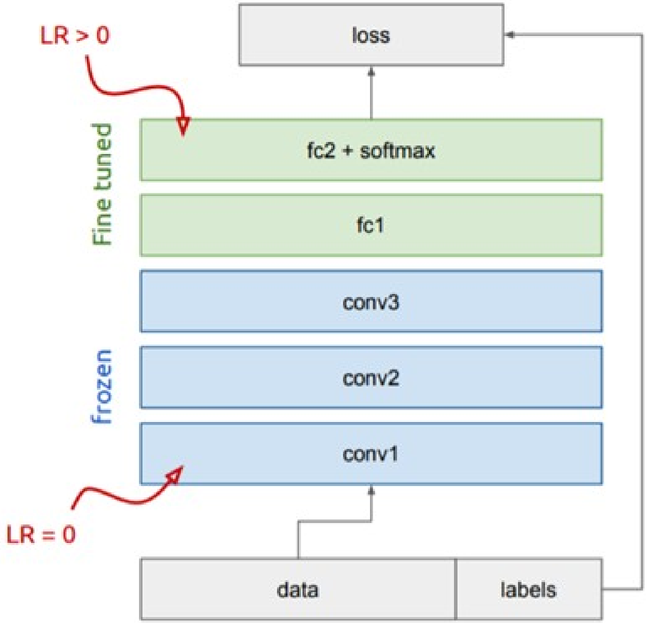
Bottom *n* layers can be frozen or fine tuned.

- **Frozen**: not updated during backprop
- **Fine-tuned**: updated during backprop

Which to do depends on target task:

- **Freeze**: target task labels are scarce, and we want to avoid overfitting
- **Fine-tune**: target task labels are more plentiful

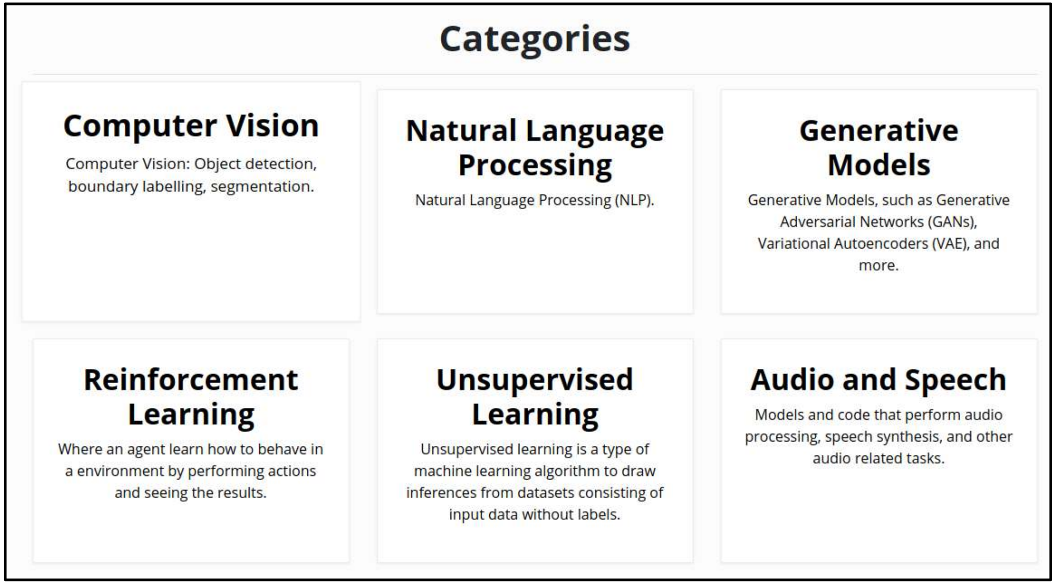
In general, we can set learning rates to be different for each layer to find a tradeoff between freezing and fine tuning



In practice – Transfer learning

Which backbone for which task? (1/2)

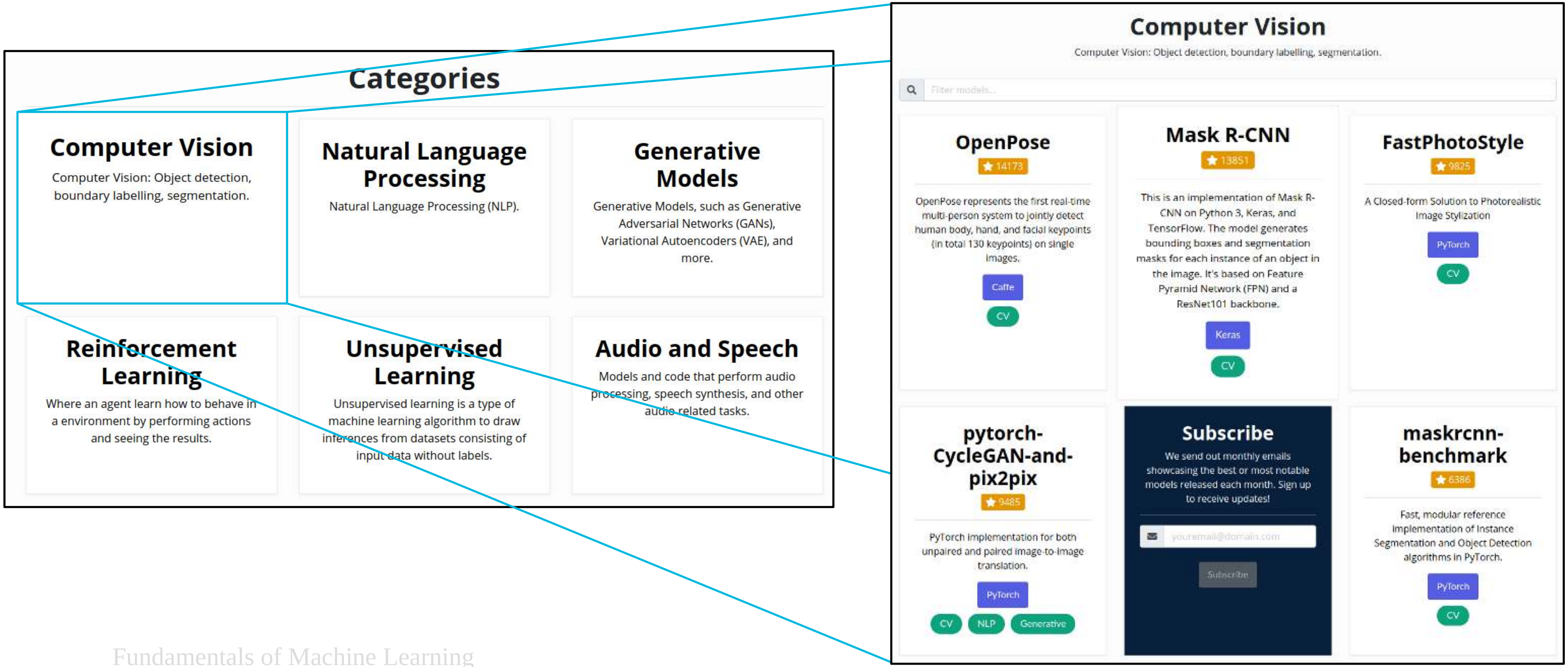
https://modelzoo.co/



In practice – Transfer learning

└ Which backbone for which task? (2/2)

https://modelzoo.co/



OUTLINE

- Introduction
- From neural networks to deep learning
- In practice:
  - Transfer learning
  - Standard libraries
- Limitations & future
- Focus: few-shot learning

In practice – Standard libraries

A wide catalogue (1/4)

https://en.wikipedia.org/wiki/Comparison_of_deep-learning_software

Software	Creator	Initial Release	Software license[a]	Open source	Platform	Written in	Interface	OpenMP support	OpenCL support	CUDA support	Automatic differentiation[1]	Has pretrained models	Recurrent nets	Convolutional nets	RBM/DBNs	Parallel execution (multi node)	Actively Developed
BigDL	Jason Dai (Intel)	2016	Apache 2.0	Yes	Apache Spark	Scala	Scala, Python			No		Yes	Yes	Yes			
Caffe	Berkeley Vision and Learning Center	2013	BSD	Yes	Linux, macOS, Windows[2]	C++	Python, MATLAB, C++	Yes	Under development[3]	Yes	Yes	Yes[4]	Yes	Yes	No	?	No[5]
Chainer	Preferred Networks	2015	BSD	Yes	Linux, macOS	Python	Python	No	No	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes
Deeplearning4j	Skymind engineering team; Deeplearning4j community; originally Adam Gibson	2014	Apache 2.0	Yes	Linux, macOS, Windows, Android (Cross-platform)	C++, Java	Java, Scala, Clojure, Python (Keras), Kotlin	Yes	No[6]	Yes[7][8]	Computational Graph	Yes[9]	Yes	Yes	Yes	Yes[10]	
Dlib	Davis King	2002	Boost Software License	Yes	Cross-Platform	C++	C++	Yes	No	Yes	Yes	Yes	No	Yes	Yes	Yes	
Intel Data Analytics Acceleration Library	Intel	2015	Apache License 2.0	Yes	Linux, macOS, Windows on Intel CPU[11]	C++, Python, Java	C++, Python, Java[11]	Yes	No	No	Yes	No		Yes		Yes	
Intel Math Kernel Library	Intel		Proprietary	No	Linux, macOS, Windows on Intel CPU[12]		C[13]	Yes[14]	No	No	Yes	No	Yes[15]	Yes[15]		No	
Keras	François Chollet	2015	MIT license	Yes	Linux, macOS, Windows	Python	Python, R	Only if using Theano as backend	Can use Theano, Tensorflow or PlaidML as backends	Yes	Yes	Yes[16]	Yes	Yes	No[17]	Yes[18]	Yes
MATLAB + Deep Learning Toolbox	MathWorks		Proprietary	No	Linux, macOS, Windows	C, C++, Java, MATLAB	MATLAB	No	No	Train with Parallel Computing Toolbox and generate CUDA code with GPU Coder[19]	Yes[20]	Yes[21][22]	Yes[21]	Yes[21]	Yes	With Parallel Computing Toolbox[23]	Yes
Microsoft Cognitive Toolkit (CNTK)	Microsoft Research	2016	MIT license[24]	Yes	Windows, Linux[25] (macOS via Docker on roadmap)	C++	Python (Keras), C++, Command line,[26] BrainScript[27] (.NET on roadmap[28])	Yes[29]	No	Yes	Yes	Yes[30]	Yes[31]	Yes[31]	No[32]	Yes[33]	No[34]
Apache MXNet	Apache Software Foundation	2015	Apache 2.0	Yes	Linux, macOS, Windows,[35][36] AWS, Android,[37] iOS, JavaScript[38]	Small C++ core library	C++, Python, Julia, Matlab, JavaScript, Go, R, Scala, Perl, Clojure	Yes	On roadmap[39]	Yes	Yes[40]	Yes[41]	Yes	Yes	Yes	Yes[42]	Yes
Neural Designer	Artelnics		Proprietary	No	Linux, macOS, Windows	C++	Graphical user interface	Yes	No	No	?	?	No	No	No	?	
OpenNN	Artelnics	2003	GNU LGPL	Yes	Cross-platform	C++	C++	Yes	No	Yes	?	?	No	No	No	?	
PlaidML	Vertex.AI, Intel	2017	AGPL	Yes	Linux, macOS, Windows	Python, C++, OpenCL	Python, C++	?	Some OpenCL ICDs are not recognized	No	Yes	Yes	Yes	Yes		Yes	Yes
PyTorch	Adam Paszke, Sam Gross, Soumith Chintala, Gregory Chanan (Facebook)	2016	BSD	Yes	Linux, macOS, Windows	Python, C, C++, CUDA	Python, C++	Yes	Via separately maintained package[43][44][44]	Yes	Yes	Yes	Yes	Yes		Yes	Yes
Apache SINGA	Apache Software Foundation	2015	Apache 2.0	Yes	Linux, macOS, Windows	C++	Python, C++, Java	No	Supported in V1.0	Yes	?	Yes	Yes	Yes	Yes	Yes	
TensorFlow	Google Brain	2015	Apache 2.0	Yes	Linux, macOS, Windows,[45] Android	C++, Python, CUDA	Python (Keras), C/C++, Java, Go, JavaScript, R,[46] Julia, Swift	No	On roadmap[47] but already with SYCL[48] support	Yes	Yes[49]	Yes[50]	Yes	Yes	Yes	Yes	Yes
Theano	Université de Montréal	2007	BSD	Yes	Cross-platform	Python	Python (Keras)	Yes	Under development[51]	Yes	Yes[52][53]	Through Lasagne's model zoo[54]	Yes	Yes	Yes	Yes[55]	No
Torch	Ronan Collobert, Koray Kavukcuoglu, Clement Farabet	2002	BSD	Yes	Linux, macOS, Windows,[56] Android,[57] iOS	C, Lua	Lua, LuaJIT,[58] C, utility library for C++/OpenCL[59]	Yes	Third party implementations[60][61]	Yes[62][63]	Through Twitter's Autograd[64]	Yes[65]	Yes	Yes	Yes	Yes[56]	No
Wolfram Mathematica	Wolfram Research	1988	Proprietary	No	Windows, macOS, Linux, Cloud computing	C++, Wolfram Language, CUDA	Wolfram Language	Yes	No	Yes	Yes	Yes[66]	Yes	Yes	Yes	Yes[67]	Yes

In practice – Standard libraries

A wide catalogue (2/4)

https://en.wikipedia.org/wiki/Comparison_of_deep-learning_software

Software	Creator	Initial Release	Software license[a]	Open source	Platform	Written in	Interface	OpenMP support	OpenCL support	CUDA support	Automatic differentiation[1]	Has pretrained models	Recurrent nets	Convolutional nets	RBM/DBNs	Parallel execution (multi node)	Actively Developed
BigDL	Jason Dai (Intel)	2016	Apache 2.0	Yes	Apache Spark	Scala	Scala, Python			No		Yes	Yes	Yes			
Caffe	Berkeley Vision and Learning Center	2013	BSD	Yes	Linux, macOS, Windows[2]	C++	Python, MATLAB, C++	Yes	Under development[3]	Yes	Yes	Yes[4]	Yes	Yes	No	?	No[5]
Chainer	Preferred Networks	2015	BSD	Yes	Linux, macOS	Python	Python	No	No	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes
Deeplearning4j	Skymind engineering team; Deeplearning4j community; origi... Adam Gibson			Yes	Linux, macOS, Windows, Android (Cross-platform)	C++, Java	Java, Scala, Clojure, Python (Keras), Kotlin	Yes	No[6]	Yes[7][8]	Computational Graph	Yes[9]	Yes	Yes	Yes	Yes[10]	
Dlib	Davi...				Cross-Platform	C++	C++	Yes	No	Yes	Yes	Yes	No	Yes	Yes	Yes	
Intel Data Analytics Acceleration Library					...acOS, Windows on Intel CPU[11]	C++, Python, Java	C++, Python, Java[11]	Yes	No	No	Yes	No		Yes		Yes	
Intel Math Kernel Library					...acOS, Windows on Intel CPU[12]		C[13]	Yes[14]	No	No	Yes	No	Yes[15]	Yes[15]		No	
Keras	François Ch...				Linux, macOS, Windows	Python	Python, R	Only if using Theano as backend	Can use Theano, Tensorflow or PlaidML as backends	Yes	Yes	Yes[16]	Yes	Yes	No[17]	Yes[18]	Yes
MATLAB + Deep Learning Toolbox	MathWorks		Proprietary	No	Linux, macOS, Windows	C, C++, Java, MATLAB	MATLAB	No	No	Train with Parallel Computing Toolbox and generate CUDA code with GPU Coder[19]	Yes[20]	Yes[21][22]	Yes[21]	Yes[21]	Yes	With Parallel Computing Toolbox[23]	Yes
Microsoft Cognitive Toolkit (CNTK)	Microsoft Research	2016	MIT license[24]	Yes	Windows, Linux[25] (macOS via Docker on roadmap)	C++	Python (Keras), C++, Command line,[26] BrainScript[27] (.NET on roadmap[28])	Yes[29]	No	Yes	Yes	Yes[30]	Yes[31]	Yes[31]	No[32]	Yes[33]	No[34]
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Neural Designer	Artelnics		Proprietary	No	Linux, macOS, Windows	C++	Graphical user interface	Yes	No	No	?	?	No	No	No	?	
OpenNN	Artelnics	2003	GNU LGPL	Yes	Cross-platform	C++	C++	Yes	No	Yes	?	?	No	No	No	?	
PlaidML	Vertex.AI, Intel	2017	AGPL	Yes	Linux, macOS, Windows	Python, C++, OpenCL	Python, C++	?	Some OpenCL ICDs are not recognized	No	Yes	Yes	Yes	Yes		Yes	Yes
PyTorch	Adam Paszke, Sam Gross, Soumith Chintala, Gregory Chanan (Facebook)	2016	BSD	Yes	Linux, macOS, Windows	Python, C, C++, CUDA	Python, C++	Yes	Via separately maintained package[43][44][44]	Yes	Yes	Yes	Yes	Yes		Yes	Yes
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TensorFlow	Google Brain	2015	Apache 2.0	Yes	Linux, macOS, Windows,[45] Android	C++, Python, CUDA	Python (Keras), C/C++, Java, Go, JavaScript, R,[46] Julia, Swift	No	On roadmap[47] but already with SYCL[48] support	Yes	Yes[49]	Yes[50]	Yes	Yes	Yes	Yes	Yes
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TensorFlow

(Google Brain Team)

Not easy to learn
Bigger community
TensorBoard

In practice – Standard libraries

└ A wide catalogue (3/4)

https://en.wikipedia.org/wiki/Comparison_of_deep-learning_software

Keras

TensorFlow

(Google Brain Team)

~~Not~~ easy to learn
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Software	Creator	Initial Release	Software license[a]	Open source	Platform	Written in	Interface	OpenMP support	OpenCL support	CUDA support	Automatic differentiation[1]	Has pretrained models	Recurrent nets	Convolutional nets	RBM/DBNs	Parallel execution (multi node)	Actively Developed
BigDL	Jason Dai (Intel)				Apache Spark	Scala	Scala, Python			No		Yes	Yes	Yes			
Caffe	Berkeley Vision and				Linux, macOS, Windows[2]	C++	Python, MATLAB, C++	Yes	Under development[3]	Yes	Yes	Yes[4]	Yes	Yes	No	?	No[5]
Chainer	Preferr				Linux, macOS	Python	Python	No	No	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes
Deeplearning4j	Skymin Deeplearn				OS, Windows, Android ss-platform)	C++, Java	Java, Scala, Clojure, Python (Keras), Kotlin	Yes	No[6]	Yes[7][8]	Computational Graph	Yes[9]	Yes	Yes	Yes	Yes[10]	
Dlib					Platform	C++	C++	Yes	No	Yes	Yes	Yes	No	Yes	Yes	Yes	
Intel Data Analytics Acceleration Library					Windows on Intel CPU[11]	C++, Python, Java	C++, Python, Java[11]	Yes	No	No	Yes	No		Yes		Yes	
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In practice – Standard libraries

└ A wide catalogue (4/4)

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Keras

TensorFlow

(Google Brain Team)

~~Not~~ easy to learn
Bigger community
TensorBoard

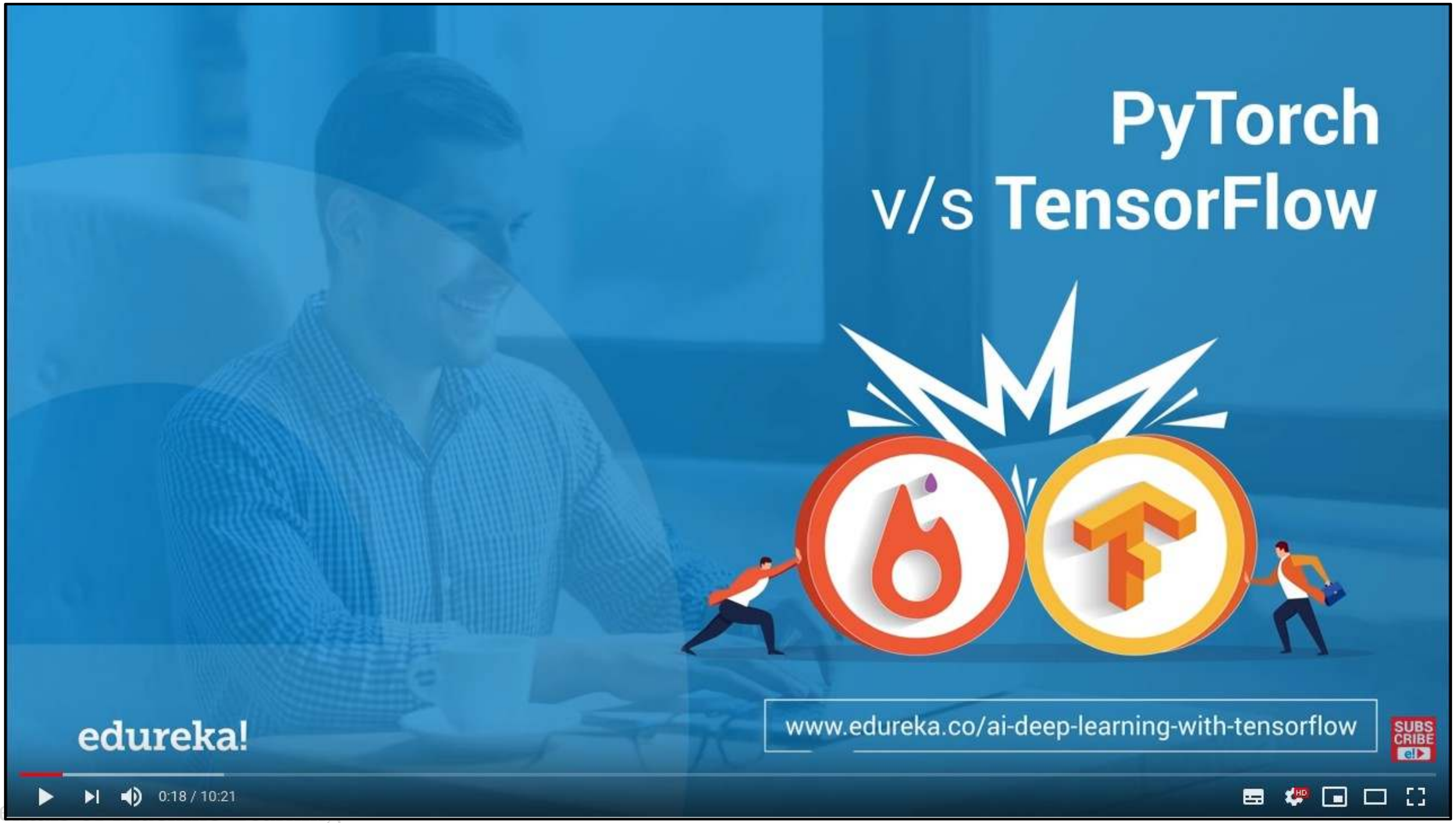
PYTORCH

(Facebook AI Research)

Production-ready
Strong community
Faster than Keras

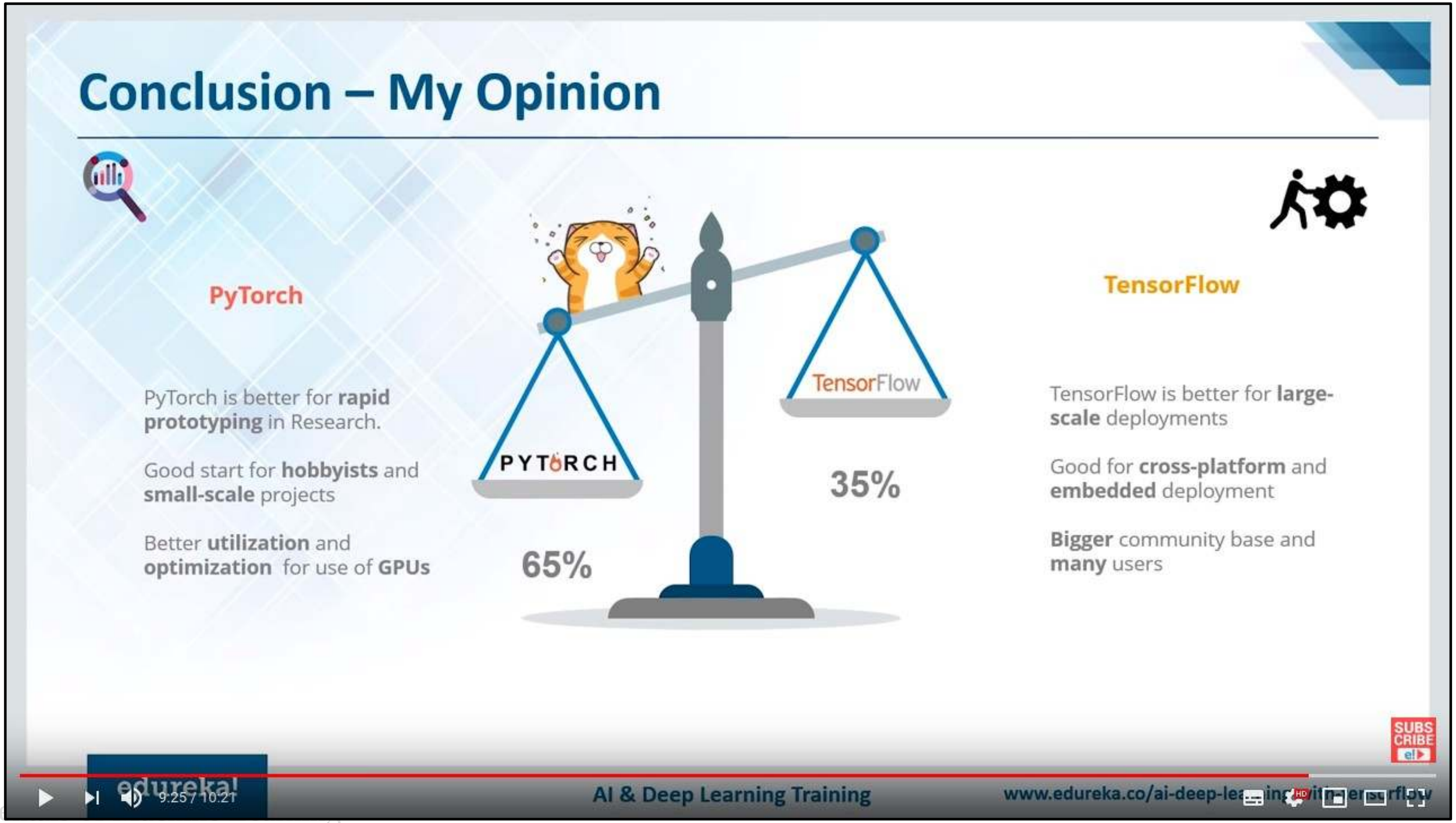
└ PyTorch vs. TensorFlow (1/3)

https://www.youtube.com/watch?v=Mn1AACvK-YI



In practice – Standard libraries
└ PyTorch vs. TensorFlow (2/3)

https://www.youtube.com/watch?v=Mn1AACvK-YI

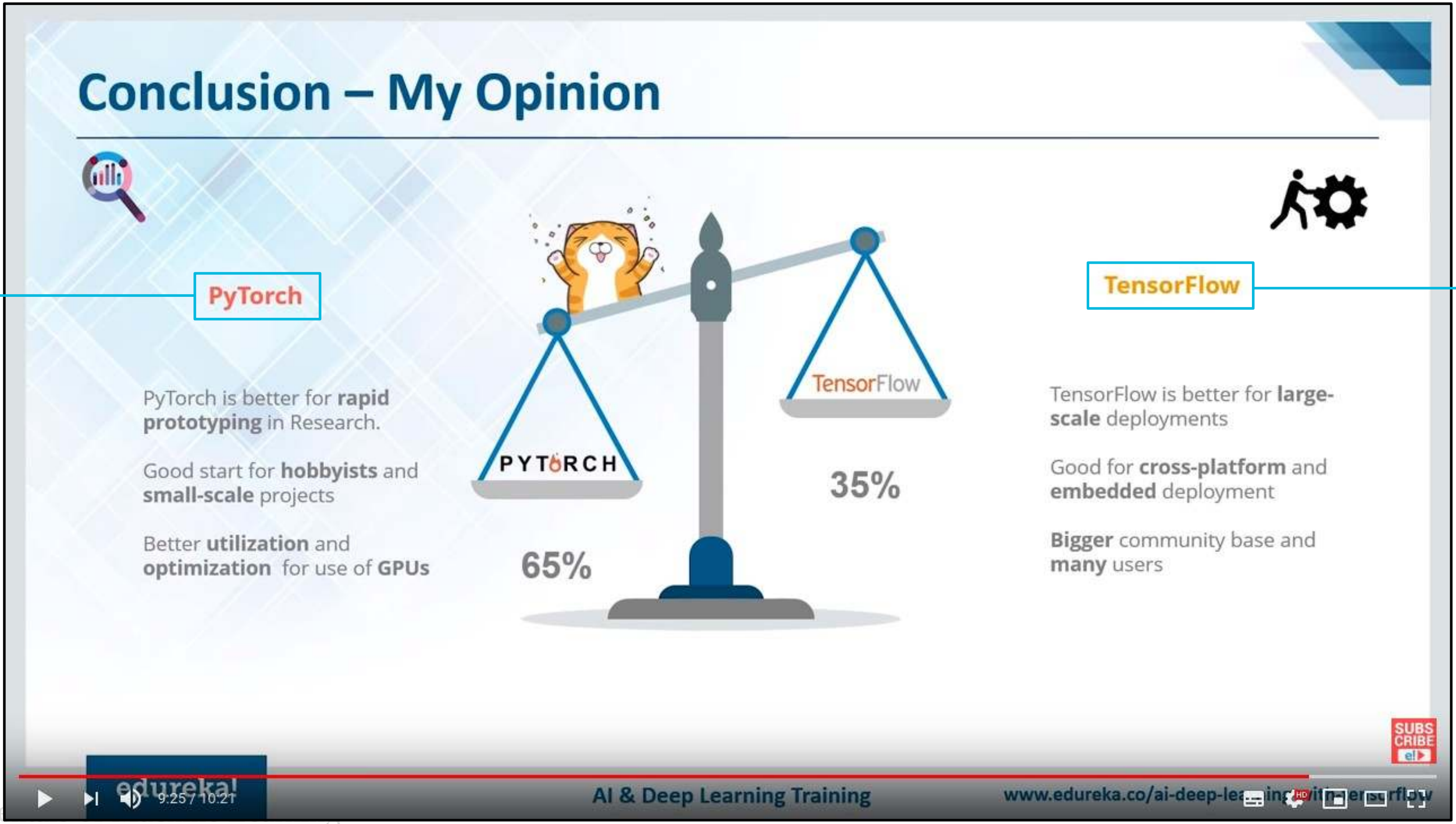


In practice – Standard libraries

└ PyTorch vs. TensorFlow (3/3)

https://www.youtube.com/watch?v=Mn1AACvK-YI

Today's
practical session



Later, when you're
a data scientist

… if you
really need it …

In practice – Standard libraries

└ A quick example in PyTorch - Step 1: prepare the dataloaders

https://pytorch.org/tutorials/beginner/blitz/cifar10_tutorial.html

```
transform = transforms.Compose(
    [transforms.ToTensor(),
     transforms.Normalize((0.5, 0.5, 0.5), (0.5, 0.5, 0.5))])

batch_size = 4

trainset = torchvision.datasets.CIFAR10(root='./data', train=True,
                                        download=True, transform=transform)
trainloader = torch.utils.data.DataLoader(trainset, batch_size=batch_size,
                                          shuffle=True, num_workers=2)

testset = torchvision.datasets.CIFAR10(root='./data', train=False,
                                       download=True, transform=transform)
testloader = torch.utils.data.DataLoader(testset, batch_size=batch_size,
                                         shuffle=False, num_workers=2)

classes = ('plane', 'car', 'bird', 'cat',
           'deer', 'dog', 'frog', 'horse', 'ship', 'truck')
```

In practice – Standard libraries

A quick example in PyTorch - Step 2: Define the model

https://pytorch.org/tutorials/beginner/blitz/cifar10_tutorial.html

```
class Net(nn.Module):
    def __init__(self):
        super().__init__()
        self.conv1 = nn.Conv2d(3, 6, 5)
        self.pool = nn.MaxPool2d(2, 2)
        self.conv2 = nn.Conv2d(6, 16, 5)
        self.fc1 = nn.Linear(16 * 5 * 5, 120)
        self.fc2 = nn.Linear(120, 84)
        self.fc3 = nn.Linear(84, 10)

    def forward(self, x):
        x = self.pool(F.relu(self.conv1(x)))
        x = self.pool(F.relu(self.conv2(x)))
        x = torch.flatten(x, 1) # flatten all dimensions except batch
        x = F.relu(self.fc1(x))
        x = F.relu(self.fc2(x))
        x = self.fc3(x)
        return x


net = Net()
```

A quick example in PyTorch - Step 3: Choose the loss function, optimizer, scheduler, etc.

https://pytorch.org/tutorials/beginner/blitz/cifar10_tutorial.html

```
import torch.optim as optim

criterion = nn.CrossEntropyLoss()
optimizer = optim.SGD(net.parameters(), lr=0.001, momentum=0.9)
```

In practice – Standard libraries

A quick example in PyTorch - Step 4: Train the model

utorials/beginner/blitz/cifar10_tutorial.html

```
for epoch in range(2):  # loop over the dataset multiple times

    running_loss = 0.0
    for i, data in enumerate(trainloader, 0):
        # get the inputs; data is a list of [inputs, labels]
        inputs, labels = data

        # zero the parameter gradients
        optimizer.zero_grad()

        # forward + backward + optimize
        outputs = net(inputs)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()


        # print statistics
        running_loss += loss.item()
        if i % 2000 == 1999:    # print every 2000 mini-batches
            print(f'[{epoch + 1}, {i + 1:5d}] loss: {running_loss / 2000:.3f}')
            running_loss = 0.0

print('Finished Training')
```

You can evaluate the model on the validation dataset here

A quick example in PyTorch - Step 5: Test the model

https://pytorch.org/tutorials/beginner/blitz/cifar10_tutorial.html

```
correct = 0
total = 0
# since we're not training, we don't need to calculate the gradients for our outputs
with torch.no_grad():
    for data in testloader:
        images, labels = data
        # calculate outputs by running images through the network
        outputs = net(images)
        # the class with the highest energy is what we choose as prediction
        _, predicted = torch.max(outputs.data, 1)
        total += labels.size(0)
        correct += (predicted == labels).sum().item()

print(f'Accuracy of the network on the 10000 test images: {100 * correct // total} %')
```

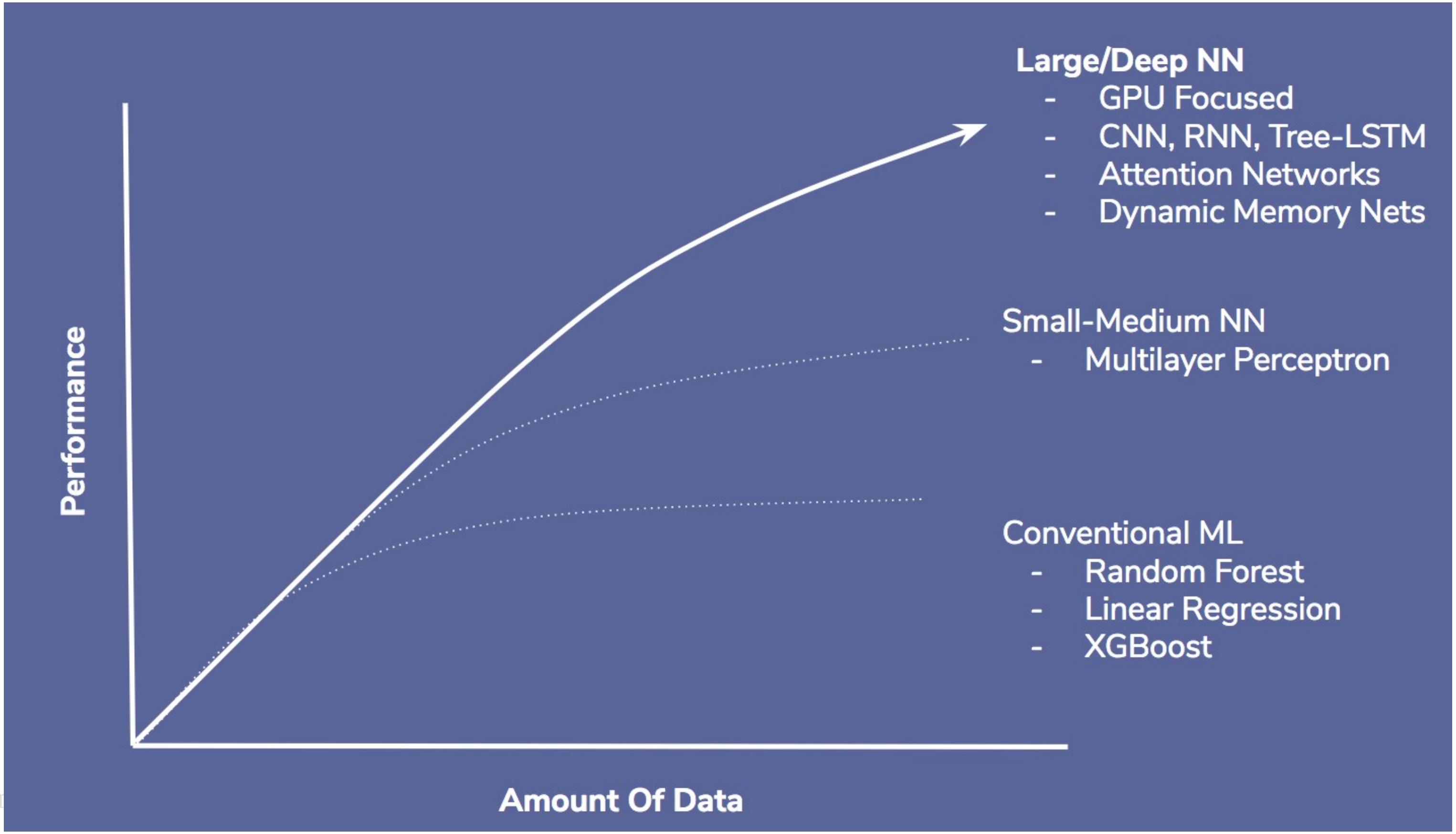
OUTLINE

- Introduction
- From neural networks to deep learning
- In practice:
  - Transfer learning
  - Standard libraries
- Limitations & future
- Focus: few-shot learning

Limitations & future

└ The limits of deep learning - It (generally) requires a lot of data

https://towardsdatascience.com/deep-misconceptions-about-deep-learning-f26c41faceec

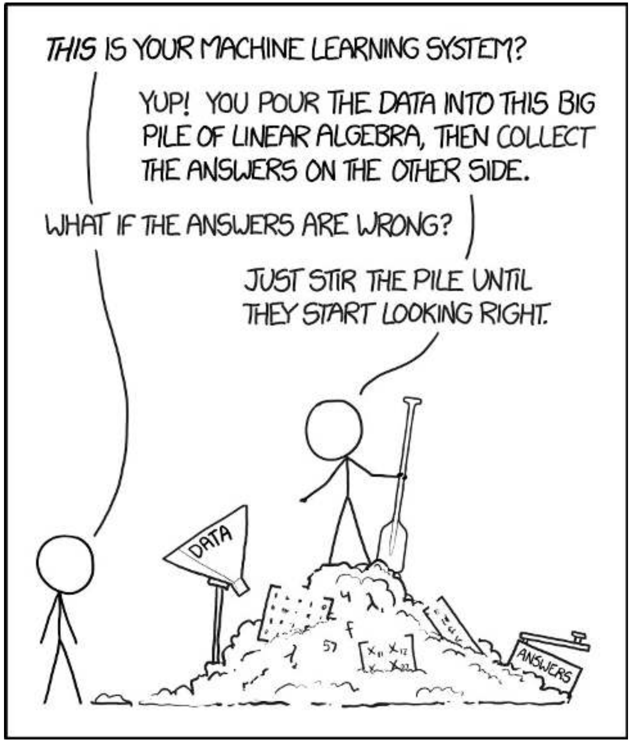


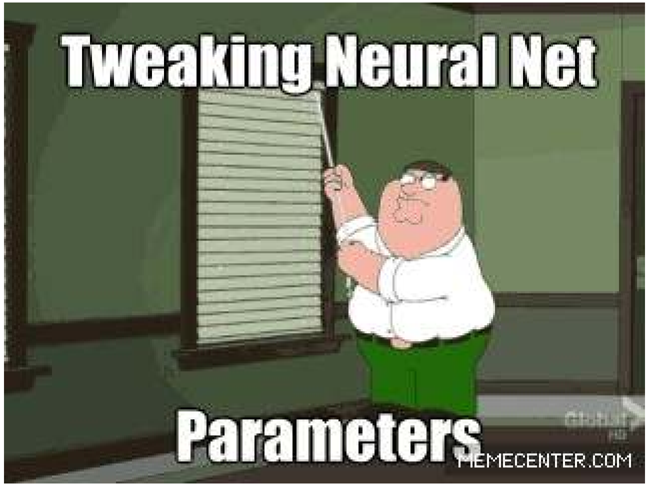
Limitations & future

└ The limits of deep learning - It requires similarities in data distributions

https://www.theverge.com/2016/4/7/11383272/google-self-driving-arizona-desert-conditions
https://www.pinterest.fr/pin/476255729316690445/

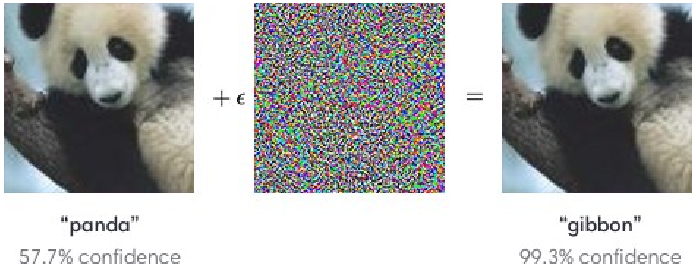
https://xkcd.com/1838/
https://towardsdatascience.com/three-reasons-that-you-should-not-use-deep-learning-15bec517b622

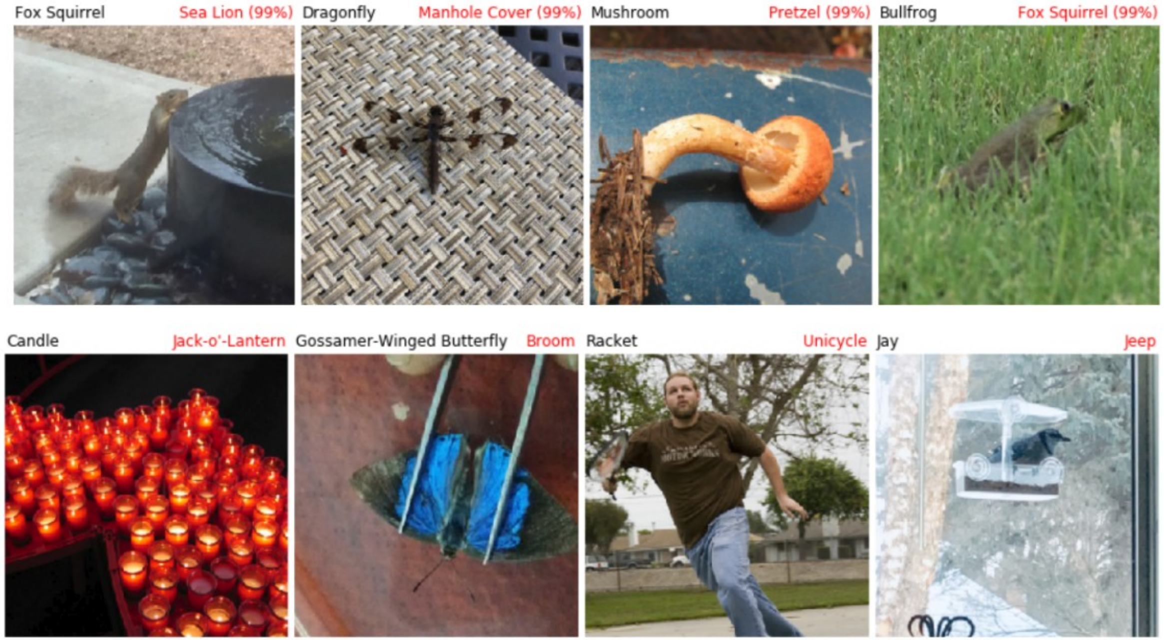




└ The limits of deep learning - Adversarial examples

https://openai.com/blog/adversarial-example-research/
https://medium.com/syncedreview/natural-adversarial-examples-slash-image-classification-accuracy-by-90-702f381acbb8
https://nicholas.carlini.com/code/audio_adversarial_examples



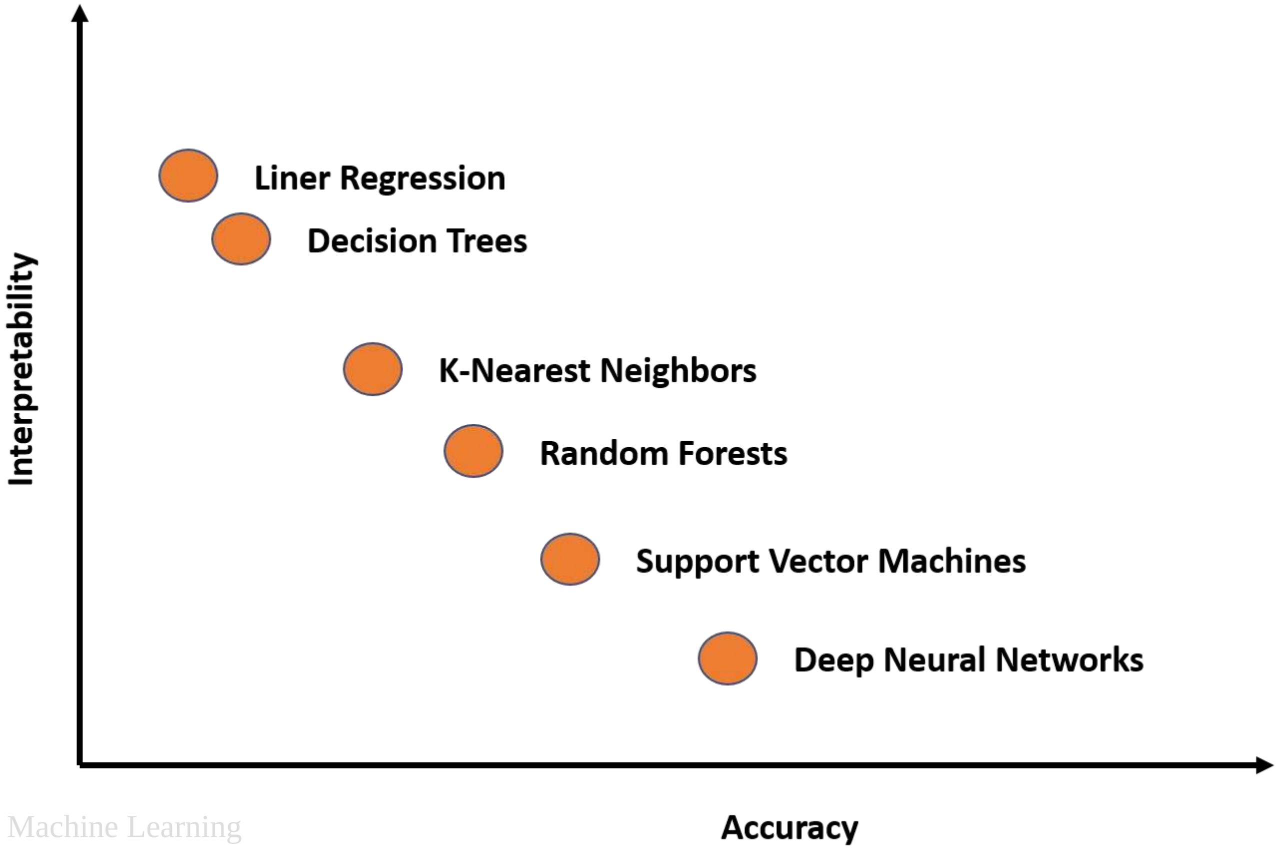


"okay google browse to evil dot com"

"without the dataset the article is useless"

The limits of deep learning - The challenge of interpretability

https://towardsdatascience.com/interpretability-vs-accuracy-the-friction-that-defines-deep-learning-dae16c84db5c

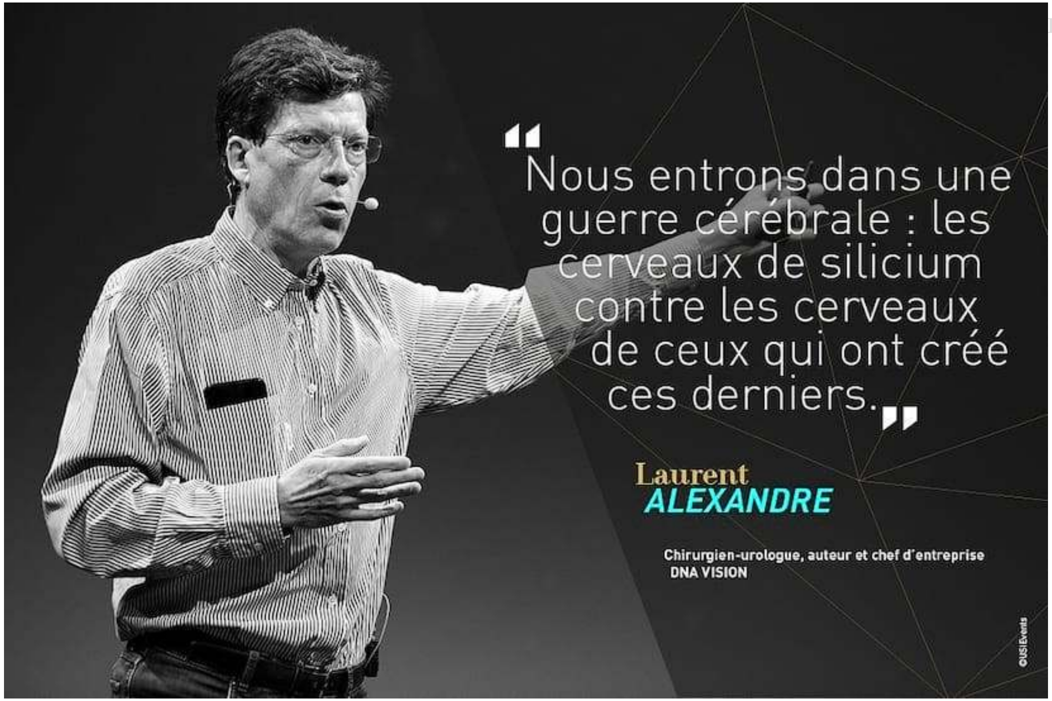


Limitations & future

The limits of deep learning - Something obscure for society, exploited by quacks

https://www.youtube.com/watch?app=desktop&v=miIy2gb-Lgc

nts.com/laurent-alexandre-premunir-contre-dangers-de-intelligence-artificielle/



└ The limits of deep learning - A strong environmental impact

https://news.mit.edu/2020/shrinking-deep-learning-carbon-footprint-0807
https://arxiv.org/pdf/1906.02243.pdf

In June, OpenAI unveiled the largest language model in the world, a text-generating tool called GPT-3 that can write creative fiction, translate legalese into plain English, and answer obscure trivia questions. It's the latest feat of intelligence achieved by deep learning, a machine learning method patterned after the way neurons in the brain process and store information.

But it came at a hefty price: at least $4.6 million and 355 years in computing time, assuming the model was trained on a standard neural network chip, or GPU. The model's colossal size — 1,000 times larger than a typical language model — is the main factor in its high cost.

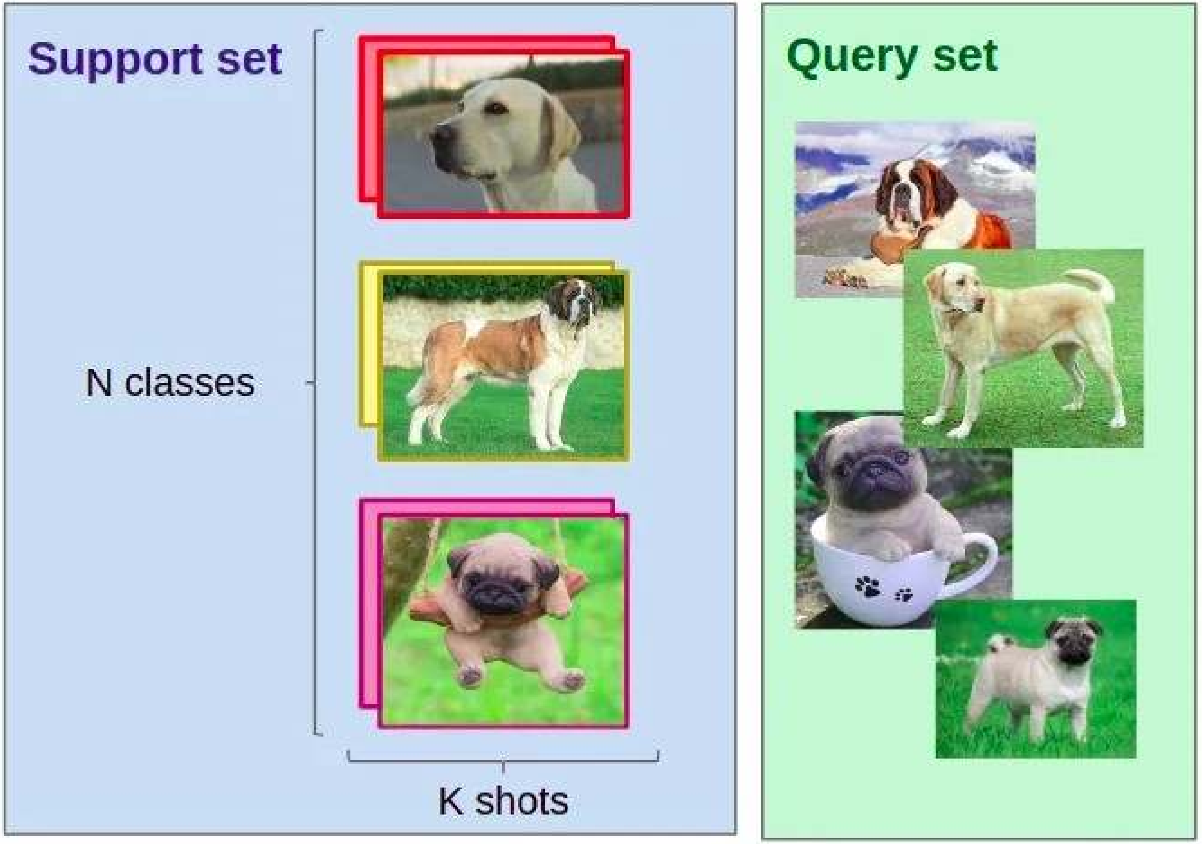
Consumption	CO_2e (lbs)
Air travel, 1 passenger, NY↔SF	1984
Human life, avg, 1 year	11,023
American life, avg, 1 year	36,156
Car, avg incl. fuel, 1 lifetime	126,000
Training one model (GPU)	
NLP pipeline (parsing, SRL)	39
w/ tuning & experimentation	78,468
Transformer (big)	192
w/ neural architecture search	626,155

Table 1: Estimated CO_2 emissions from training common NLP models, compared to familiar consumption.[1]

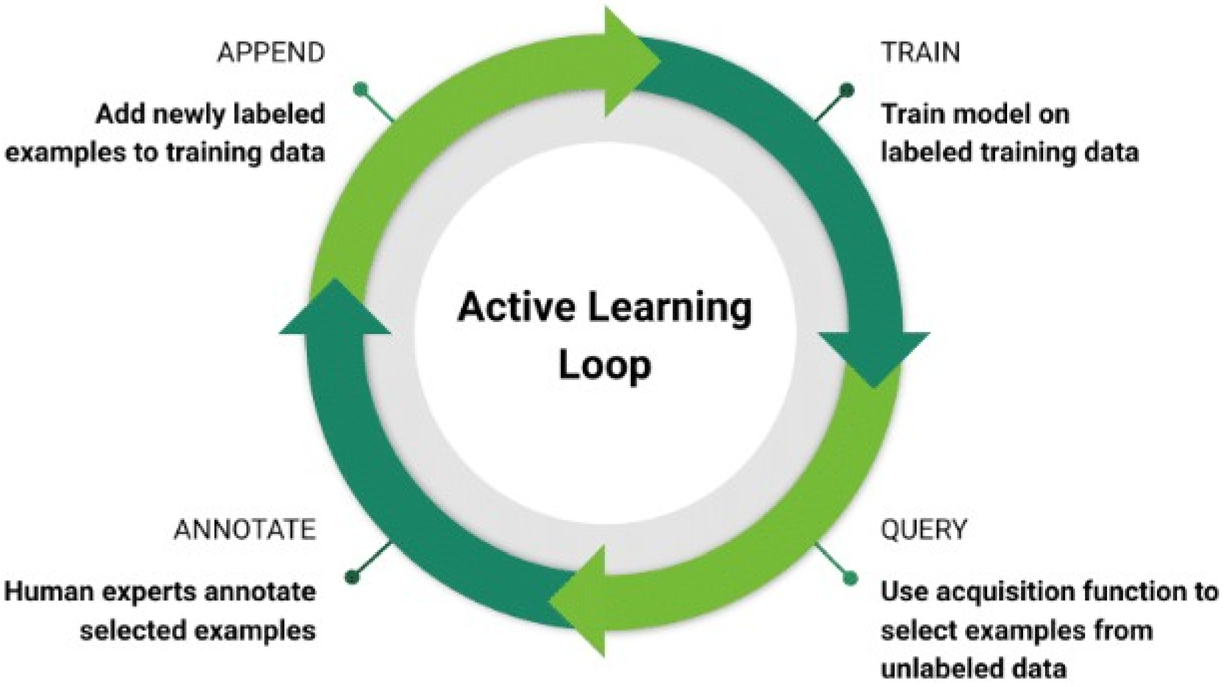
Limitations & future

└ Research in deep learning - Few-shot learning

https://neptune.ai/blog/understanding-few-shot-learning-in-computer-vision



https://blogs.nvidia.com/blog/2020/01/16/what-is-active-learning/



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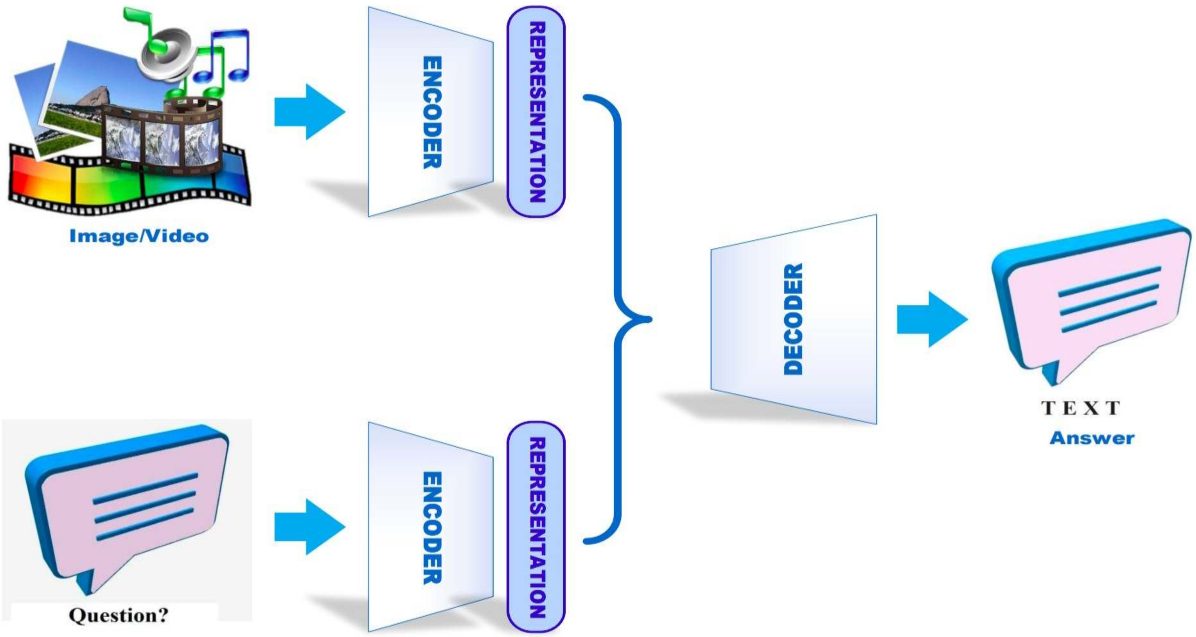
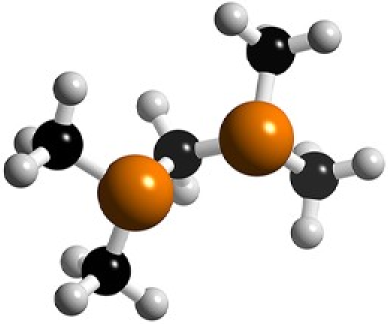


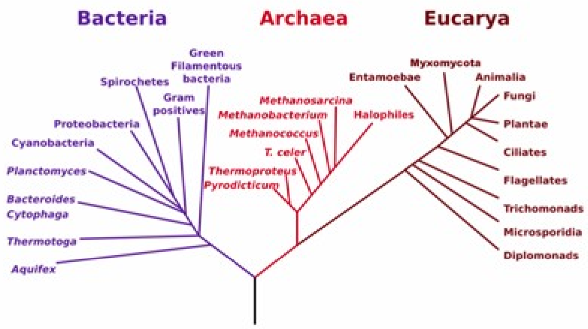
Limitations & future

└ Research in deep learning - Geometric deep learning

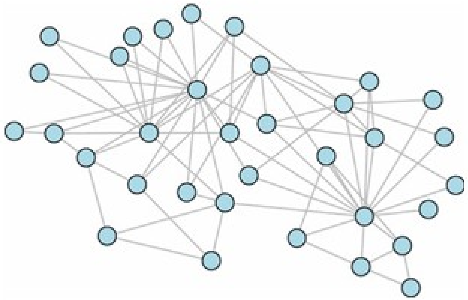
https://flawnsontong.medium.com/what-is-geometric-deep-learning-b2adb662d91d



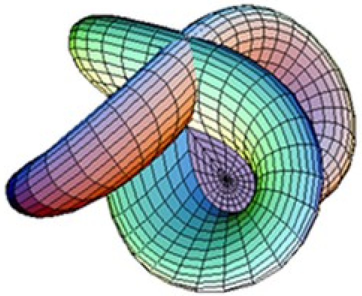
Molecules



Trees



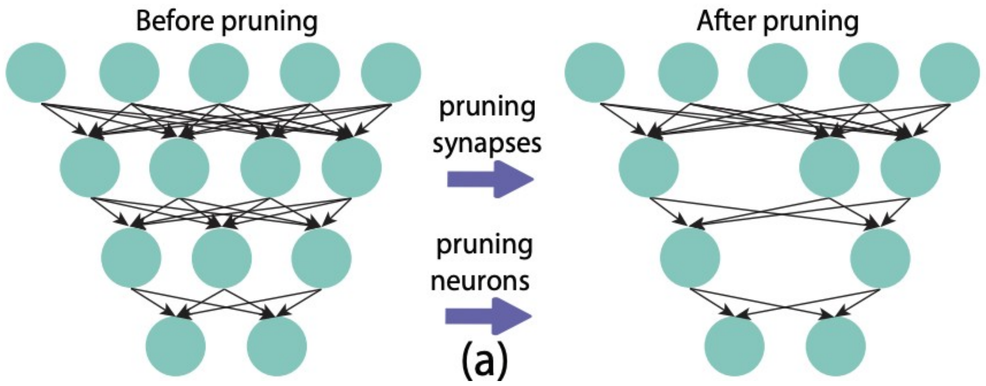
Networks

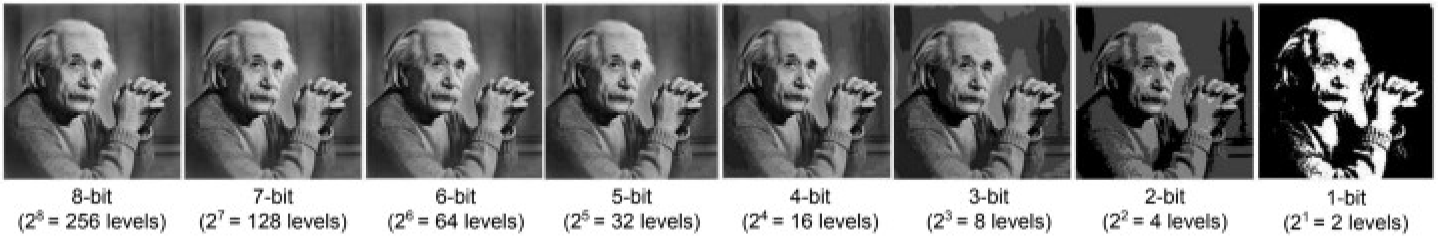


Manifolds

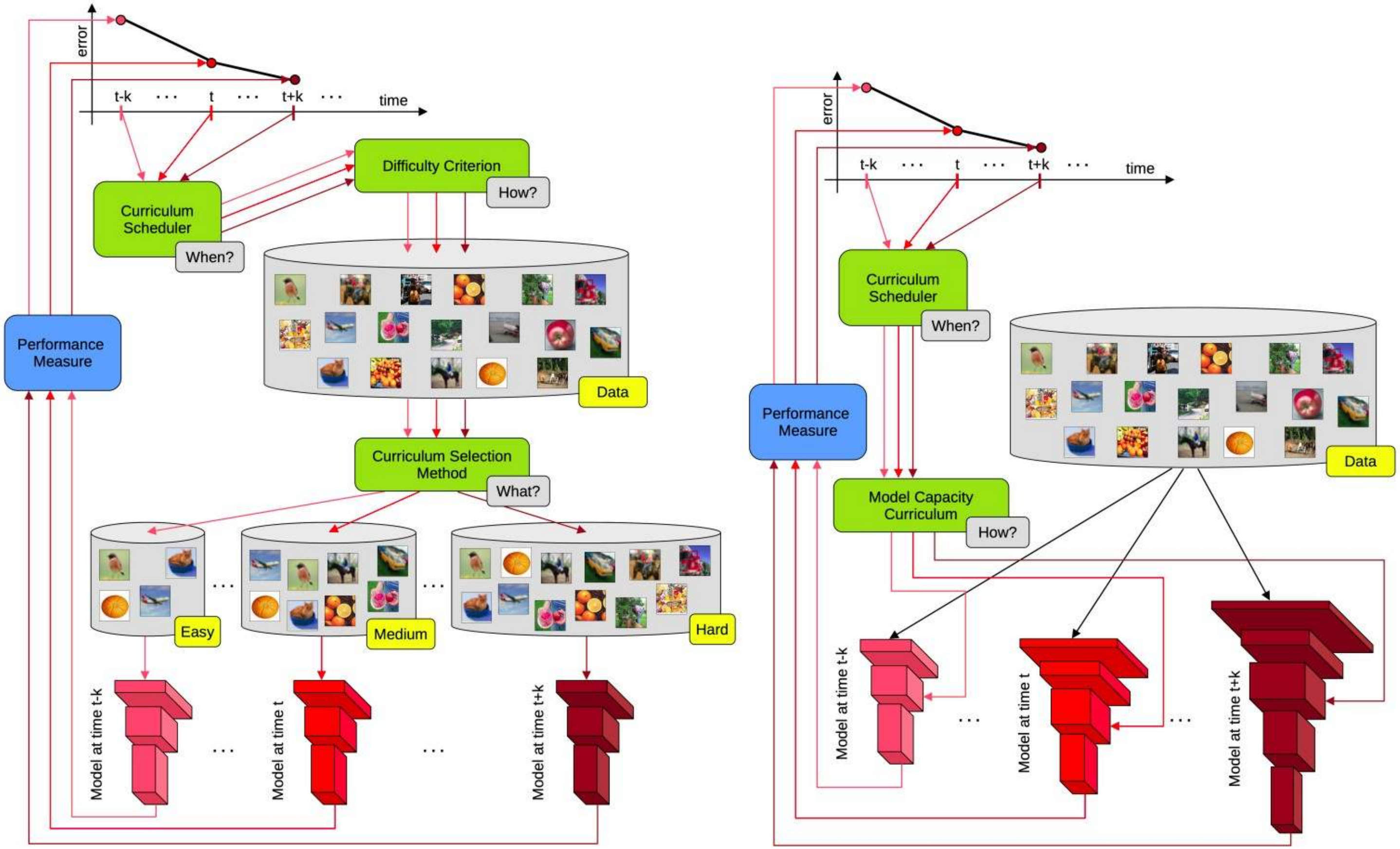
└ Research in deep learning - Compression of models

https://medium.com/gsi-technology/an-overview-of-model-compression-techniques-for-deep-learning-in-space-3fd8d4ce84e5



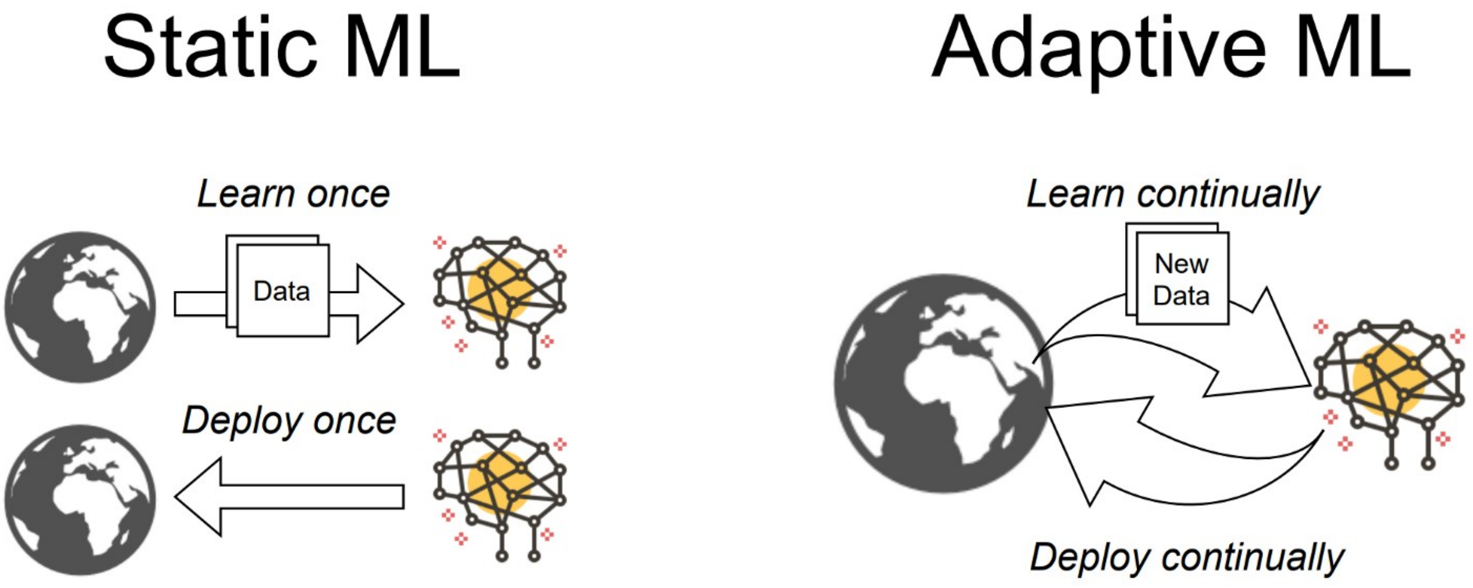


https://arxiv.org/pdf/2101.10382.pdf

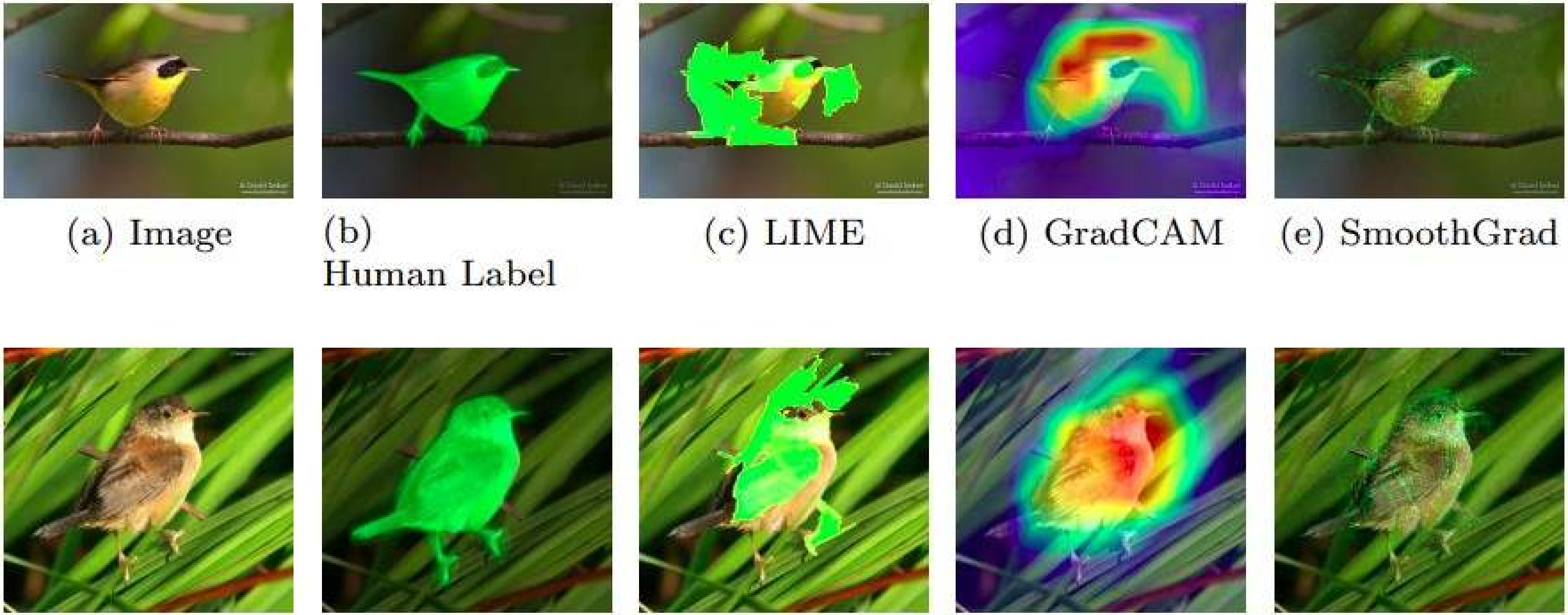


a General framework for data-level curriculum learning. b General framework for model-level curriculum.

https://medium.com/continual-ai/towards-adaptive-ai-with-continual-learning-f493fd0d698



https://arxiv.org/pdf/2103.10689.pdf



(a) Image (b) Human Label (c) LIME (d) GradCAM (e) SmoothGrad

(f) Image (g) Human Label (h) LIME (i) GradCAM (j) SmoothGrad

Limitations & future

Research in deep learning - Robustness

https://infoscience.epfl.ch/record/229872/files/spm_preprint.pdf
https://upcommons.upc.edu/bitstream/handle/2117/188799/Givaki%20et%20al.pdf

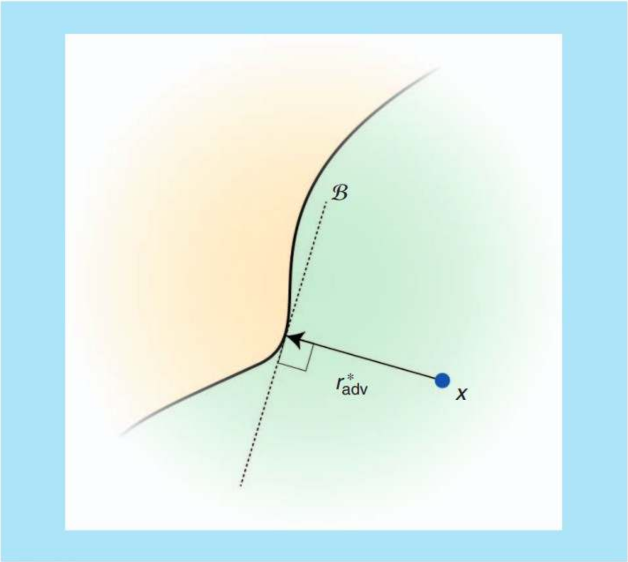
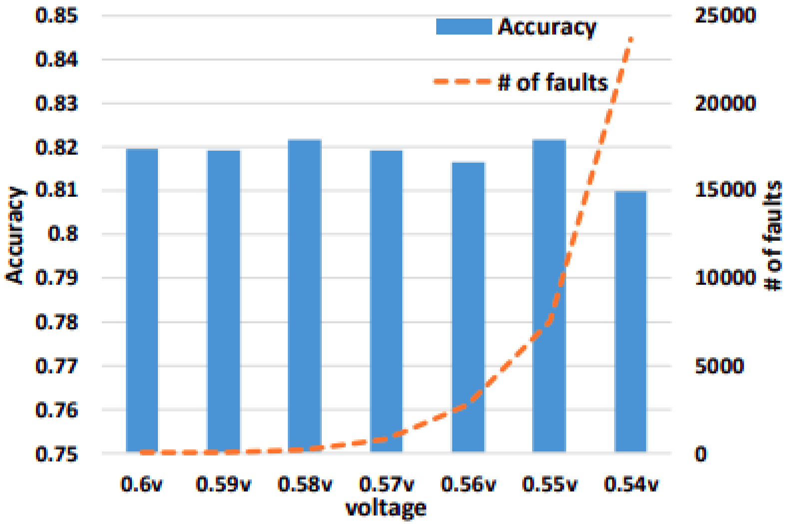


FIGURE 6. r^*_{adv} denotes the adversarial perturbation of x (with $p = 2$). Note that r^*_{adv} is orthogonal to the decision boundary $\mathcal{B}$ and $\|r^*_{adv}\|_2 = \text{dist}(x, \mathcal{B})$.



(c) (Activation Function, FPGA)= (Tanh,VC707)

Limitations & future

Research in deep learning - Etc.

https://www.google.com

research topics in deep learning

Tous | Images | Vidéos | Actualités | Livres | Plus | Outils

Environ 372 000 000 résultats (0,64 secondes)

Articles universitaires correspondant aux termes **research topics in deep learning**

Deep learning - Goodfellow - Cité 43293 fois

Deep learning in bioinformatics - Min - Cité 1286 fois

Deep learning - Bengio - Cité 1447 fois

This article features the top 10 research and thesis topics for ML projects for students to try in 2022

- Text Mining and Text Classification. ...
- Image-Based Applications. ...
- Machine Vision. ...
- Clustering. ...
- Optimization. ...
- Voice Classification. ...
- Sentiment Analysis. ...
- Recommendation Framework Project.

Plus... • 7 janv. 2022

OUTLINE

- Introduction
- From neural networks to deep learning
- In practice:
  - Transfer learning
  - Standard libraries
- Limitations & future
- Focus: few-shot learning

Few-shot learning

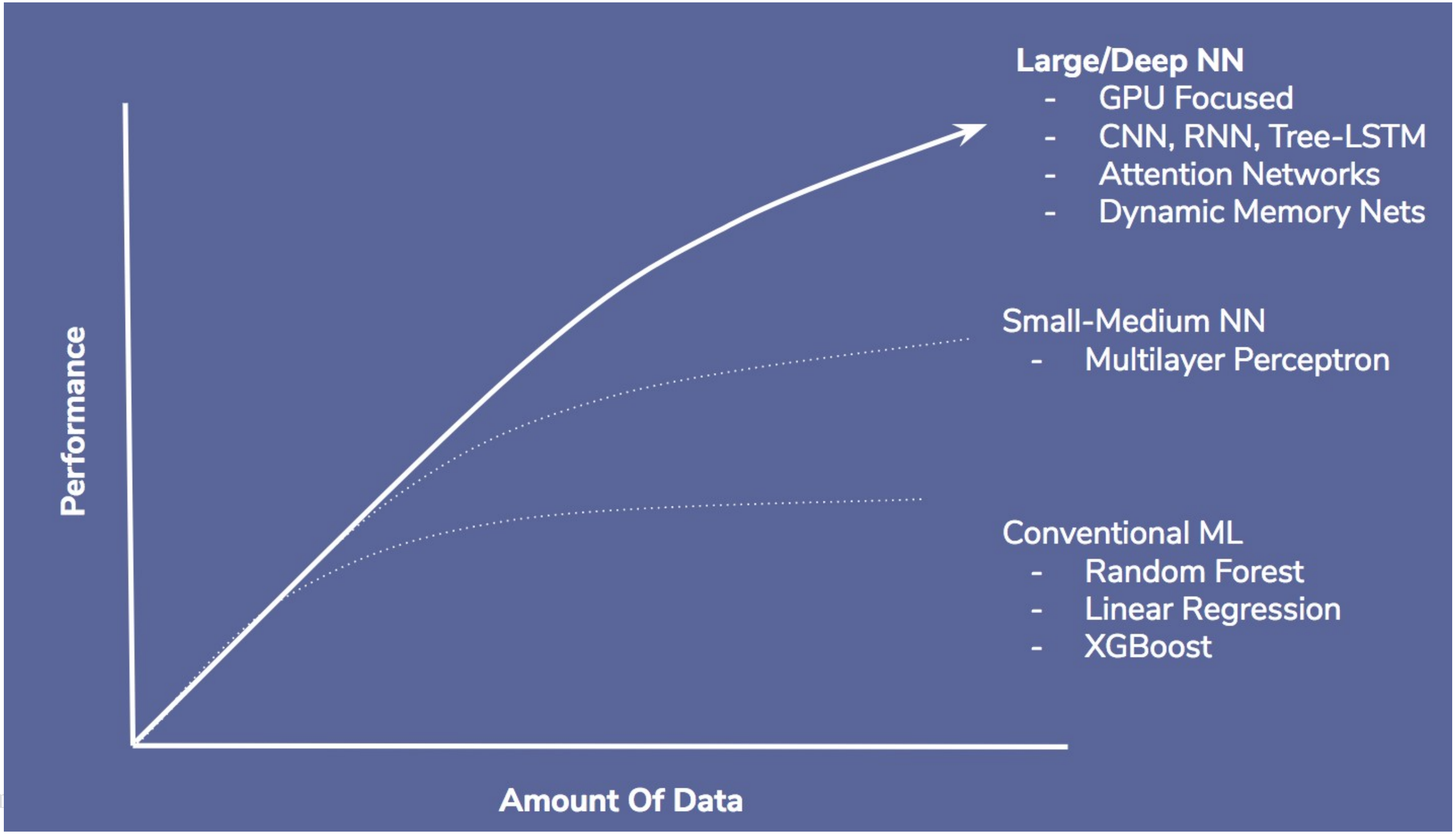
└ Motivating example

Tenenbaum, J. B., Kemp, C., Griffiths, T. L., & Goodman, N. D. (2011). How to grow a mind: Statistics, structure, and abstraction. science, 331(6022), 1279-1285.

Few-shot learning

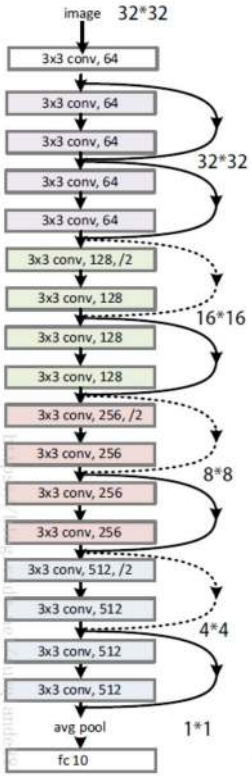
└ Recall: supervised algorithms require a lot of data

https://towardsdatascience.com/deep-misconceptions-about-deep-learning-f26c41faceec



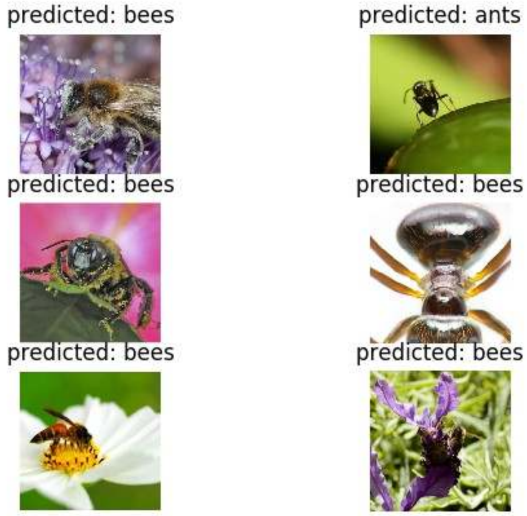
└ The difference with transfer learning (1/2)

https://hackernoon.com/traffic-sign-classification-6e7113d9c4d5
https://medium.com/zylapp/review-of-deep-learning-algorithms-for-image-classification-5fdbca4a05e2
https://pytorch.org/tutorials/beginner/transfer_learning_tutorial.html



ResNet18 is a powerful image recognition network that has been trained on Imagenet 1000

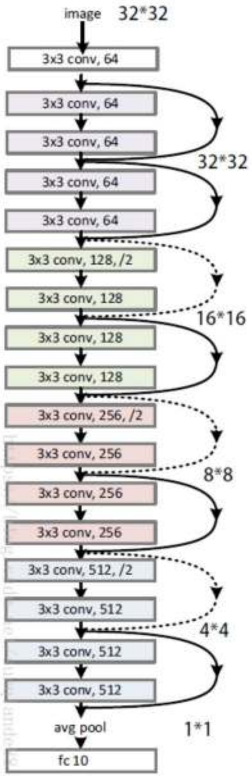
Objective: use its recognition power on our small dataset of bees and ants



Few-shot learning

└ The difference with transfer learning (2/2)

https://hackernoon.com/traffic-sign-classification-6e7113d9c4d5
https://medium.com/zylapp/review-of-deep-learning-algorithms-for-image-classification-5fdbca4a05e2
https://pytorch.org/tutorials/beginner/transfer_learning_tutorial.html



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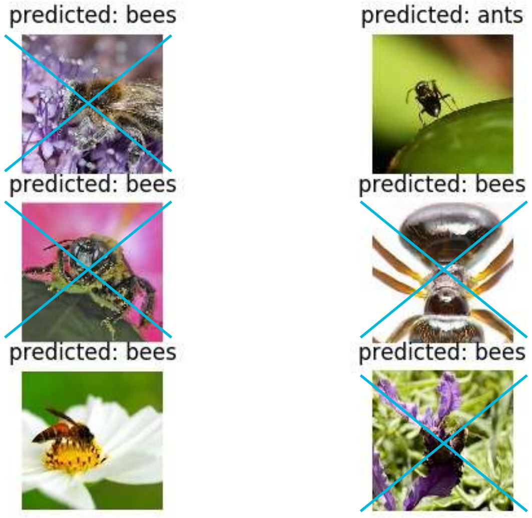
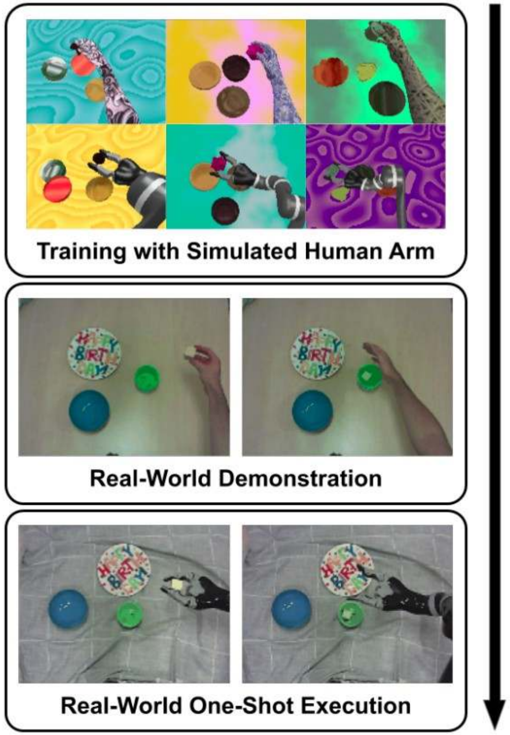


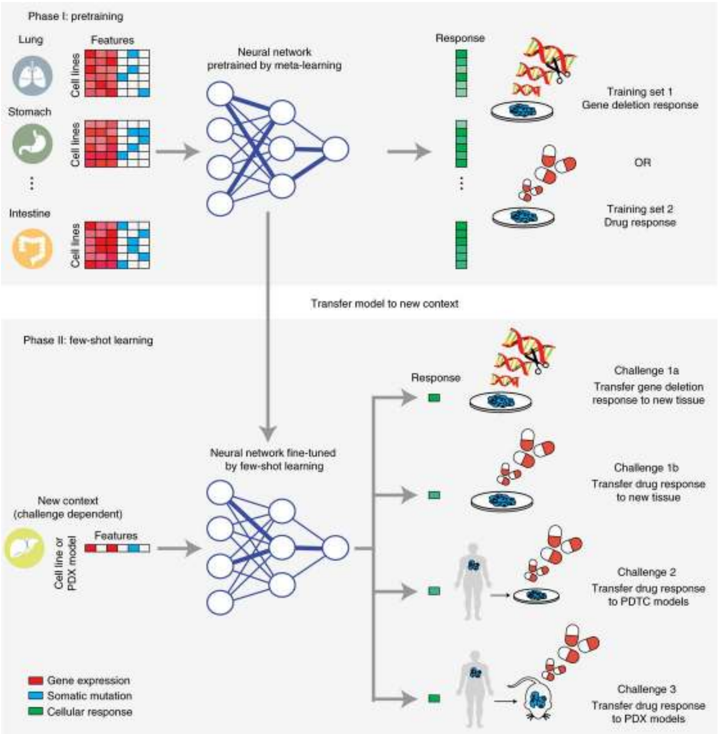
Typically 1-5 labeled samples per class

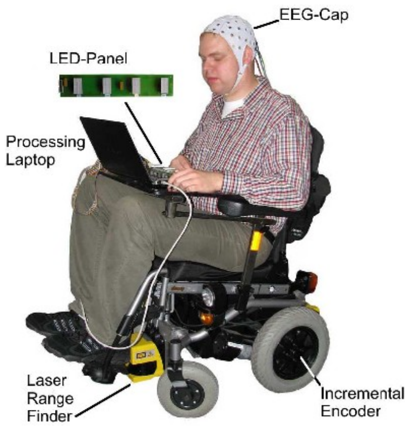
Few-shot learning
└ When does it occur?

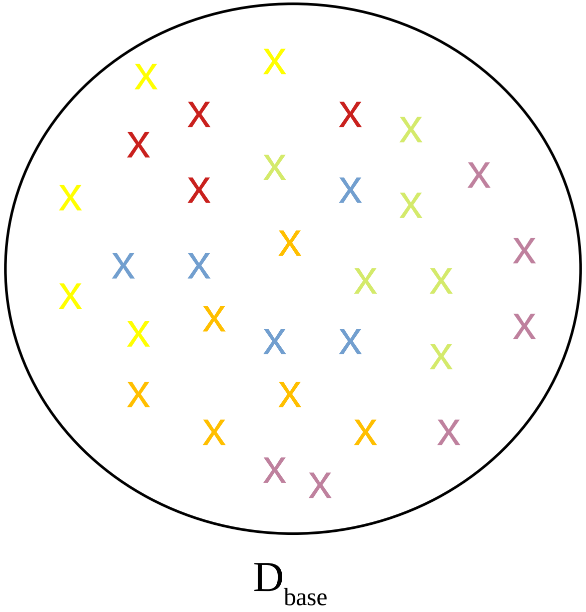
https://www.semanticscholar.org/paper/Navigating-a-smart-wheelchair-with-a-brain-computer-Mandel-L%C3%BCth/8b6c1b061bbc3c54186a06bb9cc40a0367eb9d44
https://www.nature.com/articles/s43018-020-00169-2
Bonardi et al. **Learning One-Shot Imitation from Humans without Humans**. *IEEE Robotics and Automation Letters*, *5*. 2020

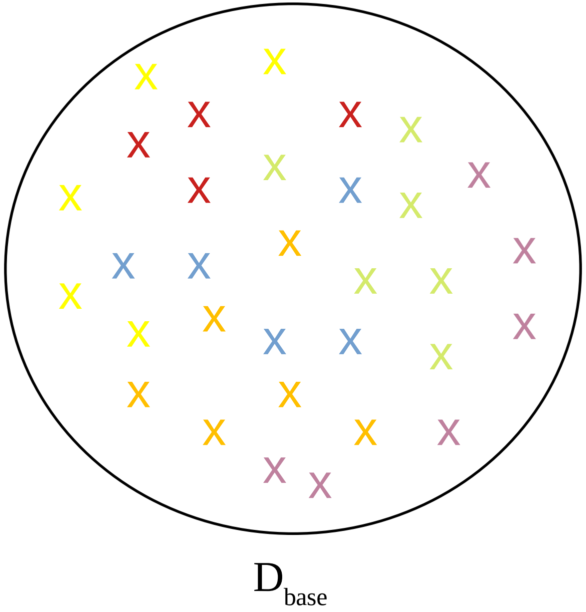


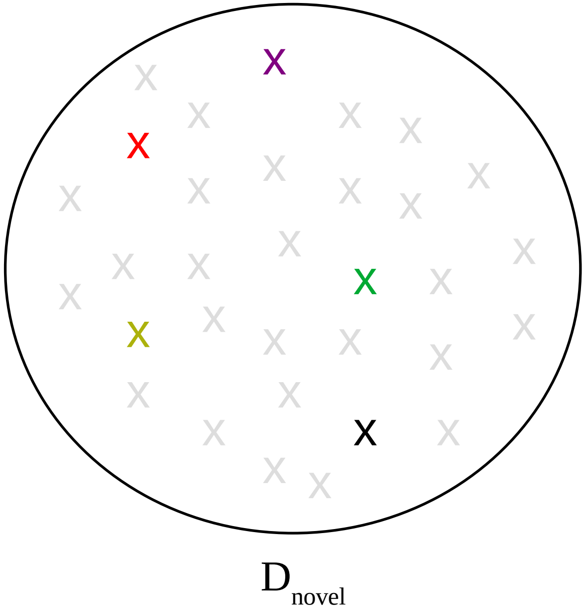
ng

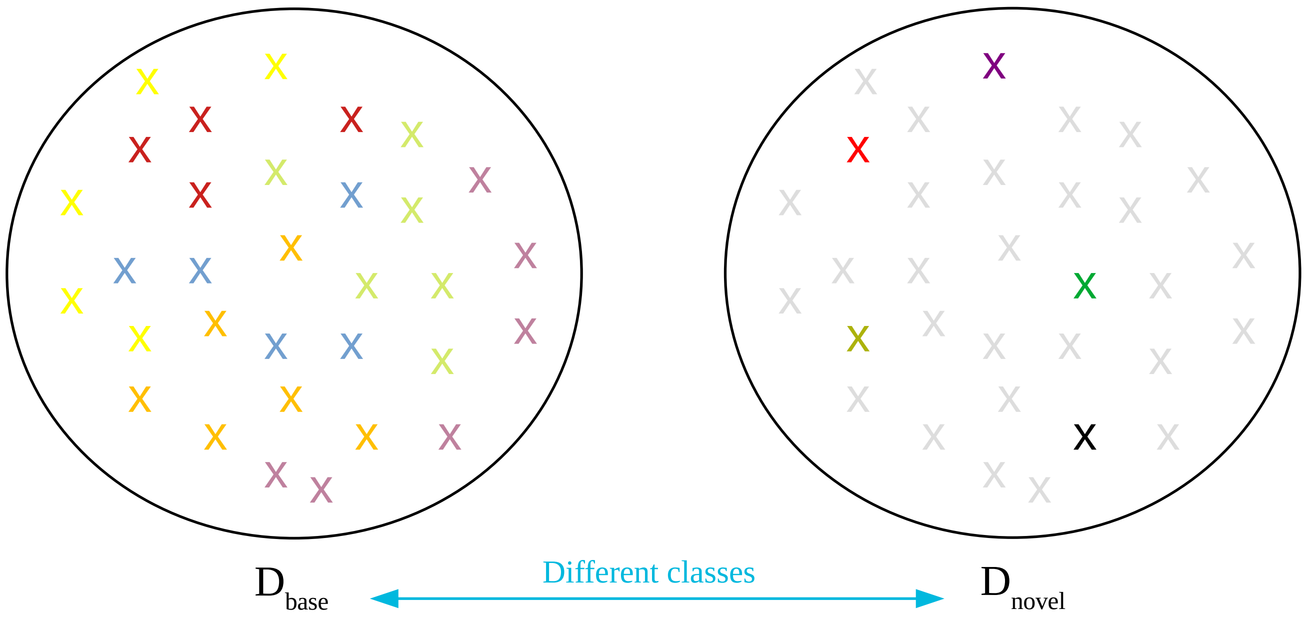




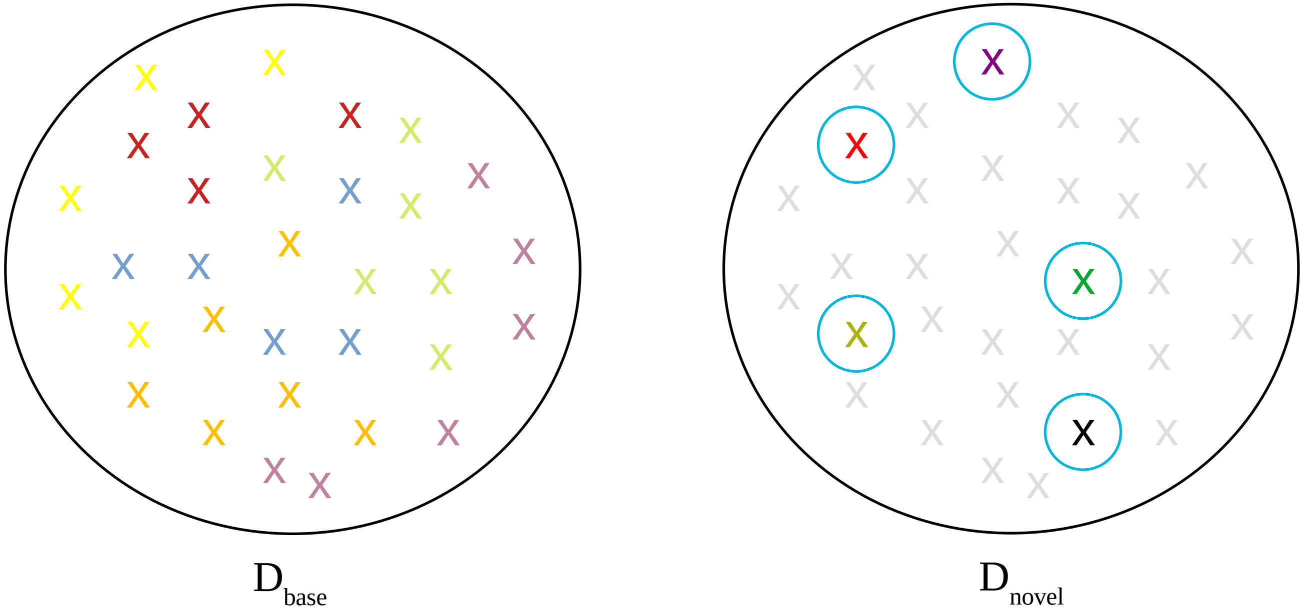

D_{base}


X
D_base

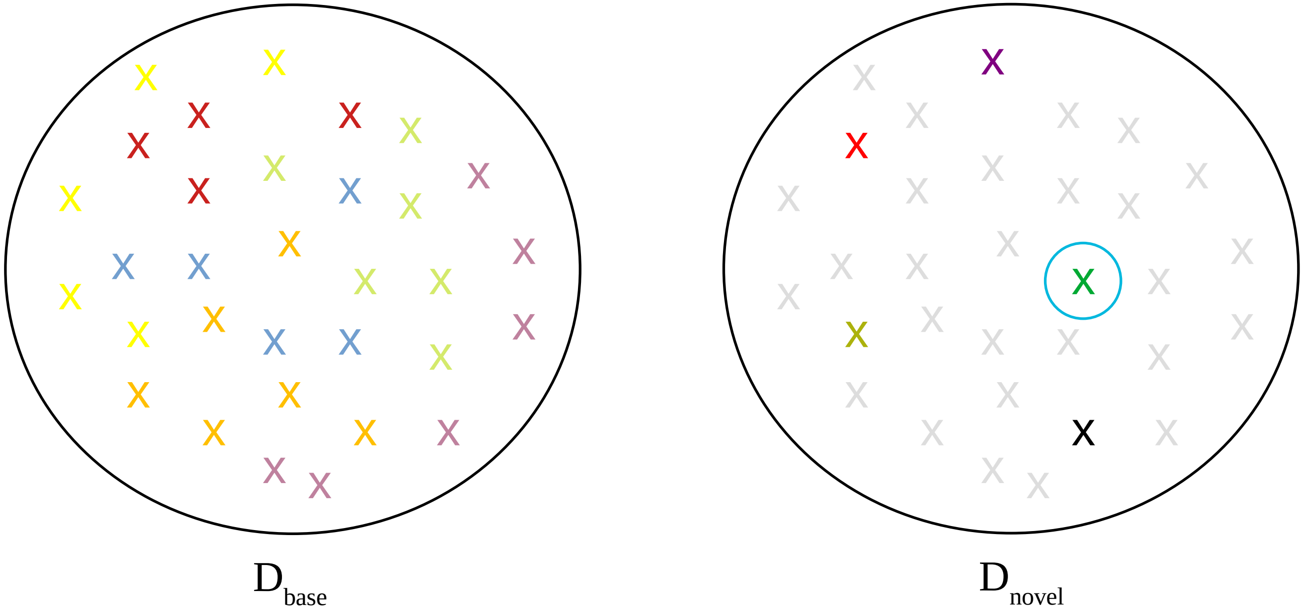

X
D_novel



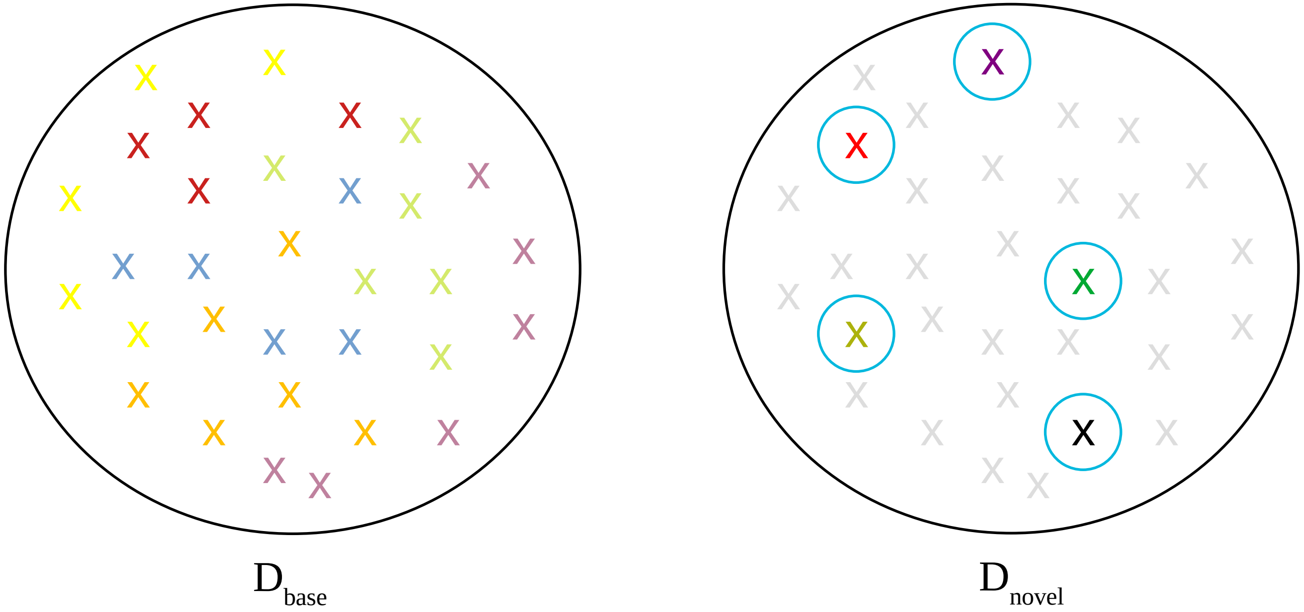
D_{base}
Different classes
D_{novel}



- N ways:
  Unseen classes
  Here, 5

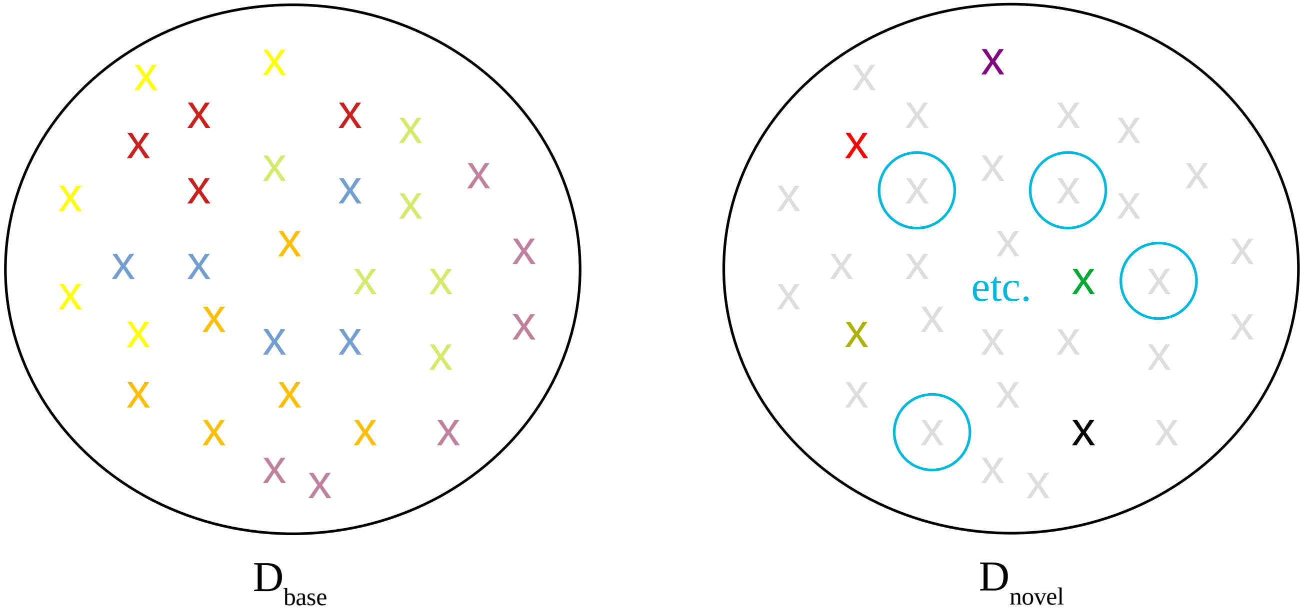


- *N* ways:
  Unseen classes
  Here, 5
- *K* shots:
  Labeled samples per class
  Here, 1



Support set of *NK* samples

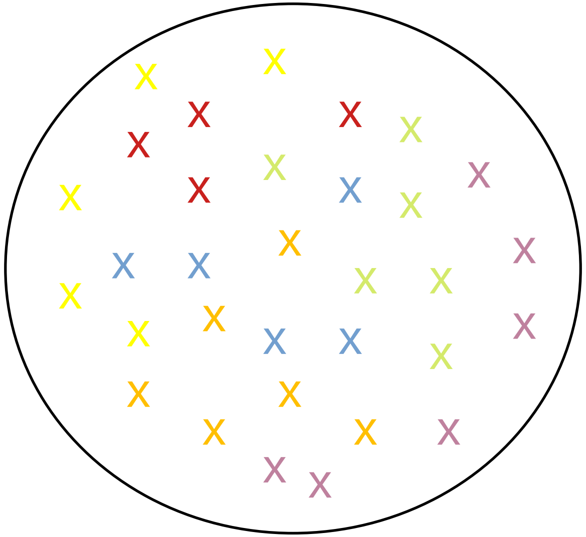
- *N* ways:
  Unseen classes
  Here, 5
- *K* shots:
  Labeled samples per class
  Here, 1



Support set of *NK* samples

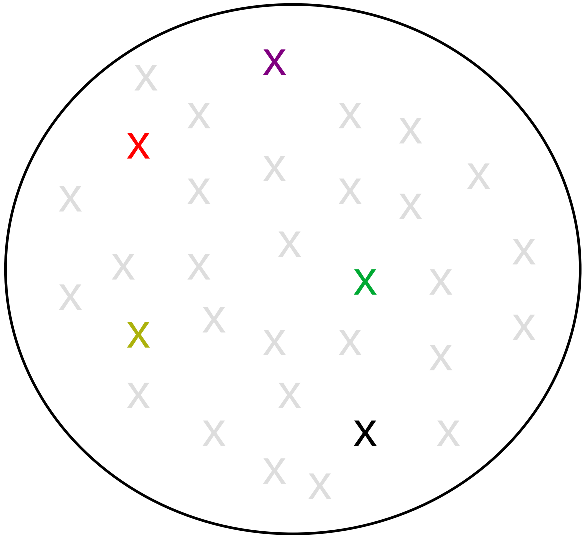
- *N* ways:
  Unseen classes
  Here, 5
- *K* shots:
  Labeled samples per class
  Here, 1

- Queries:
  Samples to classify
  From selected ways



D_{base}

Used to train model

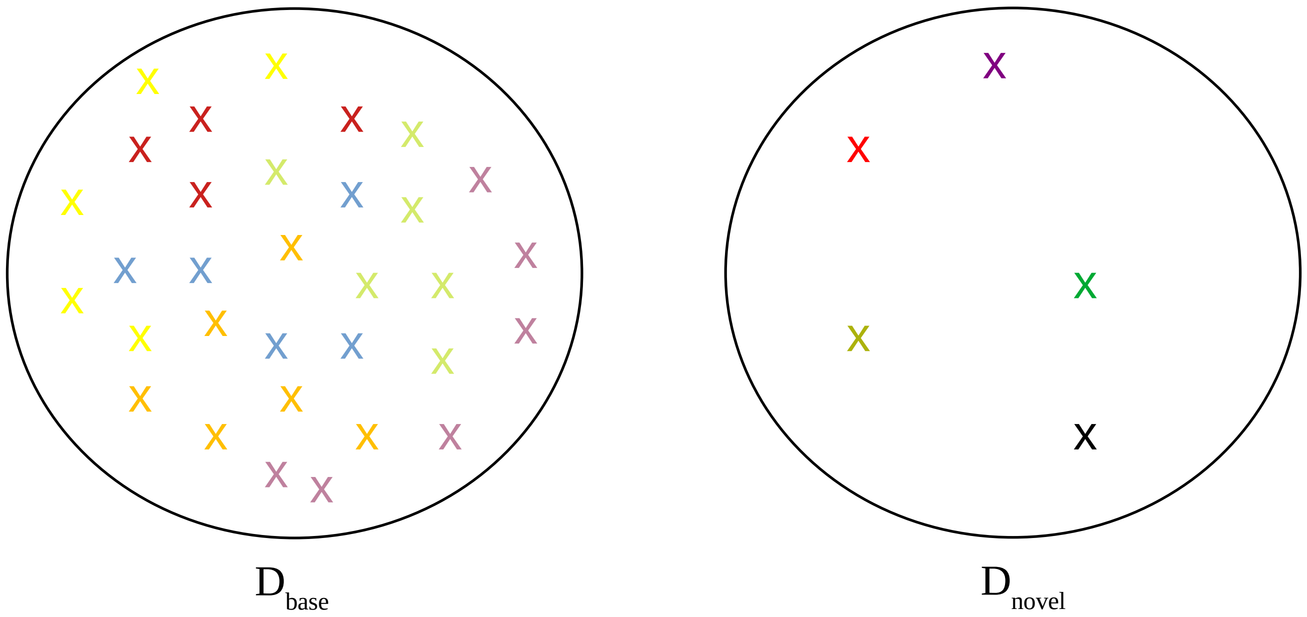


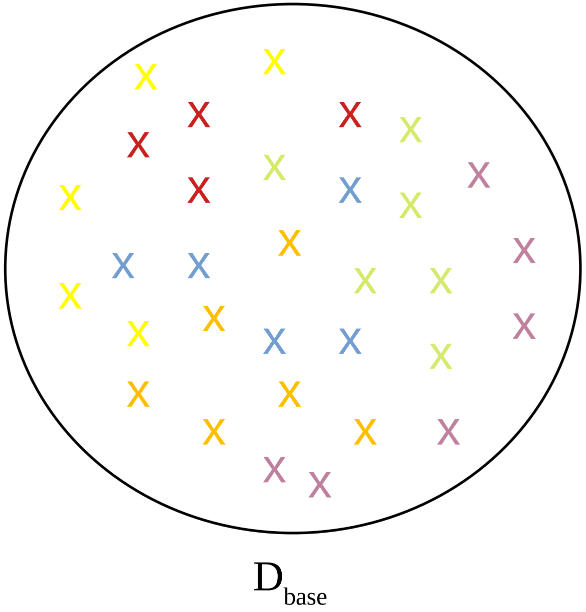
D_{novel}

Support set of *NK* samples

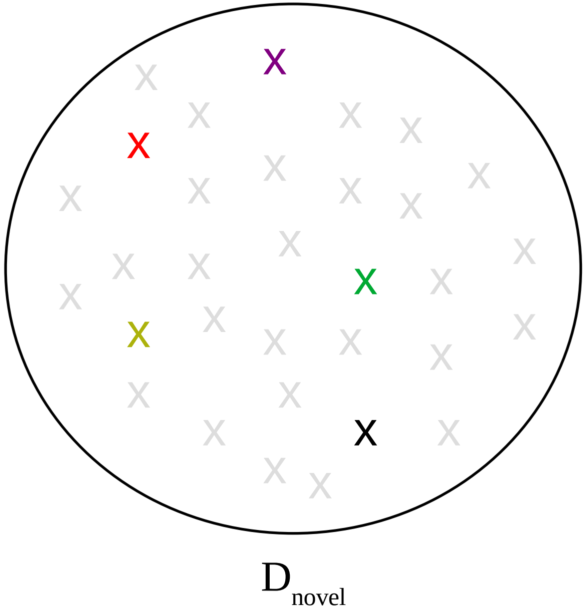
- *N* ways:
  Unseen classes
  Here, 5
- *K* shots:
  Labeled samples per class
  Here, 1
- Queries:
  Samples to classify
  From selected ways

Specific application case


D_{base}
D_{novel}



D_{base}



D_{novel}

Few-shot learning

└ The classical pipeline

Credit: Raphael Lafargue

Data
Transformations?
Hallucinations?
Distribution Calibration?

→

Backbone
What kind of NN?
Complexity?

→

Routine
Algorithm ?
Loss?

→

Preprocessing
Centering?
Projections?
Gaussian?

→

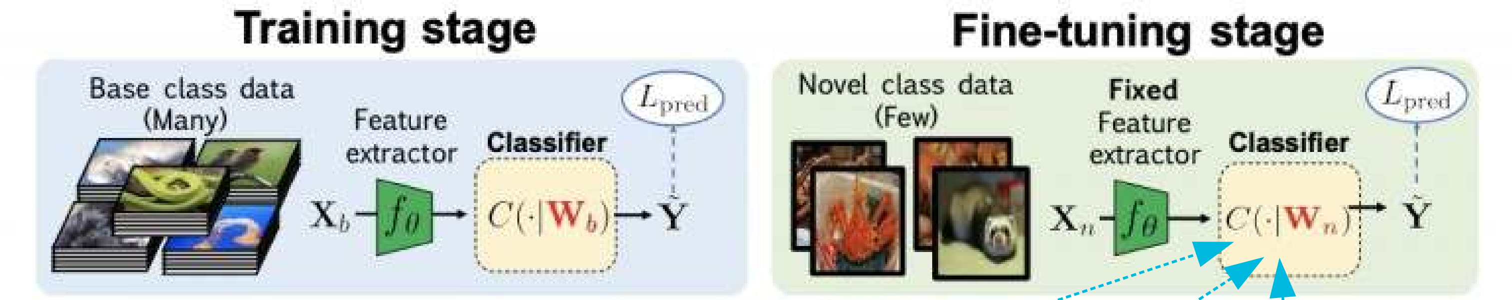
Classifier
Nearest Class Mean?
Cosine classifier?
Logistic regression?

→

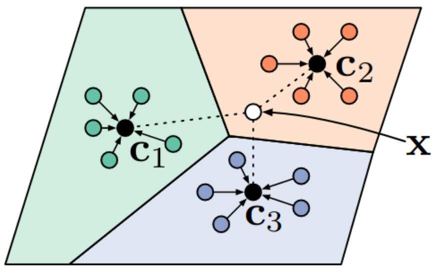
Label Prediction

Aim :
Understand and find boosts within these steps

https://opencv.org/understanding-transductive-few-shot-learning/
https://spin.atomicobject.com/2015/05/26/mean-shift-clustering/

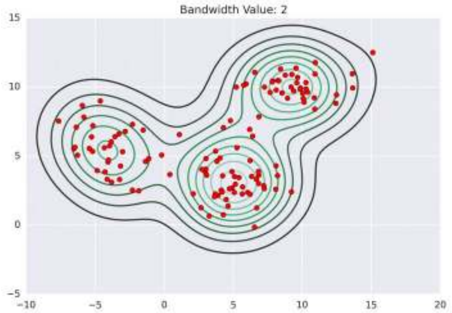


Same classifier
On validation set



Nearest class mean

Label propagation

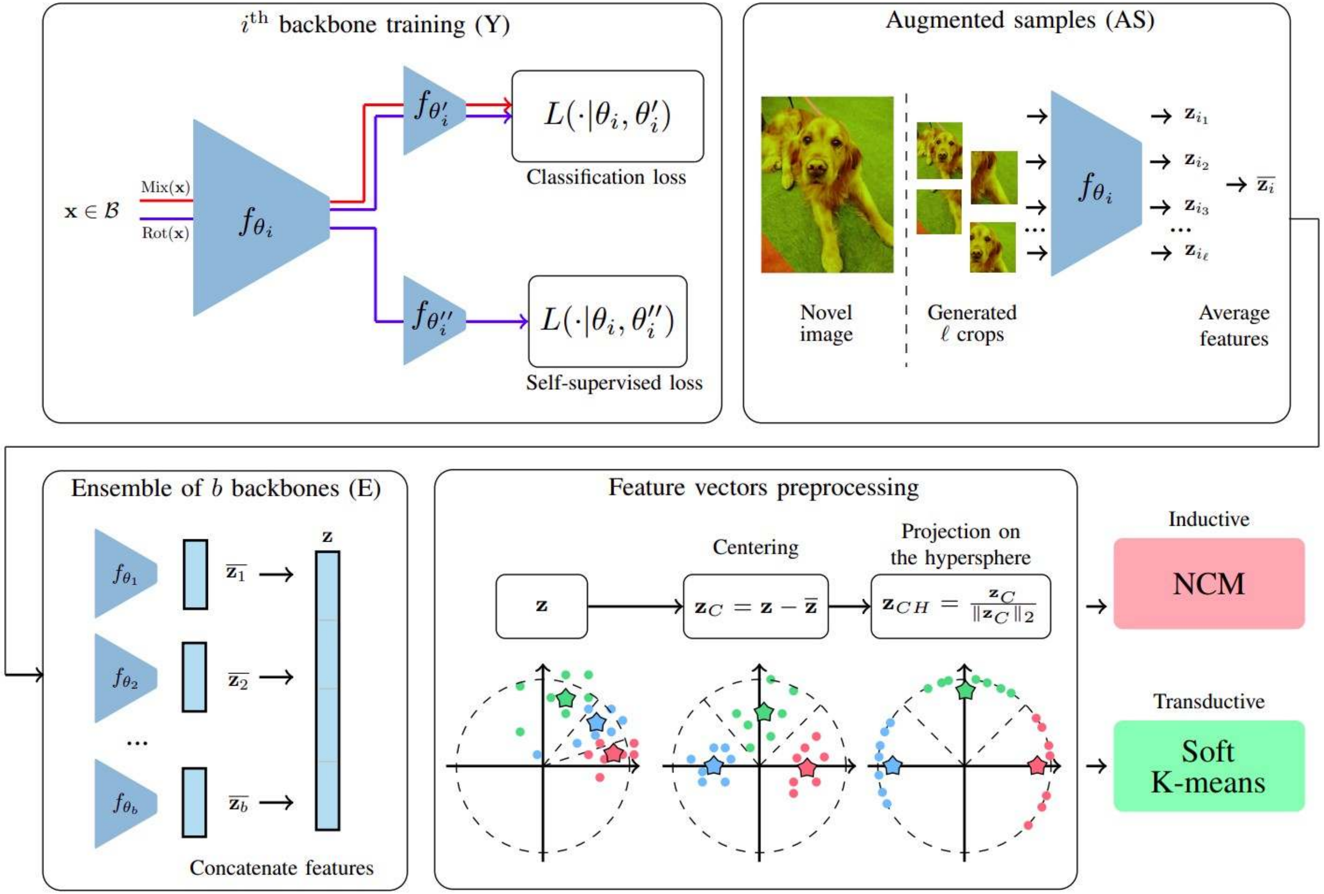


Mean shift

Few-shot learning

└ The EASY approach - Description

https://arxiv.org/pdf/2201.09699.pdf



https://arxiv.org/pdf/2201.09699.pdf

TABLE I
1-SHOT AND 5-SHOT ACCURACY OF STATE-OF-THE-ART METHODS AND PROPOSED SOLUTION ON **MINIIMAGENET** IN **INDUCTIVE** SETTING.

	Method	1-shot	5-shot
$\leq 12M$	SimpleShot [29]	62.85 ± 0.20	80.02 ± 0.14
	Baseline++ [30]	53.97 ± 0.79	75.90 ± 0.61
	TADAM [35]	58.50 ± 0.30	76.70 ± 0.30
	ProtoNet [10]	60.37 ± 0.83	78.02 ± 0.57
	R2-D2 (+ens) [20]	64.79 ± 0.45	81.08 ± 0.32
	FEAT [36]	66.78	82.05
	CNL [37]	67.96 ± 0.98	83.36 ± 0.51
	MERL [38]	67.40 ± 0.43	83.40 ± 0.28
	Deep EMD v2 [13]	68.77 ± 0.29	84.13 ± 0.53
	PAL [8]	69.37 ± 0.64	84.40 ± 0.44
	inv-equ [39]	67.28 ± 0.80	84.78 ± 0.50
	CSEI [40]	68.94 ± 0.28	85.07 ± 0.50
	COSOC [9]	69.28 ± 0.49	85.16 ± 0.42
	EASY 2×ResNet12$\left(\frac{1}{\sqrt{2}}\right)$ (ours)	$\mathbf{70.63 \pm 0.20}$	$\mathbf{86.28 \pm 0.12}$
$36M$	S2M2R [12]	64.93 ± 0.18	83.18 ± 0.11
	LR + DC [17]	68.55 ± 0.55	82.88 ± 0.42
	EASY 3×ResNet12 (ours)	$\mathbf{71.75 \pm 0.19}$	$\mathbf{87.15 \pm 0.12}$

TABLE VI
1-SHOT AND 5-SHOT ACCURACY OF STATE-OF-THE-ART METHODS AND PROPOSED SOLUTION ON **MINIIMAGENET** IN **TRANSDUCTIVE** SETTING.

	Method	1-shot	5-shot
$\leq 12M$	TIM-GD [42]	73.90	85.00
	ODC [43]	77.20 ± 0.36	87.11 ± 0.42
	PEM$_n$E-BMS* [32]	80.56 ± 0.27	87.98 ± 0.14
	SSR [44]	68.10 ± 0.60	76.90 ± 0.40
	iLPC [45]	69.79 ± 0.99	79.82 ± 0.55
	EPNet [31]	66.50 ± 0.89	81.60 ± 0.60
	DPGN [46]	67.77 ± 0.32	84.60 ± 0.43
	ECKPN [47]	70.48 ± 0.38	85.42 ± 0.46
	Rot+KD+POODLE [48]	77.56	85.81
	EASY 2×ResNet12$\left(\frac{1}{\sqrt{2}}\right)$ (ours)	$\mathbf{82.31 \pm 0.24}$	$\mathbf{88.57 \pm 0.12}$
$36M$	SSR [44]	72.40 ± 0.60	80.20 ± 0.40
	fine-tuning(train+val) [49]	68.11 ± 0.69	80.36 ± 0.50
	SIB+E^{3}BM [50]	71.40	81.20
	LR+DC [17]	68.57 ± 0.55	82.88 ± 0.42
	EPNet [31]	70.74 ± 0.85	84.34 ± 0.53
	TIM-GD [42]	77.80	87.40
	PT+MAP [51]	82.92 ± 0.26	88.82 ± 0.13
	iLPC [45]	83.05 ± 0.79	88.82 ± 0.42
	ODC [43]	80.64 ± 0.34	89.39 ± 0.39
	PEM$_n$E-BMS* [32]	83.35 ± 0.25	$\mathbf{89.53 \pm 0.13}$
	EASY 3×ResNet12 (ours)	$\mathbf{84.04 \pm 0.23}$	89.14 ± 0.11

TAKE-HOME MESSAGES

- Introduction
- From neural networks to deep learning
- In practice:
  - Transfer learning
  - Standard libraries
- Limitations & future
- Focus: few-shot learning

Deep learning is getting prevalent in society

It has many applications in many fields

It has become state of the art in most domains

TAKE-HOME MESSAGES

Various architectures tackle various tasks

Backpropagation is very important

Beware of overfitting!

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This is what most people do in practice

Allows exploiting already trained architectures

A NN consists of a backbone and a classifier

Training the classifier is necessary

TAKE-HOME MESSAGES

Powerful options such as Keras or Torch exist

Do not reinvent the wheel!

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TAKE-HOME MESSAGES

There is a big shift toward transformers

High ecological impact

Research in the field is very vast

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Sometimes, we don't have enough data even for transfer learning

Solutions exist and employ semi-supervised techniques

There are many more questions linked to few-shot learning, e.g., data selection

Workshop on FSL
IMT – 2-6/12/2024

https://wfsl.istc-conf.org/

TAKE-HOME MESSAGES

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BRAIn

https://www.imt-atlantique.fr/en/research-innovation/teams/brain

Workshop on FSL
IMT – 2-6/12/2024

https://wfsl.istc-conf.org/

(Deep) Neural Networks

IMT Atlantique
Bretagne-Pays de la Loire
École Mines-Télécom

Fundamentals of Machine Learning

22/10/2024